

Egyptian Baccalaureate Mathematics

Student Edition • Open Workbook / Guided Practice • Part2

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Concept 1 | Defining Angles and Basic Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Radians

$$\pi \text{ rad} = 180^\circ$$

Arc length

$$S = r\theta$$

Trig

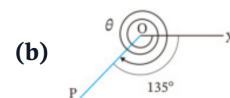
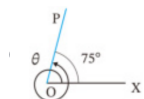
$$\sin \theta = \frac{y}{r}, \cos \theta = \frac{x}{r}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q001 Written

Find the measures of θ in the two figures: (a) one full counter-clockwise revolution plus 75° ; (b) two full clockwise revolutions and a further 135° .



Solution

1. Add $360^\circ + 75^\circ$ for (a).
2. Use $360^\circ \times (-2) - 135^\circ$ for (b).

Final answer

(a) 435° (b) -855°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q002 Written

Draw the terminal side for 300° and -450° .

Solution

1. Reduce each angle by complete turns while keeping the sign of rotation.
2. 300° ends in quadrant IV; -450° is coterminal with -90° .

Final answer

300° : QIV -450° : negative y – axis

Q003 Written

Express 420° and -210° as $\alpha + 360^\circ \times n$, where n is an integer and $0^\circ \leq \alpha < 360^\circ$.

Solution

1. Choose the coterminal angle in the interval $0^\circ \leq \alpha < 360^\circ$.
2. Write the full family by adding $360^\circ \times n$.

Final answer

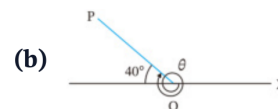
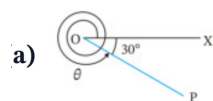
$60^\circ + 360^\circ \times n$; $150^\circ + 360^\circ \times n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q004 Written

Find the measures of θ in the two figures: (a) a clockwise turn of 30° from the positive x -axis; (b) a terminal side 40° above the negative x -axis, measured counter-clockwise from the positive x -axis).



2 Draw the terminal side for the following angles.

Solution

1. Clockwise rotation is negative, so (a) is -30° .
2. For (b), $180^\circ - 40^\circ = 140^\circ$.

Final answer

(a) -30° (b) 140°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q005 Written

Draw the terminal side for 500° and -120° .

Solution

1. Subtract or add complete turns: $500^\circ - 360^\circ = 140^\circ$; -120° is already in QIII.

Final answer

500° : *QI* -120° : *QIII*

Q006 Written

Express 520° and -140° as $\alpha + 360^\circ \times n$.

Solution

1. Reduce each angle to the principal representative in $[0^\circ, 360^\circ)$.
2. Attach $360^\circ n$.

Final answer

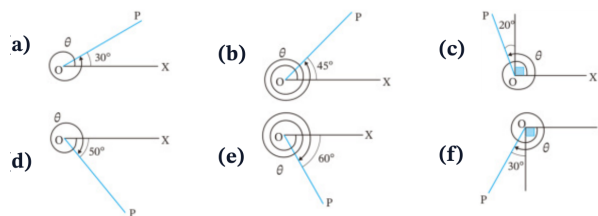
$160^\circ + 360^\circ \times n$; $220^\circ + 360^\circ \times n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q007 Written

Find the six diagram measures θ shown in the given figure (a)–(f).



2 Draw the terminal side for the following angles.

Solution

1. Read the marked acute angle and the direction of each complete turn.
2. Use $360^\circ \pm$ marked angle and $90^\circ \pm 20^\circ$ where shown.

Final answer

30° ; 405° ; 110° ; 310° ; 300° ; 240°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q008 Written

Draw the terminal sides for 470° , 780° , 225° , -380° , -990° , -200° .

Solution

1. Reduce each angle modulo 360° , preserving clockwise direction for negative angles.
2. Sketch the ray from the positive x -axis in the resulting quadrant.

Final answer

QII; QII; QIII; QI; QII; QII

Q009 Written

Express 610° , 920° , 840° , -430° , -960° , -20° as $\alpha + 360^\circ \times n$.

Solution

1. Find each representative α in $[0^\circ, 360^\circ)$.
2. The integer coefficient of 360° is absorbed into arbitrary integer n .

Final answer

$250^\circ + 360^\circ n$; $200^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $290^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $340^\circ + 360^\circ n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q010 **Written****Express 129720° and -2592100° as $\alpha + 360^\circ \times n$, with $0^\circ \leq \alpha < 360^\circ$.****Solution**

1. Divide by 360° and retain the remainder in the required interval.
2. Write each result as $\alpha + 360^\circ n$.

Final answer

$$120^\circ + 360^\circ n \quad ; \quad 260^\circ + 360^\circ n$$

Q011 **MCQ****The principal angle coterminal with 765° is:****Solution**

$$1. \quad 765^\circ - 720^\circ = 45^\circ$$

Correct option: A**Final answer**

$$45^\circ$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q012 MCQ

The principal angle coterminal with -410° is:**Solution**

$$1. -410^\circ + 720^\circ = 310^\circ$$

Correct option: C

Final answer 310°

Q013 MCQ

The terminal side of 540° lies on the:**Solution**

$$1. 540^\circ - 360^\circ = 180^\circ$$

Correct option: B

Final answernegative x -axis

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q014 MCQ

A positive angle of 315° terminates in:**Solution**1. $315^\circ \in \text{QIV}$

Correct option: D

Final answer

QIV

Q015 MCQ

Which is the general-angle form for -75° ?**Solution**1. $-75^\circ + 360^\circ = 285^\circ$

Correct option: B

Final answer $285^\circ + 360^\circ n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q016 MCQ

The direction of a negative angle is:**Solution**

1. Negative rotation is clockwise.

Correct option: B

Final answer

clockwise

Q017 MCQ

The terminal side of 1080° is the:**Solution**1. $1080^\circ = 3 \times 360^\circ$

Correct option: A

Final answerpositive x -axis

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q018 MCQ

For $\alpha = 0^\circ$, the general-angle family is:**Solution**

1. Substitute $\alpha = 0^\circ$ in $\alpha + 360^\circ n$.

Correct option: A

Final answer $360^\circ n$

Q019 MCQ

The principal angle for -990° is:**Solution**

1. $-990^\circ + 1080^\circ = 90^\circ$

Correct option: C

Final answer 90°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q020 MCQ

A ray (P) is 25° above the negative x -axis. Its standard-position angle is:

Solution

1. $180^\circ - 25^\circ = 155^\circ$

Correct option: B

Final answer

155°

Q021 Written

(1) Convert 630° to radians. (2) Convert $-\frac{7\pi}{4}$ to degrees. (3) For a sector with $r = 3$ and $\theta = \frac{4\pi}{3}$, find arc length L and area S .

Solution

1. Multiply degrees by $\pi/180$, and radians by $180/\pi$.
2. Use $L = r\theta$ and $S = \frac{1}{2}r^2\theta$ with θ in radians.

Final answer

$\frac{7\pi}{2}$; -315° ; $L = 4\pi$, $S = 6\pi$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q022 Written

Complete the standard-angle degree/radian table, then: convert $288^\circ, -400^\circ$ to radians; convert $\frac{11\pi}{6}, -\frac{3\pi}{5}$ to degrees; and find L, S for $r = 6, \theta = \frac{3\pi}{4}$.

Solution

1. Use the conversion multipliers and the standard table.
2. For a sector, substitute directly in $L = r\theta$ and $S = \frac{1}{2}r^2\theta$.

Final answer

$$\frac{8\pi}{5}, -\frac{20\pi}{9}; 330^\circ, -108^\circ; L = \frac{9\pi}{2}, S = \frac{27\pi}{2}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q023 Written

Convert $108^\circ, 36^\circ, -240^\circ, -225^\circ, 420^\circ, 400^\circ, -570^\circ, -2700^\circ$ to radians.**Solution**

1. Multiply every degree measure by $\pi/180$.
2. Reduce each fraction to lowest terms.

Final answer

$$\frac{3\pi}{5}, \frac{\pi}{5}, -\frac{4\pi}{3}, -\frac{5\pi}{4}, \frac{7\pi}{3}, \frac{20\pi}{9}, -\frac{19\pi}{6}, -15\pi$$

Q024 Written

Convert $\frac{7\pi}{6}, \frac{4\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{5\pi}{4}, -\frac{\pi}{2}, -\frac{11\pi}{2}, -\frac{7\pi}{18}$ to degrees.**Solution**

1. Multiply every radian measure by $180^\circ/\pi$.
2. Keep the sign and simplify.

Final answer

$$210^\circ, 144^\circ, 120^\circ, 240^\circ, -225^\circ, -90^\circ, -990^\circ, -70^\circ$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q025 Written

Find L and S for: (a) $r = 6, \theta = \frac{7\pi}{6}$; (b) $r = 4, \theta = \frac{2\pi}{5}$; (c) $r = 6, \theta = \frac{5\pi}{4}$.

Solution

1. Apply $L = r\theta$.
2. Apply $S = \frac{1}{2}r^2\theta$ to each sector.

Final answer

(a) $(7\pi, 21\pi)$; (b) $(\frac{8\pi}{5}, \frac{16\pi}{5})$; (c) $(\frac{15\pi}{2}, \frac{45\pi}{2})$

Q026 Written

Draw terminal sides for $\frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{3\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{6}, -\frac{\pi}{3}, -\frac{\pi}{4}$.

Solution

1. Use the standard radian-angle table and the sign of rotation.
2. Sketch each terminal ray from the positive x -axis.

Final answer

QI; QI; QII; QII; QIII; QIII; QIV; QIV

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q027 Written

A sector of radius 9 cm and central angle $4\pi/3$ is joined to form a cone. Determine the exact vertical height and total volume.

Solution

1. The sector arc becomes the cone base circumference: $2\pi R = 9(4\pi/3)$, so $R = 6$.
2. The slant height is 9, hence $h = \sqrt{9^2 - 6^2} = 3\sqrt{5}$.
3. Use $V = \frac{1}{3}\pi R^2 h$.

Final answer

$$h = 3\sqrt{5} \text{ cm}; \quad V = 36\pi\sqrt{5} \text{ cm}^3$$

Q028 MCQ

Convert 150° to radians:

Solution

1. $150 \times \pi/180 = 5\pi/6$

Correct option: A

Final answer

$$\frac{5\pi}{6}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q029 MCQ

Convert $-\frac{5\pi}{6}$ to degrees:**Solution**

$$1. -5\pi/6 \times 180/\pi = -150^\circ$$

Correct option: B

Final answer

$$-150^\circ$$

Q030 MCQ

For $r = 5, \theta = \frac{2\pi}{3}$, the arc length is:**Solution**

$$1. L = r\theta = 5(2\pi/3)$$

Correct option: B

Final answer

$$\frac{10\pi}{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q031 MCQ

For $r = 4$, $\theta = \frac{\pi}{2}$, the sector area is:

Solution

1. $S = \frac{1}{2}(4^2)(\pi/2) = 4\pi$

Correct option: B

Final answer

4π

Q032 MCQ

The radian measure of a full turn is:

Solution

1. A full circumference subtends 2π radians.

Correct option: B

Final answer

2π

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q033 MCQ

The degree measure of $\frac{7\pi}{4}$ is:

Solution

1. $7\pi/4 \times 180/\pi = 315^\circ$

Correct option: C

Final answer

315°

Q034 MCQ

The principal angle for $-\frac{11\pi}{6}$ is:

Solution

1. $-11\pi/6 + 2\pi = \pi/6$

Correct option: A

Final answer

$\frac{\pi}{6}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q035 MCQ

For $r = 3$, $\theta = \pi$, the sector area is:

Solution

1. $S = \frac{1}{2}(3^2)\pi = 9\pi/2$

Correct option: D

Final answer

$$\frac{9\pi}{2}$$

Q036 MCQ

The angle $\frac{5\pi}{3}$ lies in:

Solution

1. $5\pi/3 = 300^\circ$

Correct option: D

Final answer

QIV

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q037 Written

Convert 252° to radians.**Solution**

$$1. 252\pi/180 = 7\pi/5.$$

Final answer

$$\frac{7\pi}{5}$$

Q038 Written

A sector has $r = 8$ and $\theta = \frac{3\pi}{4}$. Find L and S .**Solution**

$$1. L = 8(3\pi/4) = 6\pi.$$

$$2. S = \frac{1}{2}(8^2)(3\pi/4) = 24\pi.$$

Final answer

$$L = 6\pi, \quad S = 24\pi$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q039 Written

If $\theta = \frac{4\pi}{3}$, find $\sin \theta$, $\cos \theta$, $\tan \theta$.

Solution

1. The terminal side is in QIII with reference angle $\pi/3$.
2. Use the signs of x, y on the unit circle.

Final answer

$$-\frac{\sqrt{3}}{2}, -\frac{1}{2}, \sqrt{3}$$

Q040 Written

Find the exact values (write undefined where appropriate): $\sin \frac{7\pi}{4}$, $\cos \frac{\pi}{3}$, $\tan \frac{\pi}{6}$, $\sin(-\frac{2\pi}{3})$, $\cos(-\pi)$, $\tan(-\frac{\pi}{2})$.

Solution

1. Locate each angle on the unit circle.
2. Read x, y , then use $\sin \theta = y$, $\cos \theta = x$, $\tan \theta = y/x$.

Final answer

$$-\frac{\sqrt{2}}{2}; \frac{1}{2}; \frac{1}{\sqrt{3}}; -\frac{\sqrt{3}}{2}; -1; \text{undefined}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q041 Written

Find the exact values of the eighteen trigonometric functions listed in the given exercise (a)–(r), including negative and undefined cases.

Find the values of the following trigonometric functions. If the value is undefined, write “undefined”.

(a) $\sin \frac{\pi}{3}$

(b) $\sin \frac{7}{6}\pi$

(c) $\sin \pi$

(d) $\cos \frac{5}{4}\pi$

(e) $\cos \frac{5}{3}\pi$

(f) $\cos \frac{7}{6}\pi$

(g) $\tan \frac{2}{3}\pi$

(h) $\tan \frac{5}{4}\pi$

(i) $\tan \pi$

(j) $\sin \left(-\frac{\pi}{6}\right)$

(k) $\sin \left(-\frac{5}{4}\pi\right)$

(l) $\sin \left(-\frac{5}{3}\pi\right)$

(m) $\cos \left(-\frac{\pi}{4}\right)$

(n) $\cos \left(-\frac{5}{6}\pi\right)$

(o) $\cos \left(-\frac{4}{3}\pi\right)$

(p) $\tan \left(-\frac{4}{3}\pi\right)$

(q) $\tan \left(-\frac{11}{6}\pi\right)$

(r) $\tan \left(-\frac{3}{2}\pi\right)$

Solution

1. Reduce each angle to a standard reference angle.
2. Apply quadrant signs and the unit-circle exact values; tangent is undefined when $x = 0$.

Final answer

$\frac{\sqrt{3}}{2}; -\frac{1}{2}; 0; -\frac{\sqrt{2}}{2}; \frac{1}{2}; -\frac{\sqrt{3}}{2}; -\sqrt{3}; 1; 0; -\frac{1}{2}; \frac{\sqrt{2}}{2}; \frac{\sqrt{3}}{2}; \frac{\sqrt{2}}{2}; -\frac{\sqrt{3}}{2}; -\frac{1}{2}; -\sqrt{3}; \frac{1}{\sqrt{3}}; \text{undefined}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q042 **Written**

The panel height is $h(t) = 3 \sin(\frac{2\pi}{3}t - \frac{\pi}{4}) + 1.5$. Find the exact height at $t = 0$.

Solution

1. Substitute $t = 0$: $h(0) = 3 \sin(-\pi/4) + 1.5$.
2. Use $\sin(-\pi/4) = -\sqrt{2}/2$.

Final answer

$$\frac{3(1 - \sqrt{2})}{2} \text{ m}$$

Q043 **MCQ**

$\sin \frac{\pi}{6}$ equals:

Solution

1. Use the QI special-angle value.

Correct option: B

Final answer

$$\frac{1}{2}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q044 MCQ

 $\cos \frac{2\pi}{3}$ equals:**Solution**1. QII has negative cosine; reference angle $\pi/3$.

Correct option: B

Final answer

$$-\frac{1}{2}$$

Q045 MCQ

 $\tan \frac{3\pi}{4}$ equals:**Solution**1. QII reference angle $\pi/4$.

Correct option: A

Final answer

$$-1$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q046 MCQ

 $\sin(-\frac{\pi}{2})$ equals:**Solution**1. The terminal point is $(0, -1)$.

Correct option: A

Final answer

-1

Q047 MCQ

 $\tan \frac{\pi}{2}$ is:**Solution**1. $\tan = y/x$ and $x = 0$.

Correct option: D

Final answer

undefined

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q048 MCQ

 $\cos \frac{5\pi}{3}$ equals:**Solution**1. $5\pi/3$ has reference angle $\pi/3$ in QIV.

Correct option: D

Final answer

$$\frac{1}{2}$$

Q049 MCQ

If $P(x, y)$ lies on the unit circle, then $\sin \theta$ is:**Solution**1. By definition, $\sin \theta = y$.

Correct option: B

Final answer

$$y$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q050 MCQ

 $\tan\left(-\frac{3\pi}{4}\right)$ equals:**Solution**1. Tangent is odd: $\tan(-3\pi/4) = -\tan(3\pi/4) = 1$.

Correct option: C

Final answer

1

Q051 MCQ

 $\sin \frac{11\pi}{6}$ equals:**Solution**1. QIV reference angle $\pi/6$, so sine is negative.

Correct option: A

Final answer $-\frac{1}{2}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q052 Written

Find $\sin \frac{5\pi}{6}, \cos \frac{4\pi}{3}, \tan \frac{7\pi}{4}$.**Solution**

1. Use reference angles $\pi/6, \pi/3, \pi/4$ and quadrant signs.

Final answer

$$\frac{1}{2}, -\frac{1}{2}, -1$$

Q053 Written

Find $\sin(-\frac{5\pi}{6}), \cos(-\frac{7\pi}{6}), \tan(-\frac{\pi}{3})$.**Solution**

1. Reduce negative angles and apply odd/even symmetry.

Final answer

$$-\frac{1}{2}, -\frac{\sqrt{3}}{2}, -\sqrt{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q054 Written

For $P = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$ on the unit circle, find the corresponding $\sin \theta$, $\cos \theta$, $\tan \theta$.

Solution

1. Read x, y directly, then compute y/x .

Final answer

$$\frac{\sqrt{3}}{2}, -\frac{1}{2}, -\sqrt{3}$$

Q055 Written

Find the exact value of $3 \sin \frac{3\pi}{2} - 2 \cos \pi$.

Solution

1. $3(-1) - 2(-1) = -1$.

Final answer

$$-1$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q056 Written

State the two standard angles where $\tan \theta$ is undefined in $[0, 2\pi)$.**Solution**

1. Tangent is undefined when the unit-circle x -coordinate is zero.

Final answer

$$\frac{\pi}{2}, \frac{3\pi}{2}$$

Q057 Written

Solve for θ , $0 \leq \theta < 2\pi$: (a) $\sin \theta = -\frac{1}{2}$; (b) $\cos \theta = \frac{1}{\sqrt{2}}$; (c) $\tan \theta = -\sqrt{3}$.**Solution**

1. Draw the unit circle and the corresponding horizontal/vertical/oblique line.
2. Use the reference angle and select the quadrants with the required sign.

Final answer

$$(a) \frac{7\pi}{6}, \frac{11\pi}{6}; (b) \frac{\pi}{4}, \frac{7\pi}{4}; (c) \frac{2\pi}{3}, \frac{5\pi}{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q058 Written

Solve the six trigonometric equations in the given Try set for $0 \leq \theta < 2\pi$: $\sin \theta = 1/\sqrt{2}$, $\cos \theta = -\sqrt{3}/2$, $\tan \theta = 1/\sqrt{3}$, $\sin \theta = 0$, $\cos \theta = 1/2$, $\tan \theta = -1$.

Solution

1. Find the reference angle from the special triangle.
2. Use the sign of the function to choose the correct quadrants.

Final answer

$$\frac{\pi}{4}, \frac{3\pi}{4}; \frac{5\pi}{6}, \frac{7\pi}{6}; \frac{\pi}{6}, \frac{7\pi}{6}; 0, \pi; \frac{\pi}{3}, \frac{5\pi}{3}; \frac{3\pi}{4}, \frac{7\pi}{4}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q059 Written

Solve the sixteen equations in the given exercise set (a)–(p) for $0 \leq \theta < 2\pi$.

Solution

1. Apply the same unit-circle/reference-angle method to each equation.
2. Remember that sine/cosine can have two solutions, while $\sin \theta = \pm 1$ and $\cos \theta = \pm 1$ have one in the interval.

Final answer

$$\frac{\pi}{6}, \frac{5\pi}{6}; \frac{\pi}{3}, \frac{2\pi}{3}; \frac{\pi}{2}, \frac{3\pi}{2}; 0; 0, \pi; \frac{\pi}{3}, \frac{4\pi}{3}; \frac{\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}; \frac{3\pi}{4}, \frac{5\pi}{4}; \frac{2\pi}{3}, \frac{4\pi}{3}; \frac{5\pi}{6}, \frac{11\pi}{6}; \frac{\pi}{4}, \frac{5\pi}{4}; \frac{4\pi}{3}, \frac{5\pi}{3}; \frac{3\pi}{2}; \pi; \frac{\pi}{6}, \frac{11\pi}{6}$$

Q060 Written

Solve $\tan^2 \theta + (1 - \sqrt{3}) \tan \theta = \sqrt{3}$ for $0 \leq \theta < 360^\circ$.

Solution

1. Let $u = \tan \theta$ and factor $u^2 + (1 - \sqrt{3})u - \sqrt{3} = 0$.
2. The two tangent values are 1 and $-\sqrt{3}$; list their angles in $[0, 360^\circ)$.

Final answer

$$60^\circ, 135^\circ, 240^\circ, 315^\circ$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q061 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = \frac{1}{2}$ gives:

Solution

1. Sine is positive in QI and QII with reference angle $\pi/6$.

Correct option: A

Final answer

$$\frac{\pi}{6}, \frac{5\pi}{6}$$

Q062 MCQ

For $0 \leq \theta < 2\pi$, $\cos \theta = -\frac{1}{2}$ gives:

Solution

1. Cosine is negative in QII and QIII.

Correct option: B

Final answer

$$\frac{2\pi}{3}, \frac{4\pi}{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q063 MCQ

For $0 \leq \theta < 2\pi$, $\tan \theta = 1$ gives:**Solution**

1. Tangent is positive in QI and QIII.

Correct option: A**Final answer**

$$\frac{\pi}{4}, \frac{5\pi}{4}$$

Q064 MCQ

The solutions of $\sin \theta = 0$ in $[0, 2\pi)$ are:**Solution**

1. The unit-circle y-coordinate is zero on the x-axis.

Correct option: A**Final answer**

$$0, \pi$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q065 MCQ

If $\cos \theta = 1$, then in $[0, 2\pi)$, θ is:**Solution**1. The point $(1, 0)$ corresponds to $\theta = 0$.**Correct option:** A**Final answer**

0

Q066 MCQ

For $0 \leq \theta < 2\pi$, $\tan \theta = -\sqrt{3}$ gives:**Solution**1. Tangent is negative in QII/QIV; reference angle $\pi/3$.**Correct option:** B**Final answer** $\frac{2\pi}{3}, \frac{5\pi}{3}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q067 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = -1$ gives:**Solution**

1. The lowest unit-circle point has y-coordinate -1 .

Correct option: B**Final answer**

$$\frac{3\pi}{2}$$

Q068 MCQ

For $0 \leq \theta < 2\pi$, $\cos \theta = 0$ gives:**Solution**

1. Cosine is the x-coordinate, zero on the y-axis.

Correct option: B**Final answer**

$$\frac{\pi}{2}, \frac{3\pi}{2}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q069 MCQ

The equation $\tan \theta = 0$ has solutions in $[0, 2\pi)$:

Solution

1. Tangent is zero when $y = 0$ and $x \neq 0$.

Correct option: A

Final answer

$0, \pi$

Q070 Written

Solve $\cos \theta = \frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$.

Solution

1. Cosine is positive in QI/QIV with reference angle $\pi/6$.

Final answer

$\frac{\pi}{6}, \frac{11\pi}{6}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q071 Written

Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$.

Solution

1. Sine is negative in QIII/QIV with reference angle $\pi/3$.

Final answer

$$\frac{4\pi}{3}, \frac{5\pi}{3}$$

Q072 Written

Solve $\tan \theta = \frac{1}{\sqrt{3}}$ for $0 \leq \theta < 2\pi$.

Solution

1. Tangent is positive in QI/QIII with reference angle $\pi/6$.

Final answer

$$\frac{\pi}{6}, \frac{7\pi}{6}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q073 Written

Solve $2 \cos \theta - 1 = 0$ for $0 \leq \theta < 2\pi$.**Solution**1. Rearrange to $\cos \theta = 1/2$, then choose QI/QIV.**Final answer**

$$\frac{\pi}{3}, \frac{5\pi}{3}$$

Q074 Written

Solve $2 \sin \theta + \sqrt{2} = 0$ for $0 \leq \theta < 2\pi$.**Solution**1. Rearrange to $\sin \theta = -\sqrt{2}/2$, then choose QIII/QIV.**Final answer**

$$\frac{5\pi}{4}, \frac{7\pi}{4}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 1 **Concept Quiz · MCQ**

The radian measure of 240° is:

Solution

1. $240\pi/180 = 4\pi/3$.

Correct option: B

Final answer

$$\frac{4\pi}{3}$$

Quiz MCQ 2 **Concept Quiz · MCQ**

If $\sin \theta = -\frac{\sqrt{3}}{2}$, $0 \leq \theta < 2\pi$, then θ is:

Solution

1. Sine is negative in QIII/QIV with reference angle $\pi/3$.

Correct option: B

Final answer

$$\frac{4\pi}{3}, \frac{5\pi}{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 3 Concept Quiz · MCQ

For a sector with $r = 6$ and $\theta = \frac{\pi}{3}$, the area is:

Solution

1. $S = \frac{1}{2}(6^2)(\pi/3) = 6\pi$.

Correct option: C

Final answer

$$6\pi$$

Quiz MCQ 4 Concept Quiz · MCQ

The general-angle family for -300° is:

Solution

1. Add 360° to obtain the principal angle 60° .

Correct option: A

Final answer

$$60^\circ + 360^\circ n$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written

Solve $\cos \theta = -\frac{\sqrt{2}}{2}$ for $0 \leq \theta < 2\pi$.

Solution

1. Cosine is negative in QII/QIII with reference angle $\pi/4$.

Final answer

$$\frac{3\pi}{4}, \frac{5\pi}{4}$$

Quiz WRITTEN 6 Concept Quiz · Written

Convert $-\frac{13\pi}{6}$ to degrees and state its principal angle.

Solution

1. Multiply by $180/\pi$: -390° . Add 720° for the principal representative 330° .

Final answer

$$-390^\circ; 330^\circ$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written

Find L and S for a sector with $r = 10$ and $\theta = \frac{2\pi}{5}$.

Solution

1. $L = 10(2\pi/5) = 4\pi$.
2. $S = \frac{1}{2}(10^2)(2\pi/5) = 20\pi$.

Final answer

$$L = 4\pi, \quad S = 20\pi$$

Quiz WRITTEN 8 Concept Quiz · Written

Evaluate $3 \sin \frac{7\pi}{6} - 2 \cos \frac{4\pi}{3}$.

Solution

1. $3(-1/2) - 2(-1/2) = -1/2$.

Final answer

$$-\frac{1}{2}$$

Answer key

Use the question number shown on each page.

Q001 (a) 435° (b) -855°

Q002 300° : QIV

-450° : *negative y - axis*

Q003 $60^\circ + 360^\circ \times n$; $150^\circ + 360^\circ \times n$

Q004 (a) -30° (b) 140°

Q005 500° : QI -120° : QIII

Q006 $160^\circ + 360^\circ \times n$; $220^\circ + 360^\circ \times n$

Q007 30° ; 405° ; 110° ; 310° ; 300° ; 240°

Q008 QII; QII; QIII; QI; QII; QII

Q009 $250^\circ + 360^\circ n$; $200^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $290^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $340^\circ + 360^\circ n$

Q010 $120^\circ + 360^\circ n$; $260^\circ + 360^\circ n$

Q011 A

Q012 C

Q013 B

Q014 D

Q015 B

Q016 B

Q017 A

Q018 A

Q019 C

Q020 B

Q021 $\frac{7\pi}{2}$; -315° ; $L = 4\pi$, $S = 6\pi$

Q022 $\frac{8\pi}{5}$, $-\frac{20\pi}{9}$; 330° , -108° ; $L = \frac{9\pi}{2}$, $S = \frac{27\pi}{2}$

Q023 $\frac{3\pi}{5}$, $\frac{\pi}{5}$, $-\frac{4\pi}{3}$, $-\frac{5\pi}{4}$, $\frac{7\pi}{3}$, $\frac{20\pi}{9}$, $-\frac{19\pi}{6}$, -15π

Q024 210° , 144° , 120° , 240° , -225° , -90° , -990° , -70°

Q025 (a) $(7\pi, 21\pi)$; (b) $(\frac{8\pi}{5}, \frac{16\pi}{5})$; (c) $(\frac{15\pi}{2}, \frac{45\pi}{2})$

Q026 QI; QI; QII; QII; QIII; QIII; QIV; QIV

Q027 $h = 3\sqrt{5}$ cm; $V = 36\pi\sqrt{5}$ cm³

Q028 A

Q029 B

Q030 B

Q031 B

Q032 B

Q033 C

Q034 A

Q035 D

Q036 D

Q037 $\frac{7\pi}{5}$

Q038 $L = 6\pi$, $S = 24\pi$

Q039 $-\frac{\sqrt{3}}{2}$, $-\frac{1}{2}$, $\sqrt{3}$

Q040 $-\frac{\sqrt{2}}{2}$; $\frac{1}{2}$; $\frac{1}{\sqrt{3}}$; $-\frac{\sqrt{3}}{2}$; -1 ; *undefined*

Q041 $\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; 0 ; $-\frac{\sqrt{2}}{2}$; $\frac{1}{2}$; $-\frac{\sqrt{3}}{2}$; $-\sqrt{3}$; 1 ; 0 ; $-\frac{1}{2}$; $\frac{\sqrt{2}}{2}$; $\frac{\sqrt{3}}{2}$; $-\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; $-\sqrt{3}$; $\frac{1}{\sqrt{3}}$; *undefined*

Q042 $\frac{3(1-\sqrt{2})}{2}$ m

Q043 B

Q044 B

Q045 A

Q046 A

Q047 D

Q048 D

Q049 B

Q050 C

Q051 A

Q052 $\frac{1}{2}$, $-\frac{1}{2}$, -1

Q053 $-\frac{1}{2}$, $-\frac{\sqrt{3}}{2}$, $-\sqrt{3}$

Q054 $\frac{\sqrt{3}}{2}$, $-\frac{1}{2}$, $-\sqrt{3}$

Q055 -1

Q056 $\frac{\pi}{2}$, $\frac{3\pi}{2}$

Q057 (a) $\frac{7\pi}{6}$, $\frac{11\pi}{6}$; (b) $\frac{\pi}{4}$, $\frac{7\pi}{4}$; (c) $\frac{2\pi}{3}$, $\frac{5\pi}{3}$

Q058 $\frac{\pi}{4}$, $\frac{3\pi}{4}$; $\frac{5\pi}{6}$, $\frac{7\pi}{6}$; $\frac{\pi}{6}$, $\frac{7\pi}{6}$; 0 , π ; $\frac{\pi}{3}$, $\frac{5\pi}{3}$; $\frac{3\pi}{4}$, $\frac{7\pi}{4}$

Q059 $\frac{\pi}{6}$, $\frac{5\pi}{6}$; $\frac{\pi}{3}$, $\frac{2\pi}{3}$; $\frac{\pi}{2}$, $\frac{3\pi}{2}$; 0 , π ; $\frac{4\pi}{3}$, $\frac{2\pi}{3}$; $\frac{\pi}{4}$, $\frac{5\pi}{4}$; $\frac{3\pi}{4}$, $\frac{5\pi}{4}$; $\frac{2\pi}{3}$, $\frac{4\pi}{3}$; $\frac{5\pi}{6}$, $\frac{7\pi}{6}$; $\frac{\pi}{6}$, $\frac{5\pi}{6}$; $\frac{\pi}{4}$, $\frac{3\pi}{4}$; $\frac{5\pi}{3}$, $\frac{4\pi}{3}$; $\frac{3\pi}{2}$, $\frac{1\pi}{2}$

Q060 60° , 135° , 240° , 315°

Q061 $\frac{1}{2}$, $\frac{\sqrt{2}}{2}$; $\frac{\sqrt{3}}{2}$; $\frac{\sqrt{2}}{2}$; $-\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; $-\sqrt{3}$; $\frac{1}{\sqrt{3}}$; *undefined*

Q062 B

Q063 A

Q064 A

Q065 A

Q066 B

Q067 B

Q068 B

Q069 A

Q070 $\frac{\pi}{6}$, $\frac{11\pi}{6}$

Q071 $\frac{4\pi}{3}$, $\frac{5\pi}{3}$

Q072 $\frac{\pi}{6}$, $\frac{7\pi}{6}$

Q073 $\frac{\pi}{3}$, $\frac{5\pi}{3}$

Q074 $\frac{5\pi}{4}$, $\frac{7\pi}{4}$

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{3\pi}{4}$, $\frac{5\pi}{4}$

Quiz WRITTEN 6 -390° ; 330°

Quiz WRITTEN 7 $L = 4\pi$, $S = 20\pi$

Quiz WRITTEN 8 $-\frac{1}{2}$

Concept 2 | Trigonometric Identities and Expression Evaluation

English Student Edition • Focused formulas and guided practice

Formula opener

Identity

$$\sin^2 \theta + \cos^2 \theta = 1$$

Tangent

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Secant

$$1 + \tan^2 \theta = \sec^2 \theta$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q001 Written

Given (a) $\sin \theta = -\frac{1}{5}$ in QIII and (b) $\tan \theta = -\sqrt{6}$ with $\frac{3\pi}{2} < \theta < 2\pi$, find the other two trigonometric values in each case.

Solution

1. Use $\sin^2 \theta + \cos^2 \theta = 1$, then choose the sign from the quadrant.
2. Use $\tan \theta = \sin \theta / \cos \theta$ (or $1 + \tan^2 \theta = 1 / \cos^2 \theta$).

Final answer

$$(a) \cos \theta = -\frac{2\sqrt{6}}{5}, \tan \theta = \frac{\sqrt{6}}{12}; \quad (b) \cos \theta = \frac{\sqrt{7}}{7}, \sin \theta = -\frac{\sqrt{42}}{7}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q002 Written

Given one value and the quadrant, find the other two: (a) $\cos \theta = -1/\sqrt{3}$, QIII; (b) $\tan \theta = -2$, QIV; (c) $\sin \theta = 3/5$, QII; (d) $\tan \theta = \sqrt{3}/2$, $\pi < \theta < 3\pi/2$.

Solution

1. Find the missing squared value from a Pythagorean identity.
2. Select signs using the stated quadrant, then form the remaining ratio.

Final answer

$$(a) \sin \theta = -\frac{\sqrt{6}}{3}, \tan \theta = \sqrt{2}; (b) \cos \theta = \frac{\sqrt{5}}{5}, \sin \theta = -\frac{2\sqrt{5}}{5}; (c) \cos \theta = -\frac{4}{5}, \tan \theta = -\frac{3}{4}; (d) \sin \theta = -\frac{\sqrt{21}}{7}, \cos \theta = -\frac{2\sqrt{7}}{7}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q003 Written

For each of the twelve numbered cases in the given Exercise, one of $\sin \theta$, $\cos \theta$, $\tan \theta$ and a quadrant/interval is given. Find the other two values.

Solution

1. Square the known ratio and use $\sin^2 \theta + \cos^2 \theta = 1$ or $1 + \tan^2 \theta = 1/\cos^2 \theta$.
2. Apply the quadrant sign before simplifying the final ratio.

Final answer

$$(a) \left(-\frac{\sqrt{7}}{4}, \frac{3\sqrt{7}}{7}\right) (b) \left(-\frac{\sqrt{5}}{5}, -\frac{1}{2}\right) (c) \left(\frac{\sqrt{2}}{4}, -\frac{\sqrt{14}}{4}\right) (d) \left(-\frac{\sqrt{5}}{3}, -\frac{2\sqrt{5}}{5}\right) (e) \left(-\frac{\sqrt{15}}{4}, -\sqrt{15}\right) (f) \left(\frac{1}{3}, -\frac{2\sqrt{2}}{3}\right) (g) \left(-\frac{3\sqrt{10}}{10}, \frac{\sqrt{10}}{10}\right) (h) \left(-\frac{\sqrt{5}}{5}, -2\right) (i) \left(-\frac{12}{13}, -\frac{12}{5}\right) (j) \left(\frac{12}{13}, -\frac{5}{12}\right) (k) \left(-\frac{3}{5}, \frac{3}{4}\right) (l) \left(\frac{2\sqrt{5}}{5}, -\frac{\sqrt{5}}{5}\right)$$

Q004 Written

Using trigonometric identities, find the other two values given $\tan \theta + \cot \theta = 12$ and $\pi < \theta < 2\pi$.

Solution

1. Use $\tan \theta + \cot \theta = (\sin^2 \theta + \cos^2 \theta)/(\sin \theta \cos \theta)$ to obtain $\sin \theta \cos \theta = 1/12$.
2. The interval gives QIII or QIV; take the pair consistent with the stated signs.

Final answer

$$\sin \theta = -\frac{1}{2}, \quad \cos \theta = -\frac{\sqrt{3}}{2} \quad \text{or the swapped pair}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q005 Written

Given $\sin \theta = \frac{5}{13}$ **in QII, find** $\cos \theta$ **and** $\tan \theta$.

Solution

1. Use $\cos^2 \theta = 1 - 25/169$ and QII signs.

Final answer

$$-\frac{12}{13}, -\frac{5}{12}$$

Q006 Written

Given $\cos \theta = -\frac{3}{5}$ **in QIII, find** $\sin \theta$ **and** $\tan \theta$.

Solution

1. Use the Pythagorean identity and QIII signs.

Final answer

$$-\frac{4}{5}, \frac{4}{3}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q007 Written

Given $\tan \theta = \frac{3}{4}$ **in QI, find** $\sin \theta$ **and** $\cos \theta$.

Solution

1. Use a 3-4-5 triangle with positive signs.

Final answer

$$\frac{3}{5}, \frac{4}{5}$$

Q008 Written

Given $\tan \theta = -\frac{5}{12}$ **in QIV, find** $\sin \theta$ **and** $\cos \theta$.

Solution

1. Use a 5-12-13 triangle and QIV signs.

Final answer

$$-\frac{5}{13}, \frac{12}{13}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q009 Written

Given $\sin \theta = -\frac{\sqrt{2}}{2}$ **in QIII, find** $\cos \theta$ **and** $\tan \theta$.

Solution

1. The reference angle is $\pi/4$; both sine and cosine are negative in QIII.

Final answer

$$-\frac{\sqrt{2}}{2}, 1$$

Q010 MCQ

If $\sin \theta = -\frac{3}{5}$ **in QIII, $\cos \theta$ is:**

Solution

1. QIII makes cosine negative.

Correct option: B**Final answer**

$$-\frac{4}{5}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q011 MCQ

If $\tan \theta = 2$ in QI, $\sin \theta$ is:**Solution**

1. Use a 1-2- $\sqrt{5}$ triangle.

Correct option: A

Final answer

$$\frac{2\sqrt{5}}{5}$$

Q012 MCQ

If $\cos \theta = \frac{1}{\sqrt{2}}$ in QIV, $\tan \theta$ is:**Solution**

1. The reference angle is $\pi/4$ and tangent is negative in QIV.

Correct option: B

Final answer

$$-1$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q013 MCQ

For $\tan \theta = -\sqrt{3}$ in QII, $\sin \theta$ is:

Solution

1. Reference angle $\pi/3$, sine positive in QII.

Correct option: C

Final answer

$$\frac{\sqrt{3}}{2}$$

Q014 MCQ

If $\sin \theta = \frac{7}{25}$ in QI, $\tan \theta$ is:

Solution

1. The triangle is 7-24-25.

Correct option: A

Final answer

$$\frac{7}{24}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q015 Written

Given $\sin \theta + \cos \theta = \frac{1}{2}$, **with** θ **in QII**, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Solution

1. Square the sum: $1 + 2 \sin \theta \cos \theta = 1/4$.
2. Use $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$, then choose the QII sign for (c).

Final answer

$$(a) -\frac{3}{8} \quad (b) \frac{11}{16} \quad (c) \frac{\sqrt{7}}{2}$$

Q016 Written

Given $\sin \theta + \cos \theta = \frac{1}{3}$, **with** θ **in QII**, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Solution

1. Square the given sum to find the product.
2. Use the sum-of-cubes identity and the quadrant sign.

Final answer

$$(a) -\frac{4}{9} \quad (b) \frac{37}{27} \quad (c) \frac{\sqrt{17}}{3}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q017 Written

Solve the three numbered Exercise problems: (1) given $\sin \theta - \cos \theta = \sqrt{3}/2$ in QI; (2) given $\sin \theta + \cos \theta = \sqrt{3}/2$ in QIV; (3) given $\sin \theta - \cos \theta = 1/2$ in QIII. Evaluate the three expressions requested in each case.

Solution

1. Square the given sum or difference to obtain the product.
2. Use the sum/difference of cubes and select the sign from the quadrant.

Final answer

$$(1) \sin \theta \cos \theta = \frac{1}{8}, \sin^3 \theta - \cos^3 \theta = \frac{7\sqrt{3}}{16}, \sin \theta + \cos \theta = \frac{\sqrt{5}}{2}; (2) -\frac{1}{8}, \frac{9\sqrt{3}}{16}, -\frac{\sqrt{5}}{2}; (3) \frac{3}{8}, \frac{11}{16}, -\frac{\sqrt{7}}{2}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q018 Written

Given $\sin \theta - \cos \theta = 0.5$ **and** θ **in** QIII, **evaluate the exact value of** $\sin^6 \theta - \cos^6 \theta$.

Solution

1. Find $\sin \theta \cos \theta = 3/8$, then $\sin^3 \theta - \cos^3 \theta = 11/16$ and $\sin^3 \theta + \cos^3 \theta = -5\sqrt{7}/16$.
2. Multiply the last two expressions.

Final answer

$$-\frac{55\sqrt{7}}{256}$$

Q019 Written

Given $\sin \theta + \cos \theta = \frac{2}{3}$ **in** QII, **find** $\sin \theta \cos \theta$ **and** $\sin \theta - \cos \theta$.

Solution

1. Square the sum and use QII signs.

Final answer

$$-\frac{5}{18}, \frac{\sqrt{14}}{3}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q020 Written

Given $\sin \theta - \cos \theta = \frac{\sqrt{5}}{2}$ **in QI, find** $\sin \theta \cos \theta$ **and** $\sin^3 \theta - \cos^3 \theta$.

Solution

1. Use $(s - c)^2 = 1 - 2sc$, then the difference-of-cubes identity.

Final answer

$$-\frac{1}{8}, \frac{7\sqrt{5}}{16}$$

Q021 Written

Given $\sin \theta + \cos \theta = -\frac{1}{2}$ **in QIII, find** $\sin^3 \theta + \cos^3 \theta$.

Solution

1. Find $sc = -3/8$, then apply the sum-of-cubes identity.

Final answer

$$-\frac{11}{16}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q022 Written

Given $\sin \theta - \cos \theta = \frac{1}{3}$ **in QIV, find** $\sin \theta + \cos \theta$.

Solution

1. Find $sc = 4/9$, so the square of the sum is $17/9$; use QIV sign.

Final answer

$$\frac{\sqrt{11}}{3}$$

Q023 Written

Given $\sin \theta - \cos \theta = 1$ **in QIII, find** $\sin^3 \theta - \cos^3 \theta$.

Solution

1. The product is zero and the difference-of-cubes factor is $1 + sc$.

Final answer

$$\frac{3}{2}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q024 MCQ

If $s + c = \frac{1}{2}$, then sc equals:

Solution

1. Square the sum and use $s^2 + c^2 = 1$.

Correct option: A

Final answer

$$-\frac{3}{8}$$

Q025 MCQ

If $s + c = \frac{1}{2}$, then $s^3 + c^3$ equals:

Solution

1. Use $(s + c)^3 - 3sc(s + c)$.

Correct option: B

Final answer

$$\frac{11}{16}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q026 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ equals:

Solution

1. The sum is negative in QIII.

Correct option: B

Final answer

$$-\frac{\sqrt{7}}{2}$$

Q027 MCQ

If $s - c = \frac{\sqrt{3}}{2}$ in QI, then sc equals:

Solution

1. Use $(s - c)^2 = 1 - 2sc$.

Correct option: A

Final answer

$$\frac{1}{8}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q028 MCQ

If $s - c = 0.5$ in QIII, then $s^6 - c^6$ equals:

Solution

1. Use the two cubic factors and multiply.

Correct option: A

Final answer

$$-\frac{55\sqrt{7}}{256}$$

Q029 Written

Prove $\frac{\sin \theta}{1 + \cos \theta} + \frac{1}{\tan \theta} = \frac{1}{\sin \theta}$.

Solution

1. Rewrite $1/\tan \theta = \cos \theta/\sin \theta$.
2. Use the common denominator $(1 + \cos \theta)\sin \theta$, then $\sin^2 \theta + \cos^2 \theta = 1$.

Final answer

$$\text{LHS} = \text{RHS}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q030 Written

Prove (a) $\frac{1-\tan \theta}{1+\tan \theta} = \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta}$; **(b)** $\frac{\cos \theta}{1-\sin \theta} - \tan \theta = \frac{1}{\cos \theta}$.

Solution

1. For (a), multiply numerator and denominator by $\cos \theta$.
2. For (b), multiply the first fraction by $(1 + \sin \theta)/(1 + \sin \theta)$, then simplify.

Final answer

Both identities verified

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q031 Written

Prove the six identities in the given Exercise: (a) $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$; (b) $\tan \theta + 1/\tan \theta = 1/(\sin \theta \cos \theta)$; (c) $\cos \theta/(1 - \sin \theta) - 1/\cos \theta = \tan \theta$; (d) $\tan \theta/\sin \theta - \sin \theta/\tan \theta = \sin \theta \tan \theta$; (e) $(1 + \sin \theta)/\cos \theta + \cos \theta/(1 + \sin \theta) = 2/\cos \theta$; (f) $(\tan^2 \theta - \sin^2 \theta)/\sin^2 \theta = \tan^2 \theta$.

Solution

1. Replace tangent by $\sin \theta / \cos \theta$ where useful.
2. Combine fractions and use $\sin^2 \theta + \cos^2 \theta = 1$.

Final answer

LHS = RHS for (a)+(f)

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q032 Written

Prove that, for all defined values of x , $\frac{1+\sin x-\cos x}{1+\sin x+\cos x} + \frac{1+\sin x+\cos x}{1+\sin x-\cos x} = 2 \csc x$.

Solution

1. Set $A = 1 + \sin(x)$ and $(B = \cos(x))$, combine the two fractions, and simplify the numerator to $(2(1 + \sin(x)))$.
2. Use $(1 - \sin^2 x = \cos^2 x)$ in the denominator to obtain $(2/\sin(x))$.

Final answer

$$\text{LHS} = 2 \csc x$$

Q033 Written

Prove $\frac{1+\sin \theta}{\cos \theta} = \frac{\cos \theta}{1-\sin \theta}$.

Solution

1. Cross-multiply and use $1 - \sin^2 \theta = \cos^2 \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q034 **Written**

Prove $\frac{1}{1+\cos \theta} + \frac{1}{1-\cos \theta} = 2 \sec^2 \theta.$

Solution

1. Combine over $1 - \cos^2 \theta = \sin^2 \theta$, then rewrite.

Final answer

Identity proved

Q035 **Written**

Prove $\frac{\tan \theta}{1+\tan \theta} = \frac{\sin \theta}{\sin \theta + \cos \theta}.$

Solution

1. Multiply numerator and denominator on the left by $\cos \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q036 Written

Prove $\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}.$

Solution

1. Multiply top and bottom by $1 + \cos \theta$, then use the Pythagorean identity.

Final answer

Identity proved

Q037 Written

Prove $\frac{\cos \theta}{1 + \sin \theta} = \frac{1 - \sin \theta}{\cos \theta}.$

Solution

1. Rationalise the left denominator and cancel $\cos^2 \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q038 MCQ

The first move in proving a trig identity is usually to:**Solution**

1. Keep one side fixed and transform the other.

Correct option: B**Final answer**

Simplify one side

Q039 MCQ

The identity used to replace $1 - \sin^2 \theta$ is:**Solution**

1. From $\sin^2 \theta + \cos^2 \theta = 1$.

Correct option: A**Final answer** $\cos^2 \theta$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q040 MCQ

$\tan \theta$ should be rewritten as:

Solution

1. Use the quotient identity.

Correct option: B

Final answer $\sin \theta / \cos \theta$

Q041 MCQ

To add two trig fractions, first find a:

Solution

1. Fraction addition requires a common denominator.

Correct option: B

Final answer

common denominator

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q042 MCQ

The result of a valid identity proof is:**Solution**

1. The two expressions are shown equal.

Correct option: B**Final answer**

LHS=RHS

Q043 MCQ

If $\cos \theta = -\frac{12}{13}$ in QII, $\sin \theta$ is:**Solution**

1. Use the 5-12-13 triangle and the QII sign.

Correct option: A**Final answer** $\frac{5}{13}$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q044 MCQ

If $\tan \theta = -\frac{1}{2}$ in QII, $\cos \theta$ is:

Solution

1. Cosine is negative in QII.

Correct option: A

Final answer

$$-\frac{2\sqrt{5}}{5}$$

Q045 MCQ

If $\sin \theta = \frac{8}{17}$ in QIV, $\tan \theta$ is:

Solution

1. The 8-15-17 triangle has negative sine in QIV.

Correct option: B

Final answer

$$-\frac{8}{15}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q046 MCQ

If $\tan \theta = \sqrt{3}$ and θ is in QIII, $\sin \theta$ is:

Solution

1. Reference angle $\pi/3$, with both ratios negative in QIII.

Correct option: D

Final answer

$$-\frac{\sqrt{3}}{2}$$

Q047 MCQ

If $s + c = 1$, then sc equals:

Solution

1. Square the sum and use $s^2 + c^2 = 1$.

Correct option: A

Final answer

$$0$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q048 MCQ

If $s - c = \sqrt{2}$, then sc equals:

Solution

1. Use $(s - c)^2 = 1 - 2sc$.

Correct option: A

Final answer

$$-\frac{1}{2}$$

Q049 MCQ

If $s + c = \frac{2}{3}$, then $s^3 + c^3$ equals:

Solution

1. Use $(s + c)^3 - 3sc(s + c)$.

Correct option: A

Final answer

$$\frac{19}{27}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q050 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s^3 - c^3$ equals:

Solution

1. Use $(s - c)(1 + sc)$ with $sc = 3/8$.

Correct option: A

Final answer

$$\frac{11}{16}$$

Q051 MCQ

Which identity converts $\frac{1}{\tan \theta}$ into sine and cosine?

Solution

1. Take the reciprocal of tangent.

Correct option: B

Final answer

$$\frac{\cos \theta}{\sin \theta}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q052 MCQ

The identity $1 - \cos^2 \theta$ simplifies to:**Solution**

1. Rearrange the Pythagorean identity.

Correct option: A**Final answer**

$$\sin^2 \theta$$

Q053 MCQ

When proving an identity, which side should normally be transformed?**Solution**

1. Keep one side fixed and simplify the other to it.

Correct option: B**Final answer**

One side only

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q054 MCQ

After combining trig fractions, the next step is usually to use:

Solution

1. Replace $1 - \sin^2\theta$ or $1 - \cos^2\theta$ using the identity.

Correct option: B

Final answer

The Pythagorean identity

Quiz MCQ 1 Concept Quiz · MCQ

If $\sin\theta + \cos\theta = \frac{1}{2}$, then $\sin\theta \cos\theta$ is:

Solution

1. Square the sum and use $s^2 + c^2 = 1$.

Correct option: A

Final answer

$-\frac{3}{8}$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz MCQ 2 [Concept Quiz · MCQ](#)

If $\tan \theta = \frac{3}{4}$ in QI, then $\cos \theta$ is:

Solution

1. Use a 3-4-5 triangle.

Correct option: B

Final answer

$$\frac{4}{5}$$

Quiz MCQ 3 [Concept Quiz · MCQ](#)

Which identity is always valid where defined?

Solution

1. Use the fundamental quotient identity.

Correct option: B

Final answer

$$1 + \tan^2 \theta = \frac{1}{\cos^2 \theta}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz MCQ 4 Concept Quiz · MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ is:

Solution

1. Square the difference and use the QIII sign.

Correct option: B

Final answer

$$-\frac{\sqrt{7}}{2}$$

Quiz WRITTEN 5 Concept Quiz · Written

Given $\cos \theta = -\frac{5}{13}$ in QIII, find $\sin \theta$ and $\tan \theta$.

Solution

1. Use the 5-12-13 triangle and QIII signs.

Final answer

$$-\frac{12}{13}, \frac{12}{5}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 6 Concept Quiz · Written

Given $s + c = \frac{2}{3}$, **find** sc and $s^3 + c^3$.

Solution

1. Square the sum to find sc , then apply the sum-of-cubes identity.

Final answer

$$-\frac{5}{18}, \frac{19}{27}$$

Quiz WRITTEN 7 Concept Quiz · Written

Prove $\frac{1-\sin \theta}{\cos \theta} = \frac{\cos \theta}{1+\sin \theta}$.

Solution

1. Cross-multiply and use $1 - \sin^2 \theta = \cos^2 \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 8 Concept Quiz · Written

Given $\tan \theta = -2$ **in QIV, find** $\sin \theta$ **and** $\cos \theta$, **then verify** $1 + \tan^2 \theta = 1/\cos^2 \theta$.

Solution

1. Use a 1-2- $\sqrt{5}$ triangle and QIV signs.
2. Substitute into both sides of the identity.

Final answer

$$\sin \theta = -\frac{2\sqrt{5}}{5}, \cos \theta = \frac{\sqrt{5}}{5}; 6 = 6$$

Concept 3 | Properties and Graphs of Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Amplitude

$$y = a \sin(bx + c) + d \Rightarrow |a|$$

Period

$$T = \frac{2\pi}{|b|}$$

Range

$$d - |a| \leq y \leq d + |a|$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q001 Written

(1) Express $\sin \frac{9\pi}{5}$ as a trigonometric function of an acute angle. (2) Simplify $\sin \theta + \sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi) + \sin(\theta + \frac{3\pi}{2})$.

Solution

1. Reduce the angle to its reference angle and keep the quadrant sign.
2. Use the quarter-turn identities: $\sin(\theta + \frac{\pi}{2}) = \cos \theta$, $\sin(\theta + \pi) = -\sin \theta$, and $\sin(\theta + \frac{3\pi}{2}) = -\cos \theta$.

Final answer

(1) $-\sin \frac{\pi}{5}$ (2) 0

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q002 Written

(1) Express $\cos(-\frac{\pi}{6})$ as a trigonometric function of an acute angle. (2) Express $\sin \frac{8\pi}{7}$ as a trigonometric function of an acute angle. (3) Simplify $\cos(\frac{\pi}{2} - \theta) \sin(\pi - \theta) - \sin(\frac{3\pi}{2} + \theta) \cos(\pi - \theta)$.

Solution

1. Use evenness of cosine and the reference-angle rules.
2. Replace shifted functions by $\sin \theta$ or $\cos \theta$, then simplify.

Final answer

(1) $\cos \frac{\pi}{6}$ (2) $-\sin \frac{\pi}{7}$ (3) $\sin^2 \theta - \cos^2 \theta$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q003 Written

Solve all three given demands: (1) express $\cos \frac{17\pi}{5}$, $\cos \frac{5\pi}{7}$, $\sin(-\frac{7\pi}{8})$, $\sin(-\frac{19\pi}{8})$ using acute angles; (2) find $\tan(-\frac{\pi}{6})$ and $\tan(-\frac{\pi}{4})$; (3) simplify (a) $\sin \theta + \cos(\pi + \theta) - \cos(\pi - \theta) + \sin(\pi + \theta)$, (b) $\sin(\theta + \frac{\pi}{2}) \cos(\theta + \pi) + \sin(-\theta) \cos(\frac{\pi}{2} - \theta)$.

Solution

1. Reduce each angle to an acute reference angle and apply the sign in its quadrant.
2. For (3), use the quarter-turn and half-turn identities before collecting terms.

Final answer

$$(1) -\cos \frac{2\pi}{5}, -\cos \frac{2\pi}{7}, -\sin \frac{\pi}{8}, -\sin \frac{3\pi}{8} \quad (2) -\frac{1}{\sqrt{3}}, -1 \quad (3a) 0, (3b) -1$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q004 Written

Simplify $\sin(\frac{3\pi}{2} - \theta) \sec(\pi + \theta) - \tan(\theta - \frac{\pi}{2}) \cot(\frac{5\pi}{2} + \theta)$.

Solution

1. Use $\sin(\frac{3\pi}{2} - \theta) = -\cos \theta$, $\sec(\pi + \theta) = -\sec \theta$, $\tan(\theta - \frac{\pi}{2}) = -\cot \theta$, and $\cot(\frac{5\pi}{2} + \theta) = -\tan \theta$.
2. Both products equal 1, so their difference is 0.

Final answer

0

Q005 Written

Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, **and** $y = \tan \theta$, **and state each period and range.**

Solution

1. Mark one standard cycle, then repeat it periodically.
2. For tangent, draw vertical asymptotes at $\theta = \frac{\pi}{2} + k\pi$.

Final answer

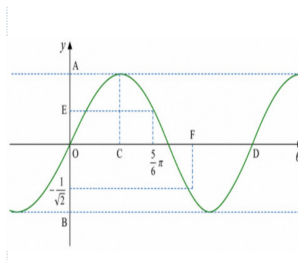
$\sin, \cos : T = 2\pi, -1 \leq y \leq 1;$ $\tan : T = \pi, y \in \mathbb{R}$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q006 Written

For the given graph of $y = \sin \theta$, determine the labelled scales A to F .



Solution

1. Read the extrema and the x-axis scale from the standard sine cycle.
2. Use $\sin \frac{5\pi}{6} = \frac{1}{2}$ and the descending branch to locate F .

Final answer

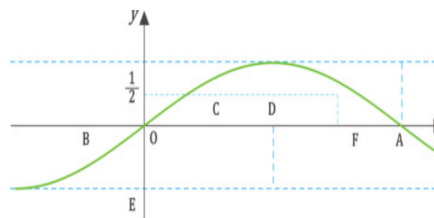
$$A = 1, B = -1, C = \frac{\pi}{2}, D = 2\pi, E = \frac{1}{2}, F = \frac{5\pi}{6}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q007 **Written**

(1) Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, $y = \tan \theta$, and state their periods. (2) For the given graph of $y = \cos \theta$, determine the labelled scales A to F .

**Solution**

1. Use the standard parent graph and mark quarter-period points.
2. The vertical values are the cosine range $[-1, 1]$; the horizontal values follow the 2π cycle.

Final answer

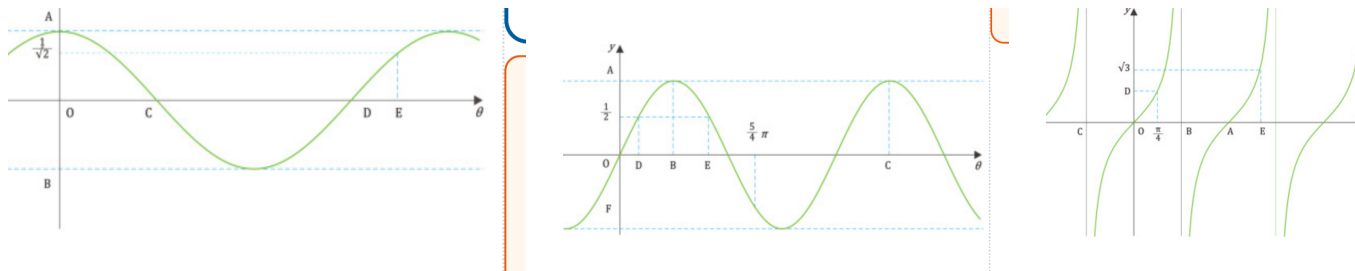
(1) $T_{\sin} = 2\pi$, $T_{\cos} = 2\pi$, $T_{\tan} = \pi$; (2) read the marked extrema, intercepts and quarter-period points from the graph

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q008 Written

Complete all four given graph tasks: draw the three parent graphs and find their periods; read the labelled scales on the given cosine graph; read the labelled scales on the given sine graph; and read the labelled scales on the given tangent graph.



Solution

1. Draw axes and the standard cycle first.
2. Read each label from an intercept, extremum, asymptote or quarter-period tick; do not change the graph scale.

Final answer

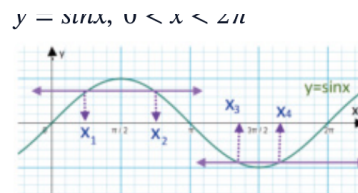
Use the standard cycles: sine/cosine have period 2π and range $[-1, 1]$; tangent has period π , range $\mathbb{R}$, and asymptotes at odd multiples of $\frac{\pi}{2}$. The requested A+F labels are read from the marked ticks and extrema in each supplied graph.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q009 Written

The given graph represents $y = \sin x$ for $0 < x < 2\pi$. Find $x_1 + x_2 + x_3 + x_4$ from the four marked x-coordinates.



Then find: $x_1 + x_2 + x_3 + x_4$

Solution

1. Identify the y-level associated with each marked point and use the sine cycle on $(0, 2\pi)$.
2. Add the four x-coordinates only after they have been read from the same horizontal scale.

Final answer

Read x_1, x_2, x_3, x_4 from the supplied graph and add the four marked coordinates; the result is the graph-reading total.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q010 Written

Draw the graphs of $y = 2 \cos \theta$ and $y = \tan \frac{\theta}{2}$, and find their periods.

Solution

1. The coefficient 2 outside cosine doubles the y-values but not the period.
2. For $\tan(\theta/2)$, multiply the parent tangent period by the reciprocal of $1/2$.

Final answer

$$y = 2 \cos \theta : T = 2\pi, -2 \leq y \leq 2; \quad y = \tan \frac{\theta}{2} : T = 2\pi$$

Q011 Written

Draw the graphs of $y = 2 \sin \theta$, $y = \tan 2\theta$, and $y = \cos \frac{\theta}{2}$, and find their periods.

Solution

1. Use the vertical factor for amplitude/range and the reciprocal of the coefficient of θ for horizontal scale.
2. For tangent, the parent period is π ; for sine/cosine it is 2π .

Final answer

$$T = 2\pi, \frac{\pi}{2}, 4\pi \text{ respectively; ranges } [-2, 2], \mathbb{R}, [-1, 1]$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q012 Written

Draw the nine given graphs and find each period: (a) $3 \sin \theta$, (b) $3 \cos \theta$, (c) $\frac{1}{2} \cos \theta$, (d) $\sin 3\theta$, (e) $\cos 2\theta$, (f) $\tan 3\theta$, (g) $\sin \frac{\theta}{2}$, (h) $\cos \frac{\theta}{3}$, (i) $\tan \frac{\theta}{3}$.

Solution

1. Read the vertical coefficient as the amplitude/range factor.
2. Divide the parent period by the absolute coefficient of θ .

Final answer

(a) 2π (b) 2π (c) 2π (d) $\frac{2\pi}{3}$ (e) π (f) $\frac{\pi}{3}$ (g) 4π (h) 6π (i) 3π ; ranges follow the outside coefficient

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q013 Written

Draw $y = -2|\sin(3x)|$ and find its period.

Solution

1. The absolute value makes the sine pattern repeat after π in its input, so the x-period is $\pi/3$.
2. The negative outside factor reflects the nonnegative $|\sin(3x)|$ graph below the x-axis.

Final answer

$$T = \frac{\pi}{3}, \quad 0 \geq y \geq -2$$

Q014 Written

Draw $y = \sin(\theta - \frac{\pi}{6})$ and find its period.

Solution

1. Start the standard sine cycle at $\theta = \pi/6$.
2. A horizontal shift does not change the parent period.

Final answer

$$\text{shift right by } \frac{\pi}{6}, \quad T = 2\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q015 **Written**

Draw the graphs and find their periods: (1) $y = \sin(\theta - \frac{\pi}{3})$, (2) $y = \cos(\theta + \frac{\pi}{4})$, (3) $y = \tan(\theta - \frac{\pi}{2})$.

Solution

1. Read $\theta - p$ as a shift right by p and $\theta + p$ as a shift left by p .
2. Keep the parent period unchanged.

Final answer

Shifts: right $\frac{\pi}{3}$, left $\frac{\pi}{4}$, right $\frac{\pi}{2}$; periods $2\pi, 2\pi, \pi$.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q016 Written

Draw all given graphs and find their periods: sine group $y = \sin(\theta + \frac{\pi}{4}), \sin(\theta - \frac{\pi}{2}), \sin(\theta + \frac{\pi}{3})$; **cosine group** $y = \cos(\theta - \frac{\pi}{3}), \cos(\theta - \frac{\pi}{4}), \cos(\theta + \frac{\pi}{6})$; **tangent group** $y = \tan(\theta + \frac{\pi}{6}), \tan(\theta - \frac{\pi}{3}), \tan(\theta - \frac{\pi}{4})$.

Solution

1. Translate the parent graph horizontally by the signed amount inside the bracket.
2. Use the same vertical range and asymptote spacing as the parent function.

Final answer

All sine/cosine periods are 2π ; all tangent periods are π . The shifts are read directly from the brackets.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q017 Written

Draw $y = 3 \cos |x + \frac{\pi}{3}|$ and find its period.

Solution

1. Use $\cos |u| = \cos(u)$, then read the horizontal shift and vertical scale.
2. The period of cosine remains 2π .

Final answer

$$y = 3 \cos(x + \frac{\pi}{3}) \text{ since cosine is even; } T = 2\pi$$

Q018 Written

Draw the graphs and find their periods: (1) $y = 2 \cos(2\theta - \frac{\pi}{2})$, (2) $y = \tan(\frac{\theta}{2} - \frac{\pi}{6})$.

Solution

1. Factor the coefficient of θ from the bracket before reading the shift.
2. Apply shift, horizontal scale, then vertical scale.

Final answer

$$(1) y = 2 \cos 2(\theta - \frac{\pi}{4}), T = \pi, -2 \leq y \leq 2; \quad (2) y = \tan \frac{1}{2}(\theta - \frac{\pi}{3}), T = 2\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q019 Written

Draw the graphs and find their periods: (a) $y = 3 \sin \frac{\theta}{2}$, (b) $y = \tan(2\theta - \frac{\pi}{2})$, (c) $y = 2 \cos(2\theta - \frac{\pi}{3})$.

Solution

1. Identify the coefficient b , factor the angle, then read the shift q/b .
2. Use the parent period divided by $|b|$ and the outside amplitude $|a|$.

Final answer

$$(a) T = 4\pi, -3 \leq y \leq 3; \quad (b) T = \frac{\pi}{2}; \quad (c) T = \pi, -2 \leq y \leq 2$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q020 Written

Draw all nine given graphs and find their periods: (a) $2 \cos \frac{\theta}{2}$, (b) $3 \sin 2\theta$, (c) $2 \cos 2\theta$, (d) $2 \cos(\theta - \frac{\pi}{4})$, (e) $2 \sin(\theta - \frac{\pi}{3})$, (f) $\tan(2\theta - \frac{\pi}{3})$, (g) $3 \cos(2\theta - \frac{\pi}{2})$, (h) $3 \sin(2\theta - \frac{2\pi}{3})$, (i) $2 \cos(\frac{\theta}{2} - \frac{\pi}{4})$.

Solution

1. Rewrite every angle as $b(\theta - q/b)$.
2. Apply the transformations from the inside out and record period, range and shift.

Final answer

Periods: (a) 4π ; (b) π ; (c) π ; (d) 2π ; (e) 2π ; (f) $\frac{\pi}{2}$; (g) π ; (h) π ; (i) 4π . Ranges are set by the outside factors.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q021 Written

Draw $y = 3 \sin |2x - \pi|$ and find its period.**Solution**

1. The absolute value makes the sine pattern repeat after π in its input; with coefficient 2 the x-period is $\pi/2$.
2. The graph is nonnegative with maximum 3.

Final answer

$$y = 3|\sin(2x - \pi)|, \quad T = \frac{\pi}{2}, \quad 0 \leq y \leq 3$$

Q022 MCQ

 $\sin(\theta + \pi)$ equals:**Solution**

1. A half-turn changes the sign of sine.

Correct option: B**Final answer**

$$-\sin \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q023 MCQ

 $\cos(\theta + \pi)$ equals:**Solution**

1. A half-turn changes the sign of cosine.

Correct option: B

Final answer

$$-\cos \theta$$

Q024 MCQ

 $\sin\left(\frac{\pi}{2} - \theta\right)$ equals:**Solution**

1. Use the complementary-angle identity.

Correct option: C

Final answer

$$\cos \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q025 MCQ

 $\cos(-\theta)$ equals:**Solution**

1. Cosine is even.

Correct option: A

Final answer

$$\cos \theta$$

Q026 MCQ

 $\sin(-\theta)$ equals:**Solution**

1. Sine is odd.

Correct option: B

Final answer

$$-\sin \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q027 MCQ

 $\cos(\pi - \theta)$ equals:**Solution**

1. Use the supplementary-angle identity.

Correct option: B

Final answer

$$-\cos \theta$$

Q028 MCQ

 $\sin\left(\frac{3\pi}{2} + \theta\right)$ equals:**Solution**

1. Shift by three quarter-turns.

Correct option: D

Final answer

$$-\cos \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q029 MCQ

 $\tan(\theta - \frac{\pi}{2})$ equals:**Solution**

1. Use the quarter-turn tangent identity.

Correct option: D

Final answer $-\cot \theta$

Q030 MCQ

The period of $y = \sin \theta$ is:**Solution**1. One sine cycle spans 2π .

Correct option: C

Final answer 2π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q031 MCQ

The range of $y = \cos \theta$ is:**Solution**

1. Cosine is bounded by -1 and 1.

Correct option: B

Final answer

$$-1 \leq y \leq 1$$

Q032 MCQ

The period of $y = \tan \theta$ is:**Solution**1. Tangent repeats every π .

Correct option: B

Final answer

$$\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q033 MCQ

A vertical asymptote of $y = \tan \theta$ occurs at:**Solution**

1. Cosine is zero at odd multiples of $\pi/2$.

Correct option: B**Final answer**

$$\theta = \frac{\pi}{2} + k\pi$$

Q034 MCQ

The maximum value of $\sin \theta$ is:**Solution**

1. Read the top of the standard sine graph.

Correct option: C**Final answer**

1

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q035 MCQ

The x-coordinate of the first positive sine maximum is:**Solution**

1. Sine first reaches 1 at $\pi/2$.

Correct option: B**Final answer** $\pi/2$

Q036 MCQ

At $\theta = 0$, $\cos \theta$ equals:**Solution**

1. The cosine graph starts at its maximum.

Correct option: C**Final answer**

1

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q037 MCQ

At $\theta = 0$, $\tan \theta$ equals:**Solution**

1. Tangent passes through the origin.

Correct option: B**Final answer**

0

Q038 MCQ

The range of $y = 3 \sin \theta$ is:**Solution**

1. Multiply the sine range by 3.

Correct option: B**Final answer** $-3 \leq y \leq 3$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q039 MCQ

The period of $y = \sin 4\theta$ is:**Solution**1. Use $2\pi/|b|$.**Correct option:** B**Final answer** $\pi/2$

Q040 MCQ

The period of $y = \cos(\theta/3)$ is:**Solution**1. The coefficient is $1/3$, so multiply the parent period by 3.**Correct option:** D**Final answer** 6π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q041 MCQ

The period of $y = \tan 5\theta$ is:**Solution**

1. Use $\pi/|b|$.

Correct option: A**Final answer**

$$\pi/5$$

Q042 MCQ

The range of $y = \frac{1}{2} \cos \theta$ is:**Solution**

1. The outside factor halves the cosine values.

Correct option: C**Final answer**

$$-\frac{1}{2} \leq y \leq \frac{1}{2}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q043 MCQ

The graph of $y = -2 \sin \theta$ is the parent sine graph:**Solution**

1. The negative sign reflects; 2 stretches.

Correct option: B**Final answer**

reflected and stretched vertically

Q044 MCQ

The period of $y = \tan(\theta/2)$ is:**Solution**1. Multiply the tangent period by the reciprocal of $1/2$.**Correct option:** C**Final answer** 2π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q045 MCQ

The period of $y = \sin(3\theta)$ is:**Solution**

1. Use $2\pi/3$.

Correct option: B**Final answer** $2\pi/3$

Q046 MCQ

The graph of $y = \sin(\theta - \pi/4)$ is shifted:**Solution**

1. $\theta - p$ means shift right by p .

Correct option: B**Final answer**right by $\pi/4$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q047 MCQ

The graph of $y = \cos(\theta + \pi/3)$ is shifted:**Solution**1. $\theta + p$ means shift left by p .

Correct option: B

Final answerleft by $\pi/3$

Q048 MCQ

A horizontal shift changes the sine period from 2π to:**Solution**

1. Translation does not change period.

Correct option: B

Final answer 2π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q049 MCQ

A horizontal shift changes the tangent period from π to:**Solution**

1. Translation does not change period.

Correct option: B**Final answer** π

Q050 MCQ

The graph of $y = \sin(\theta + \pi/2)$ equals:**Solution**

1. Use the quarter-turn identity.

Correct option: C**Final answer** $\cos \theta$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q051 MCQ

The graph of $y = \cos(\theta - \pi)$ equals:**Solution**1. A π shift reverses cosine.

Correct option: B

Final answer

$$-\cos \theta$$

Q052 MCQ

The graph of $y = \tan(\theta - \pi/2)$ has its central zero at:**Solution**1. Set $\theta - \pi/2 = 0$.

Correct option: B

Final answer

$$\theta = \pi/2$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q053 MCQ

Since cosine is even, $\cos |x|$ equals:**Solution**

1. Evenness removes the absolute value inside cosine.

Correct option: A**Final answer** $\cos x$

Q054 MCQ

In $y = a \sin(b\theta - q)$, the horizontal shift is:**Solution**

1. Factor b from the angle first.

Correct option: B**Final answer** q/b

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q055 MCQ

The period of $y = 2 \cos(3\theta - \pi)$ is:**Solution**

1. The coefficient of θ is 3.

Correct option: B**Final answer** $2\pi/3$

Q056 MCQ

The range of $y = 4 \sin(2\theta - 1)$ is:**Solution**

1. The outside factor controls the range.

Correct option: C**Final answer** $[-4, 4]$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q057 MCQ

For $y = \tan(2\theta - \pi/3)$, the shift is:**Solution**

1. Rewrite $2(\theta - \pi/6)$.

Correct option: B**Final answer**right by $\pi/6$

Q058 MCQ

The period of $y = 3 \sin(\theta/2)$ is:**Solution**

1. The coefficient is $1/2$.

Correct option: D**Final answer** 4π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q059 MCQ

The correct transformation order for $y = af(b\theta - q)$ starts with:**Solution**

1. Factor b and read q/b first.

Correct option: B**Final answer**

horizontal shift

Q060 MCQ

The period of $y = \tan(4\theta + q)$ is:**Solution**

1. Tangent period is $\pi/|4|$.

Correct option: A**Final answer** $\pi/4$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q061 MCQ

The range of $y = 3 \sin |2x - \pi|$ is:**Solution**

1. The absolute sine is nonnegative and the factor is 3.

Correct option: B**Final answer**

$$0 \leq y \leq 3$$

Q062 Written

Express $\cos \frac{11\pi}{6}$ as a function of an acute angle.**Solution**

1. Use the QIV reference angle and cosine's positive sign.

Final answer

$$\cos \frac{\pi}{6}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q063 Written

Simplify $\sin(\theta + \pi) - \cos(\frac{\pi}{2} - \theta)$.**Solution**

1. Replace the shifted functions by $-\sin \theta$ and $\sin \theta$.

Final answer

$$-\sin \theta - \sin \theta = -2 \sin \theta$$

Q064 Written

Simplify $\cos(\theta + \pi) + \cos(\frac{\pi}{2} - \theta)$.**Solution**

1. Use the half-turn and complementary-angle identities.

Final answer

$$-\cos \theta + \sin \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q065 Written

Evaluate $\tan(-\frac{3\pi}{4})$ using an acute reference angle.**Solution**

1. The reference angle is $\pi/4$ and tangent is positive in QIII.

Final answer

1

Q066 Written

State the period and range of $y = \cos \theta$, then list its key points on one cycle.**Solution**

1. Read the standard cosine cycle at quarter-period intervals.

Final answer

$$T = 2\pi, -1 \leq y \leq 1; (0, 1), (\frac{\pi}{2}, 0), (\pi, -1), (\frac{3\pi}{2}, 0), (2\pi, 1)$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q067 Written

State the period and asymptotes of $y = \tan \theta$ on $[-\pi, \pi]$.**Solution**

1. Tangent is undefined at odd multiples of $\pi/2$.

Final answer

$$T = \pi; \theta = \pm \frac{\pi}{2}$$

Q068 Written

For the parent sine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.**Solution**

1. Sine crosses the x-axis at integer multiples of π .

Final answer

$$0, \pi, 2\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q069 Written

For the parent cosine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Solution

1. Cosine is zero at odd quarter-turns.

Final answer

$$\frac{\pi}{2}, \frac{3\pi}{2}$$

Q070 Written

Find the period and range of $y = 4 \cos \theta$.

Solution

1. The outside factor changes the range, not the period.

Final answer

$$T = 2\pi, -4 \leq y \leq 4$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q071 Written

Find the period and range of $y = \sin(5\theta)$.**Solution**

1. Divide the sine period by 5.

Final answer

$$T = \frac{2\pi}{5}, -1 \leq y \leq 1$$

Q072 Written

Find the period and range of $y = \tan(\theta/4)$.**Solution**

1. Multiply the parent tangent period by 4.

Final answer

$$T = 4\pi, y \in \mathbb{R}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q073 Written

Describe the transformations from $y = \cos \theta$ to $y = -\frac{1}{2} \cos(2\theta)$.

Solution

1. Read the inside coefficient and outside factor separately.

Final answer

horizontal compression by $1/2$, reflection in x-axis and vertical scale $1/2$; $T = \pi$

Q074 Written

Find the shift and period of $y = \sin(\theta + \frac{\pi}{6})$.

Solution

1. $\theta + p$ shifts left and does not alter period.

Final answer

left by $\frac{\pi}{6}$, $T = 2\pi$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q075 Written

Find the shift and period of $y = \cos(\theta - \frac{\pi}{2})$.**Solution**

1. θ -p shifts right by p.

Final answerright by $\frac{\pi}{2}$, $T = 2\pi$

Q076 Written

Find the central zero and period of $y = \tan(\theta + \frac{\pi}{4})$.**Solution**

1. Set the bracket equal to zero and retain tangent's period.

Final answer $\theta = -\frac{\pi}{4}$; $T = \pi$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q077 Written

For $y = 2 \sin(4\theta - \pi)$, state the shift, period and range.

Solution

- Factor 4 from the angle: $4(\theta - \pi/4)$.

Final answer

right by $\frac{\pi}{4}$, $T = \frac{\pi}{2}$, $-2 \leq y \leq 2$

Q078 Written

For $y = \cos(\theta/3 + \pi/6)$, state the shift and period.

Solution

- Rewrite the angle as $\frac{1}{3}(\theta + \pi/2)$.

Final answer

left by $\frac{\pi}{2}$, $T = 6\pi$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q079 Written

For $y = 3 \tan(2\theta + \pi)$, state the shift and period.**Solution**

1. Rewrite as $3 \tan 2(\theta + \pi/2)$.

Final answerleft by $\frac{\pi}{2}$, $T = \frac{\pi}{2}$

Q080 Written

For $y = 2 \cos(\theta/2 - \pi/8)$, state the shift, period and range.**Solution**

1. Rewrite the angle as $\frac{1}{2}(\theta - \pi/4)$.

Final answerright by $\frac{\pi}{4}$, $T = 4\pi$, $-2 \leq y \leq 2$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

The period of $y = \sin(2\theta - \frac{\pi}{3})$ is:

Solution

1. Use $2\pi/2$.

Correct option: B

Final answer

π

Quiz MCQ 2 [Concept Quiz](#) · [MCQ](#)

$\sin(\theta + \pi)$ equals:

Solution

1. Use the half-turn identity.

Correct option: B

Final answer

$-\sin \theta$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz MCQ 3 Concept Quiz · MCQ

The range of $y = 2 \cos \theta$ is:

Solution

1. Multiply the cosine range by 2.

Correct option: B

Final answer

$$-2 \leq y \leq 2$$

Quiz MCQ 4 Concept Quiz · MCQ

The graph of $y = \cos(\theta - \frac{\pi}{4})$ is shifted:

Solution

1. $\theta - p$ shifts right by p .

Correct option: B

Final answer

right by $\frac{\pi}{4}$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written

For $y = 3 \sin(2\theta - \pi)$, state the horizontal shift, period and range.

Solution

1. Rewrite as $3 \sin 2(\theta - \pi/2)$.

Final answer

right by $\frac{\pi}{2}$, $T = \pi$, $-3 \leq y \leq 3$

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi)$.

Solution

1. Use the quarter-turn and half-turn identities.

Final answer

$\cos \theta - \sin \theta$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written

Draw one cycle of $y = \tan(\theta - \frac{\pi}{3})$ and state its period and asymptotes.

Solution

1. Shift the parent tangent graph right by $\pi/3$; shift its asymptotes with it.

Final answer

$$T = \pi; \theta = \frac{5\pi}{6} + k\pi$$

Quiz WRITTEN 8 Concept Quiz · Written

For $y = -2|\sin(3x)|$, state its period and range and describe the reflection.

Solution

1. Absolute sine has input period π ; the coefficient 3 gives $\pi/3$, and -2 reflects and scales.

Final answer

$$T = \frac{\pi}{3}, -2 \leq y \leq 0; \text{ reflected below the x-axis}$$

Answer key

Use the question number shown on each page.

- Q001** (1) $-\sin \frac{\pi}{5}$ (2) 0
- Q002** (1) $\cos \frac{\pi}{6}$ (2) $-\sin \frac{\pi}{7}$ (3) $\sin^2 \theta - \cos^2 \theta$
- Q003** (1) $-\cos \frac{2\pi}{5}, -\cos \frac{2\pi}{7}, -\sin \frac{\pi}{8}, -\sin \frac{3\pi}{8}$ (2) $-\frac{1}{\sqrt{3}}, -1$ (3a) 0, (3b) $-\frac{1}{\sqrt{3}}$
- Q004** 0
- Q005** $\sin, \cos$: $T = 2\pi, -1 \leq y \leq 1$; $\tan$: $T = \pi, y \in \mathbb{R}$
- Q006** $A = 1, B = -1, C = \frac{\pi}{2}, D = 2\pi, E = \frac{1}{2}, F = \frac{5\pi}{4}$
- Q007** (1) $T_{\min} = 2\pi, T_{\max} = 2\pi, T_{\min} = \pi$; (2) read the marked extrema, intercepts and quarter-period points from the graph
- Q008** Use the standard cycles: sine/cosine have period 2π and range $[-1, 1]$; tangent has period π , range $\mathbb{R}$, and asymptotes at odd multiples of $\frac{\pi}{2}$. The requested A+F labels are read from the marked ticks and extrema in each supplied graph.
- Q009** Read x_1, x_2, x_3, x_4 from the supplied graph and add the four marked coordinates; the result is the graph-reading total.
- Q010** $y = 2 \cos \theta$: $T = 2\pi, -2 \leq y \leq 2$; $y = \tan \frac{\theta}{2}$: $T = 2\pi$
- Q011** $T = 2\pi, \frac{\pi}{2}, 4\pi$ respectively; ranges $[-2, 2], \mathbb{R}, [-1, 1]$
- Q012** (a) 2π (b) 2π (c) 2π (d) $\frac{2\pi}{3}$ (e) π (f) $\frac{\pi}{3}$ (g) 4π (h) 6π (i) 3π ; ranges follow the outside coefficient
- Q013** $T = \frac{\pi}{3}, 0 \geq y \geq -2$
- Q014** shift right by $\frac{\pi}{6}, T = 2\pi$
- Q015** Shifts: right $\frac{\pi}{3}$, left $\frac{\pi}{4}$, right $\frac{\pi}{2}$; periods $2\pi, 2\pi, \pi$.
- Q016** All sine/cosine periods are 2π ; all tangent periods are π . The shifts are read directly from the brackets.
- Q017** $y = 3 \cos(x + \frac{\pi}{3})$ since cosine is even; $T = 2\pi$
- Q018** (1) $y = 2 \cos 2(\theta - \frac{\pi}{4}), T = \pi, -2 \leq y \leq 2$; (2) $y = \tan \frac{1}{2}(\theta - \frac{\pi}{3}), T = 2\pi$
- Q019** (a) $T = 4\pi, -3 \leq y \leq 3$; (b) $T = \frac{\pi}{2}$; (c) $T = \pi, -2 \leq y \leq 2$
- Q020** Periods: (a) 4π ; (b) π ; (c) π ; (d) 2π ; (e) 2π ; (f) $\frac{\pi}{2}$; (g) π ; (h) π ; (i) 4π . Ranges are set by the outside factors.
- Q021** $y = 3|\sin(2x - \pi)|, T = \frac{\pi}{2}, 0 \leq y \leq 3$
- Q022** B
- Q023** B
- Q024** C
- Q025** A
- Q026** B
- Q027** B
- Q028** D
- Q029** D
- Q030** C
- Q031** B
- Q032** B
- Q033** B
- Q034** C
- Q035** B
- Q036** C
- Q037** B
- Q038** B
- Q039** B
- Q040** D
- Q041** A
- Q042** C
- Q043** B
- Q044** C
- Q045** B
- Q046** B
- Q047** B
- Q048** B
- Q049** B
- Q050** C
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Q055** B
- Q056** C
- Q057** B
- Q058** D
- Q059** B
- Q060** A
- Q061** B
- Q062** $\cos \frac{\pi}{6}$
- Q063** $-\sin \theta - \sin \theta = -2 \sin \theta$
- Q064** $-\cos \theta + \sin \theta$
- Q065** 1
- Q066** $T = 2\pi, -1 \leq y \leq 1$; (0, 1), $(\frac{\pi}{2}, 0)$, $(\pi, -1)$, $(\frac{3\pi}{2}, 0)$, $(2\pi, 1)$
- Q067** $T = \pi; \theta = \pm \frac{\pi}{2}$
- Q068** 0, $\pi, 2\pi$
- Q069** $\frac{\pi}{2}, \frac{3\pi}{2}$
- Q070** $T = 2\pi, -4 \leq y \leq 4$
- Q071** $T = \frac{2\pi}{5}, -1 \leq y \leq 1$
- Q072** $T = 4\pi, y \in \mathbb{R}$
- Q073** horizontal compression by 1/2, reflection in x-axis and vertical scale 1/2; $T = \pi$
- Q074** left by $\frac{\pi}{6}, T = 2\pi$
- Q075** right by $\frac{\pi}{2}, T = 2\pi$
- Q076** $\theta = -\frac{\pi}{4}; T = \pi$
- Q077** right by $\frac{\pi}{4}, T = \frac{\pi}{2}, -2 \leq y \leq 2$
- Q078** left by $\frac{\pi}{2}, T = 6\pi$
- Q079** left by $\frac{\pi}{2}, T = \frac{\pi}{2}$
- Q080** right by $\frac{\pi}{4}, T = 4\pi, -2 \leq y \leq 2$
- Quiz MCQ 1** B
- Quiz MCQ 2** B
- Quiz MCQ 3** B

Quiz MCQ 4 B**Quiz WRITTEN 6** $\cos \theta - \sin \theta$ **Quiz WRITTEN 7** $T = \pi; \theta = \frac{5\pi}{6} + k\pi$ **Quiz WRITTEN 8** $T = \frac{\pi}{3}, -2 \leq y \leq 0$; reflected below the**Quiz WRITTEN 5** right by $\frac{\pi}{2}$, $T = \pi, -3 \leq y \leq 3$

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Sine

$$\sin x = \sin \alpha$$

Cosine

$$\cos x = \cos \alpha$$

Tangent

$$\tan x = \tan \alpha$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q001 Written

Solve (a) $-2 \cos \theta = \sqrt{2}$ **and (b)** $\tan \theta + \sqrt{3} = 0$, giving the general solution for (b).

Solution

1. Rewrite as $\cos \theta = -\frac{\sqrt{2}}{2}$ and use QII/QIII.
2. For tangent, the reference angle is $\pi/3$ and the period is π .

Final answer

$$(a) \theta = \frac{3\pi}{4}, \frac{5\pi}{4} \quad (b) \theta = \frac{2\pi}{3} + n\pi, n \in \mathbb{Z}$$

Q002 Written

Solve (a) $2 \cos \theta - 1 = 0$ **and (b)** $\tan \theta - 1 = 0$ for $0 \leq \theta \leq 2\pi$.

Solution

1. Use the unit circle for cosine and include both turns in the stated interval.
2. For tangent, repeat the $\pi/4$ reference angle after one period π .

Final answer

$$(a) \theta = \frac{\pi}{3}, \frac{5\pi}{3} \quad (b) \theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q003 Written

Solve all given equations on $0 \leq \theta \leq 2\pi$: (a) $2 \sin \theta + 1 = 0$, (b) $2 \sin \theta = \sqrt{3}$, (c) $3 \sin \theta + 3 = 0$, (d) $\sqrt{2} \cos \theta + 1 = 0$, (e) $2 \cos \theta = \sqrt{3}$, (f) $2 \cos \theta - \sqrt{2} = 0$, (g) $\tan \theta + 1 = 0$, (h) $\sqrt{3} \tan \theta = 1$, (i) $\sqrt{3} \tan \theta = 3$, **and give general solutions for** (j) $2 \sin \theta + \sqrt{3} = 0$, (k) $\cos \theta + 1 = 0$, (l) $\sqrt{3} \tan \theta + 3 = 0$.

Solution

1. Put each equation into a single-function form, find the reference angle, then select the quadrants from the sign.
2. Use period 2π for sine/cosine and π for tangent; reject the endpoint 2π when the interval is half-open.

Final answer

(a) $\frac{7\pi}{6}, \frac{11\pi}{6}$ (b) $\frac{\pi}{3}, \frac{2\pi}{3}$ (c) $\frac{3\pi}{2}$ (d) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (e) $\frac{\pi}{6}, \frac{11\pi}{6}$ (f) $\frac{\pi}{4}, \frac{7\pi}{4}$ (g) $\frac{3\pi}{4}, \frac{7\pi}{4}$ (h) $\frac{\pi}{6}, \frac{7\pi}{6}$ (i) $\frac{\pi}{3}, \frac{4\pi}{3}$ (j) $\frac{4\pi}{3} + 2n\pi, \frac{5\pi}{3} + 2n\pi$ (k) $\pi + 2n\pi$ (l) $\frac{2\pi}{3} + n\pi$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q004 **Written****Solve** $\tan^2 \theta - (2 + \sqrt{3}) \tan \theta + 2\sqrt{3} = 0$ **and give the general solution.****Solution**

1. Factor the quadratic as $(\tan \theta - 2)(\tan \theta - \sqrt{3}) = 0$.
2. Apply the tangent period π to both roots.

Final answer

$$\tan \theta = 2 \text{ or } \tan \theta = \sqrt{3}; \quad \theta = \arctan(2) + n\pi \text{ or } \theta = \frac{\pi}{3} + n\pi$$

Q005 **Written****Solve for** $0 \leq \theta < 2\pi$: **(a)** $2 \cos \theta > -\sqrt{3}$, **(b)** $\tan \theta \leq \frac{1}{\sqrt{3}}$.**Solution**

1. Solve the boundary equation first, then shade the correct arcs on the unit circle.
2. Tangent is undefined at odd multiples of $\pi/2$; those points are never included.

Final answer

$$(a) 0 \leq \theta < \frac{5\pi}{6} \text{ or } \frac{7\pi}{6} < \theta < 2\pi \quad (b) 0 \leq \theta \leq \frac{\pi}{6} \text{ or } \frac{\pi}{2} < \theta \leq \frac{7\pi}{6} \text{ or } \frac{3\pi}{2} < \theta < 2\pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q006 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \sin \theta \leq -1$, (b) $\cos \theta > \frac{1}{2}$, (c) $\tan \theta > 1$.

Solution

1. Mark the equality angles and choose the arcs where the function has the required sign.
2. Open endpoints correspond to strict inequalities and to tangent asymptotes.

Final answer

$$(a) \frac{7\pi}{6} \leq \theta \leq \frac{11\pi}{6} \quad (b) 0 \leq \theta < \frac{\pi}{3} \text{ or } \frac{5\pi}{3} < \theta < 2\pi \quad (c) \frac{\pi}{4} < \theta < \frac{\pi}{2} \text{ or } \frac{5\pi}{4} < \theta < \frac{3\pi}{2}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q007 Written

Solve all given inequalities for $0 \leq \theta < 2\pi$: (a) $\sin \theta > \frac{\sqrt{3}}{2}$, (b) $\sin \theta - \frac{\sqrt{2}}{2} > 0$, (c) $2 \sin \theta + \sqrt{3} \leq 0$, (d) $\cos \theta > 0$, (e) $\cos \theta - \frac{1}{2} \leq 0$, (f) $2 \cos \theta + \sqrt{2} \leq 0$, (g) $\tan \theta \leq -\frac{1}{\sqrt{3}}$, (h) $\tan \theta + \sqrt{3} > 0$, (i) $\tan \theta + 1 \geq 0$.

Solution

- For each item, solve the equality, draw the relevant horizontal line, and select the arcs satisfying the sign.
- Keep the distinction between open and closed endpoints and remove all tangent asymptotes.

Final answer

$$(a) \left(\frac{\pi}{3}, \frac{2\pi}{3}\right) (b) \left(\frac{\pi}{4}, \frac{3\pi}{4}\right) (c) \left[\frac{4\pi}{3}, \frac{5\pi}{3}\right] (d) \left[0, \frac{\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right) (e) \left[\frac{\pi}{3}, \frac{5\pi}{3}\right] (f) \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right] (g) \left(\frac{\pi}{2}, \frac{5\pi}{6}\right] \cup \left(\frac{3\pi}{2}, \frac{11\pi}{6}\right] (h) \left[0, \frac{\pi}{2}\right) \cup \left(\frac{2\pi}{3}, \frac{3\pi}{2}\right) \cup \left(\frac{5\pi}{3}, 2\pi\right) (i) \left[0, \frac{\pi}{2}\right) \cup \left[\frac{3\pi}{4}, \frac{3\pi}{2}\right) \cup \left[\frac{7\pi}{4}, 2\pi\right)$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q008 Written

Solve $2 \sin^2 \theta - \cos \theta - 1 > 0$ for $0 \leq \theta < 2\pi$.**Solution**

1. Use $\sin^2 \theta = 1 - \cos^2 \theta$ to obtain $(1 - 2 \cos \theta)(1 + \cos \theta) > 0$.
2. Since $-1 \leq \cos \theta \leq 1$, the solution is $-1 < \cos \theta < 1/2$; exclude $\theta = \pi$ where the product is zero.

Final answer

$$\frac{\pi}{3} < \theta < \pi \text{ or } \pi < \theta < \frac{5\pi}{3}$$

Q009 Written

Solve for $0 \leq \theta < 2\pi$: **(a)** $2 \cos^2 \theta - 3 \sin \theta = 0$, **(b)** $2 \sin^2 \theta - 3 \cos \theta - 3 \leq 0$.**Solution**

1. For (a), replace $\cos^2 \theta$ and solve $2s^2 + 3s - 2 = 0$; reject $s = -2$.
2. For (b), replace $\sin^2 \theta$, solve $(2c + 1)(c + 1) \geq 0$, and intersect with $[-1, 1]$.

Final answer

$$(a) \theta = \frac{\pi}{6}, \frac{5\pi}{6} \quad (b) 0 \leq \theta \leq \frac{2\pi}{3} \text{ or } \theta = \pi \text{ or } \frac{4\pi}{3} \leq \theta < 2\pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q010 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \sin^2 \theta - \sin \theta = 0$, (b) $2 \cos^2 \theta - \sin \theta = 2$, (c) $2 \sin^2 \theta - 4 < 5 \cos \theta$.

Solution

1. Set $s = \sin \theta$ or $c = \cos \theta$ after using $s^2 + c^2 = 1$.
2. Solve the resulting quadratic, reject impossible trig values, then read the unit-circle intervals.

Final answer

$$(a) \theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi \quad (b) \theta = 0, \pi, \frac{7\pi}{6}, \frac{11\pi}{6} \quad (c) 0 \leq \theta < \frac{2\pi}{3} \text{ or } \frac{4\pi}{3} < \theta < 2\pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q011 Written

Solve all given equations and inequalities for $0 \leq \theta < 2\pi$: (a) $2 \sin^2 \theta + \sin \theta - 1 = 0$, (b) $2 \sin^2 \theta + 3 \cos \theta = 0$, (c) $2 \sin^2 \theta - \cos \theta = 2$, (d) $2 \cos^2 \theta - 5 \cos \theta - 3 = 0$, (e) $2 \cos^2 \theta + \sin \theta - 1 = 0$, (f) $2 \cos^2 \theta + 3 \sin \theta = 3$, (g) $2 \cos^2 \theta + 3 \sin \theta \leq 0$, (h) $2 \sin^2 \theta + \cos \theta > 1$, (i) $2 \cos^2 \theta - 4 \leq 5 \sin \theta$.

Solution

1. Convert to one trig function, solve the quadratic, and apply $-1 \leq \sin \theta, \cos \theta \leq 1$.
2. For inequalities, use the sign chart in the trig variable and then translate the accepted interval(s) to θ .

Final answer

(a) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (b) $\frac{2\pi}{3}, \frac{4\pi}{3}$ (c) $\frac{2\pi}{3}, \frac{\pi}{2}, \frac{4\pi}{3}, \frac{3\pi}{2}$ (d) $\frac{2\pi}{3}, \frac{4\pi}{3}$ (e) $\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (f) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}$ (g) $\frac{7\pi}{6} \leq \theta \leq \frac{11\pi}{6}$ (h) $0 < \theta < \frac{2\pi}{3}$ or $\frac{4\pi}{3} < \theta < 2\pi$ (i) $0 \leq \theta \leq \frac{7\pi}{6}$ or $\frac{11\pi}{6} \leq \theta < 2\pi$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q012 Written

Solve $\frac{1-2\sin^2\theta+\sin\theta}{\cos^2\theta} \geq 1$ for $0 \leq \theta < 2\pi$.

Solution

1. Subtract 1 and simplify the numerator to $\sin\theta(1 - \sin\theta)$.
2. The denominator is positive but undefined at $\theta = \pi/2$; hence $0 \leq \sin\theta \leq 1$ with that point removed.

Final answer

$$0 \leq \theta < \frac{\pi}{2} \text{ or } \frac{\pi}{2} < \theta \leq \pi$$

Q013 Written

Solve for $0 \leq \theta < 2\pi$: **(a)** $\sin(\theta + \frac{\pi}{3}) = \frac{1}{2}$, **(b)** $\sin(\theta + \frac{\pi}{6}) > -\frac{1}{2}$.

Solution

1. Let $t = \theta + \alpha$, write the range of t , solve there, then subtract α .
2. For the inequality, shade one complete revolution in the shifted t-range and preserve strict endpoints.

Final answer

$$(a) \theta = \frac{\pi}{2}, \frac{11\pi}{6} \quad (b) 0 \leq \theta < \pi \text{ or } \frac{5\pi}{3} < \theta < 2\pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q014 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{6}) = \frac{1}{2}$, (b) $\tan(\theta + \frac{\pi}{4}) = \sqrt{3}$, (c) $\cos(\theta + \frac{\pi}{4}) < -\frac{\sqrt{2}}{2}$.

Solution

1. Find each shifted argument's exact range before selecting the unit-circle arcs.
2. For tangent, exclude asymptotes automatically; for strict cosine inequality, endpoints are open.

Final answer

$$(a) \theta = 0, \frac{2\pi}{3} \quad (b) \theta = \pi, 2\pi, \frac{13\pi}{12} \quad (c) \frac{\pi}{2} < \theta < \pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q015 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\cos(\theta + \frac{\pi}{3}) = -\frac{\sqrt{3}}{2}$, (b) $\cos(\theta + \frac{\pi}{6}) = -\frac{1}{2}$, (c) $\sin(\theta + \frac{\pi}{4}) = \frac{\sqrt{3}}{2}$, (d) $\tan(\theta + \frac{\pi}{4}) = -1$, (e) $\tan(\theta + \frac{\pi}{3}) = \frac{1}{\sqrt{3}}$, (f) $\tan(\theta + \frac{\pi}{6}) = 1$, (g) $\sin(\theta + \frac{\pi}{6}) \leq \frac{\sqrt{3}}{2}$, (h) $\sin(\theta + \frac{\pi}{4}) \geq \frac{\sqrt{2}}{2}$, (i) $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$.

Solution

1. Set $t = \theta + \alpha$, record its interval, solve the standard trig relation in that interval, then subtract α .
2. Check every endpoint against the original strict/non-strict sign and the domain $0 \leq \theta < 2\pi$.

Final answer

(a) $\frac{\pi}{2}, \frac{5\pi}{6}$ (b) $\frac{\pi}{2}, \frac{7\pi}{6}$ (c) $\pi, 2\pi$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{5\pi}{6}, \frac{11\pi}{6}$ (f) $\pi, 2\pi$ (g) $[0, \frac{\pi}{6}] \cup [\frac{\pi}{2}, 2\pi)$ (h) $[0, \frac{\pi}{2}]$ (i) $\frac{4\pi}{3} < \theta < 2\pi$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q016 Written

Solve $4 \sin^2(\theta + \frac{\pi}{6}) \geq 1$ for $0 \leq \theta < 2\pi$.**Solution**

1. Let $t = \theta + \pi/6$; then $|\sin(t)| \geq 1/2$.
2. On the shifted range $[\pi/6, 13\pi/6)$, the accepted arcs give the stated closed θ -intervals.

Final answer

$$0 \leq \theta \leq \frac{2\pi}{3} \text{ or } \pi \leq \theta \leq \frac{5\pi}{3}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q017 Written

A student divides $\tan x \times \cos x = \frac{1}{2}$ by $\cos x$, obtaining $\tan x = \frac{1}{2 \cos x}$. Evaluate this strategy and propose a better substitution on $[0^\circ, 360^\circ[$.

Solution

1. Replace $\tan(x)$ by $\sin(x)/\cos(x)$ before cancelling; the original tangent already excludes $\cos(x) = 0$.
2. Solve the resulting sine equation on the stated degree interval.

Final answer

Dividing by $\cos(x)$ separates the interacting factors and is not helpful; use $\tan(x) \cos(x) = \sin(x)$ (with $\cos(x) \neq 0$), so $\sin(x) = \frac{1}{2}$ and $x = 30^\circ, 150^\circ$.

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q018 MCQ

The period used for a general tangent solution is:**Solution**

1. Tangent repeats after π .

Correct option: B**Final answer** π

Q019 MCQ

The solutions of $2 \cos \theta = 1$ on $[0, 2\pi]$ are:**Solution**

1. Cosine is $1/2$ in QI and QIV.

Correct option: B**Final answer** $\pi/3, 5\pi/3$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q020 MCQ

Which root is impossible when solving $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$?

Solution

1. A sine value must lie in $[-1, 1]$.

Correct option: B

Final answer

$$\sin \theta = -2$$

Q021 MCQ

The quadratic challenge factors into roots:

Solution

1. Use $(u - 2)(u - \sqrt{3})$ with $u = \tan \theta$.

Correct option: B

Final answer

$$2, \sqrt{3}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q022 MCQ

For $2 \cos \theta > -\sqrt{3}$, the excluded arc is:

Solution

1. Cosine is at most $-\sqrt{3}/2$ on the central QII arc.

Correct option: B

Final answer

$[5\pi/6, 7\pi/6]$

Q023 MCQ

The inequality $\tan \theta > 1$ on $[0, 2\pi)$ holds on:

Solution

1. Use the increasing tangent branches and omit asymptotes.

Correct option: A

Final answer

$(\pi/4, \pi/2) \cup (5\pi/4, 3\pi/2)$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q024 MCQ

A closed endpoint in a trig inequality is used when the sign is:**Solution**

1. Equality is included for $\leq$ or $\geq$.

Correct option: C**Final answer**

greater than or equal to

Q025 MCQ

The 5-14 challenge reduces to which cosine condition?**Solution**

1. Factor the expression after replacing $\sin^2\theta$.

Correct option: B**Final answer** $-1 < \cos \theta < 1/2$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q026 MCQ

When a quadratic gives $\sin \theta = -2$, we:**Solution**1. Sine cannot be outside $[-1, 1]$.

Correct option: B

Final answer

reject it

Q027 MCQ

The first step for $2 \cos^2 \theta - 3 \sin \theta = 0$ is to use:**Solution**

1. Rewrite in one trig function.

Correct option: A

Final answer $\sin^2 \theta + \cos^2 \theta = 1$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q028 MCQ

The solution of $2 \sin^2 \theta + \sin \theta - 1 = 0$ includes:**Solution**

1. The valid root $\sin \theta = 1/2$ gives $\pi/6, 5\pi/6$.

Correct option: A**Final answer** $\pi/6$

Q029 MCQ

In the rational challenge, which angle must be excluded?**Solution**

1. The denominator $\cos^2 \theta$ is zero there.

Correct option: C**Final answer** $\pi/2$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q030 MCQ

After setting $t = \theta + \pi/3$ with $0 \leq \theta < 2\pi$, the t-range is:

Solution

1. Add $\pi/3$ to both domain endpoints.

Correct option: B

Final answer

$[\pi/3, 7\pi/3)$

Q031 MCQ

The solutions of $\tan(\theta + \pi/4) = \sqrt{3}$ include:

Solution

1. Set the shifted argument to $\pi/3$.

Correct option: A

Final answer

$\pi/12$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q032 MCQ

For $\sin(\theta + \pi/4) \geq \sqrt{2}/2$, the θ -solution is:

Solution

1. The shifted sine is at least $\sqrt{2}/2$ from $\pi/4$ to $3\pi/4$.

Correct option: A

Final answer

$$[0, \pi/2]$$

Q033 MCQ

The 5-16 challenge is equivalent to:

Solution

1. Divide by 4 and take the square-root inequality.

Correct option: A

Final answer

$$|\sin t| \geq 1/2$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q034 MCQ

In the Reflect problem, $\tan x \cos x$ simplifies to:**Solution**1. Use $\tan(x) = \sin(x)/\cos(x)$ where defined.

Correct option: B

Final answer $\sin x$

Quiz MCQ 1 Concept Quiz · MCQ

The general solution of $\tan \theta = -1$ is:**Solution**1. Tangent has period π .

Correct option: B

Final answer

$$\theta = -\frac{\pi}{4} + n\pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz MCQ 2 Concept Quiz · MCQ

The solution of $\cos \theta > 0$ on $[0, 2\pi)$ is:

Solution

1. Cosine is positive in QI and QIV.

Correct option: B

Final answer

$$[0, \pi/2) \cup (3\pi/2, 2\pi)$$

Quiz MCQ 3 Concept Quiz · MCQ

The valid sine root of $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$ is:

Solution

1. The other algebraic root is outside the sine range.

Correct option: B

Final answer

$$\sin \theta = \frac{1}{2}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz MCQ 4 **Concept Quiz · MCQ**

The first operation in a shifted-argument problem is to:

Solution

1. The t-range prevents losing or adding solutions.

Correct option: B

Final answer

set t equal to the whole argument and find its range

Quiz WRITTEN 5 **Concept Quiz · Written**

Solve $2 \cos \theta = \sqrt{3}$ **for** $0 \leq \theta < 2\pi$.

Solution

1. Cosine is $\sqrt{3}/2$ in QI and QIV.

Final answer

$$\theta = \frac{\pi}{6}, \frac{11\pi}{6}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz WRITTEN 6 Concept Quiz · Written**Solve** $\tan \theta \leq -1$ for $0 \leq \theta < 2\pi$.**Solution**

1. Use the two tangent branches and include the equality angles.

Final answer

$$\frac{\pi}{2} < \theta \leq \frac{3\pi}{4} \text{ or } \frac{3\pi}{2} < \theta \leq \frac{7\pi}{4}$$

Quiz WRITTEN 7 Concept Quiz · Written**Solve** $2 \sin^2 \theta + \sin \theta - 1 = 0$ for $0 \leq \theta < 2\pi$.**Solution**

1. Factor to $(2 \sin \theta - 1)(\sin \theta + 1) = 0$, then use the unit circle.

Final answer

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz WRITTEN 8 Concept Quiz · Written

Solve $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$ **for** $0 \leq \theta < 2\pi$.

Solution

1. Set $t = \theta + \pi/3$, solve $\cos(t) > 1/2$ in $[\pi/3, 7\pi/3)$, then subtract.

Final answer

$$\frac{4\pi}{3} < \theta < 2\pi$$

Answer key

Use the question number shown on each page.

- Q001** (a) $\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$ (b) $\theta = \frac{2\pi}{3} + n\pi, n \in \mathbb{Z}$ **Q012** $0 \leq \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta \leq \pi$ **Q020** B
- Q002** (a) $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$ (b) $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$ **Q013** (a) $\theta = \frac{\pi}{2}, \frac{11\pi}{6}$ (b) $0 \leq \theta < \pi$ or $\frac{5\pi}{3} < \theta < 2\pi$ **Q021** B
- Q003** (a) $\frac{3\pi}{4}, \frac{11\pi}{4}$ (b) $\frac{\pi}{2}, \frac{3\pi}{2}$ (c) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (d) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q014** (a) $\theta = 0, \frac{2\pi}{3}$ (b) $\theta = \pi, \frac{13\pi}{12}$ (c) $\frac{\pi}{2} < \theta < \pi$ **Q022** B
- Q004** $\tan \theta = 2$ or $\tan \theta = \sqrt{3}$; $\theta = \arctan(2) + n\pi$ or $\theta = \frac{\pi}{3} + n\pi$ **Q015** (a) $\frac{\pi}{2}, \frac{5\pi}{6}$ (b) $\frac{\pi}{2}, \frac{7\pi}{6}$ (c) $\pi, \frac{5\pi}{12}$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q023** A
- Q005** (a) $0 \leq \theta < \frac{5\pi}{6}$ or $\frac{7\pi}{6} < \theta < 2\pi$ (b) $0 \leq \theta \leq \frac{\pi}{6}$ or $\frac{\pi}{2} < \theta \leq \frac{7\pi}{6}$ or $\frac{3\pi}{2} < \theta < 2\pi$ **Q016** $0 \leq \theta \leq \frac{2\pi}{3}$ or $\pi \leq \theta \leq \frac{5\pi}{3}$ **Q024** C
- Q006** (a) $\frac{7\pi}{6} \leq \theta \leq \frac{11\pi}{6}$ (b) $0 \leq \theta < \frac{\pi}{3}$ or $\frac{5\pi}{3} < \theta < 2\pi$ (c) $\frac{\pi}{4} < \theta < \frac{\pi}{2}$ or $\frac{5\pi}{4} < \theta < \frac{3\pi}{2}$ **Q017** Dividing by $\cos(x)$ separates the interacting factors and is not helpful, use $\tan(x) \cos(x) = \sin(x)$ (with $\cos(x) \neq 0$), so $\sin(x) = \frac{1}{2}$ and $x = 30^\circ, 150^\circ$. **Q025** B
- Q007** (a) $\frac{\pi}{2}, \frac{3\pi}{2}$ (b) $\frac{\pi}{2}, \frac{3\pi}{2}$ (c) $\frac{\pi}{2}, \frac{3\pi}{2}$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q026** B
- Q008** $\frac{\pi}{3} < \theta < \pi$ or $\pi < \theta < \frac{5\pi}{3}$ **Q027** A
- Q009** (a) $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$ (b) $0 \leq \theta \leq \frac{2\pi}{3}$ or $\theta = \pi$ or $\frac{4\pi}{3} \leq \theta < 2\pi$ **Q028** A
- Q010** (a) $\theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi$ (b) $\theta = 0, \pi, \frac{7\pi}{6}, \frac{11\pi}{6}$ (c) $0 \leq \theta < \frac{2\pi}{3}$ or $\frac{4\pi}{3} < \theta < 2\pi$ **Q029** C
- Q011** (a) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (b) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (c) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (d) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (e) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (f) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (g) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (h) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (i) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (j) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (k) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (l) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (m) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (n) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (o) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (p) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (q) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (r) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (s) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (t) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (u) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (v) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (w) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (x) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (y) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (z) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ **Q030** B
- Q018** B **Q031** A
- Q019** B

Q032 A

Q033 A

Q034 B

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $\theta = \frac{\pi}{6}, \frac{11\pi}{6}$

Quiz WRITTEN 6 $\frac{\pi}{2} < \theta \leq \frac{3\pi}{4}$ or $\frac{3\pi}{2} < \theta \leq \frac{7\pi}{4}$

Quiz WRITTEN 7 $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

Quiz WRITTEN 8 $\frac{4\pi}{3} < \theta < 2\pi$

Concept 5 | Advanced Theorems and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Addition

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

Double angle

$$\sin 2A = 2 \sin A \cos A$$

Cosine

$$\cos 2A = \cos^2 A - \sin^2 A$$

Worked Solutions

Advanced Theorems and Applications

Q001 Written

Find the maximum and minimum of $y = \sin^2\theta + \cos\theta + 1$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Solution

1. Put $t = \cos\theta$: $y = -t^2 + t + 2 = -(t - \frac{1}{2})^2 + \frac{9}{4}$.
2. Check the vertex and the endpoints $t = -1, 1$, then return to θ .

Final answer

$$\max y = \frac{9}{4} \text{ at } \theta = \frac{\pi}{3}, \frac{5\pi}{3}; \quad \min y = 0 \text{ at } \theta = \pi$$

Q002 Written

Find the maximum and minimum of $y = \cos^2\theta - \sin\theta$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Solution

1. Put $t = \sin\theta$: $y = 1 - t^2 - t = -(t + \frac{1}{2})^2 + \frac{5}{4}$.
2. The endpoint $t = 1$ gives the minimum -1 .

Final answer

$$\max y = \frac{5}{4} \text{ at } \theta = \frac{7\pi}{6}, \frac{11\pi}{6}; \quad \min y = -1 \text{ at } \theta = \frac{\pi}{2}$$

Worked Solutions

Advanced Theorems and Applications

Q003 Written

Solve the four given maximum/minimum problems on $0 \leq \theta < 2\pi$, including $y = \sin^2 \theta + \sin \theta - 1$, $y = \cos^2 \theta - \cos \theta + 1$, $y = 2 \sin \theta + \cos^2 \theta - 1$, and $y = -2 \sin^2 \theta - 2 \cos \theta + 1$. Include the given challenge $y = \cos^2 \theta - \sin \theta$.

Solution

1. Choose the single trig variable that makes the expression quadratic.
2. The vertex is valid only when it lies in $[-1, 1]$; convert every extremal t back to the corresponding angles.

Final answer

See the numbered solution: substitute $t = \sin \theta$ or $t = \cos \theta$, complete the square, then test $t = \pm 1$.

Worked Solutions

Advanced Theorems and Applications

Q004 Written

Use addition theorems to find $\sin 75^\circ$ and $\cos(\pi/12)$.

Solution

1. Rewrite $75^\circ = 45^\circ + 30^\circ$ and $\pi/12 = 45^\circ - 30^\circ$.
2. Apply the sine-sum and cosine-difference identities.

Final answer

$$\sin 75^\circ = \cos \pi/12 = \frac{\sqrt{6} + \sqrt{2}}{4}$$

Q005 Written

Given $\sin \alpha = \frac{\sqrt{3}}{2}$ in QII and $\cos \beta = -\frac{3}{5}$ in QIII, find $\sin(\alpha + \beta)$ and $\cos(\alpha - \beta)$.

Solution

1. The quadrant signs give $\cos \alpha = -1/2$ and $\sin \beta = -4/5$.
2. Substitute in the addition identities without a calculator.

Final answer

$$\sin \alpha + \beta = \frac{4 - 3\sqrt{3}}{10}; \quad \cos \alpha - \beta = \frac{3 - 4\sqrt{3}}{10}$$

Worked Solutions

Advanced Theorems and Applications

Q006 Written

Complete the given addition-theorem exercises *exact values, quadrant – controlled α, β , and the three – angle challenge* ($\sin \alpha = 0.6, \cos \beta = 0.8, \tan \gamma = 0.75$).

Solution

1. Convert radians to degrees only when it clarifies the reference angle.
2. Recover the missing sine/cosine value from the quadrant before applying the sum or difference identity.

Final answer

See the numbered solution; the three-angle challenge gives $\sin(\alpha + \beta + \gamma) = \frac{117}{125}$.

Q007 Written

Find $\tan 105^\circ$, then find the angle between $y = 2x + 1$ and $y = -3x + 2$.

Solution

1. Use $105^\circ = 45^\circ + 60^\circ$ and the tangent addition theorem.
2. For the lines, use $m_1 = 2, m_2 = -3$ in $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$.

Final answer

$$\tan 105^\circ = -2 + \sqrt{3}; \quad \theta = \frac{\pi}{4}$$

Worked Solutions

Advanced Theorems and Applications

Q008 Written

Find the angle between each given pair of straight lines, including $y = \frac{1}{3}x$ with $y = 2x$, and $8x + 2y - 3 = 0$ with $-5x + 3y + 1 = 0$.

Solution

1. Convert each line to slope-intercept form.
2. Use the acute angle convention $0 < \theta < \pi/2$, replacing θ_1 by $\pi - \theta_1$ when necessary.

Final answer

The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.

Worked Solutions

Advanced Theorems and Applications

Q009 Written

Find the exact angles for all given line pairs and solve the challenge: a line through $P(2, 3)$ makes acute angle $\tan \theta = 3/4$ with $x + 2y - 4 = 0$; give both line equations and the areas of the triangles with the y -axis.

Solution

1. Solve the two linear equations produced by the tangent formula for the unknown slope.
2. Use the y -intercepts and the line intersection to obtain each triangle area.

Final answer

See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.

Worked Solutions

Advanced Theorems and Applications

Q010 Written

Given $\sin \alpha = 1/3$ **with** α **in** QII, **find** $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Solution

1. The quadrant gives $\cos \alpha = -2\sqrt{2}/3$.
2. Apply $\sin 2\alpha = 2 \sin \alpha \cos \alpha$, $\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$, then divide.

Final answer

$$\sin 2\alpha = -\frac{4\sqrt{2}}{9}, \quad \cos 2\alpha = \frac{7}{9}, \quad \tan 2\alpha = -\frac{4\sqrt{2}}{7}$$

Q011 Written

Given $\sin \alpha = -2/3$ **with** $2\pi < \alpha < 3\pi$, **find** $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Solution

1. The interval places α in QIII, so $\cos \alpha = -\sqrt{5}/3$.
2. Substitute into the three double-angle formulas.

Final answer

$$\sin 2\alpha = \frac{4\sqrt{5}}{9}, \quad \cos 2\alpha = \frac{1}{9}, \quad \tan 2\alpha = 4\sqrt{5}$$

Worked Solutions

Advanced Theorems and Applications

Q012 Written

Solve the given double-angle exercises for given sine/cosine values and the challenge $\cos \theta = 0.6, \pi < \theta < 2\pi$, evaluating $\cos 3\theta \tan 2\theta$.

Solution

1. Determine the missing trig value from the interval before using the identities.
2. For 3θ , use $4\cos^3\theta - 3\cos\theta$; for 2θ , use the tangent double-angle identity.

Final answer

For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.

Q013 Written

Given $\cos \alpha = 3/5, 0 < \alpha < \pi$, find $\sin(\alpha/2), \cos(\alpha/2), \tan(\alpha/2)$, and evaluate $\cos(5\pi/8)$.

Solution

1. Use the half-angle formulas and choose signs from $0 < \alpha/2 < \pi/2$.
2. For $(5\pi/8)$, use $\cos(\pi - 3\pi/8)$.

Final answer

$$\sin \frac{\alpha}{2} = \frac{1}{\sqrt{5}}, \quad \cos \frac{\alpha}{2} = \frac{2}{\sqrt{5}}, \quad \tan \frac{\alpha}{2} = \frac{1}{2}, \quad \cos \frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$$

Worked Solutions

Advanced Theorems and Applications

Q014 Written

Given $\cos \alpha = -1/3$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\sin(3\pi/8)$.

Solution

1. Since $\alpha/2$ is in QI, both half-angle roots are positive.
2. Use $3\pi/8 = \pi/2 - \pi/8$ or the half-angle identity.

Final answer

$$\sin \frac{\alpha}{2} = \sqrt{\frac{2}{3}}, \quad \cos \frac{\alpha}{2} = \frac{1}{\sqrt{3}}, \quad \tan \frac{\alpha}{2} = \sqrt{2}, \quad \sin \frac{3\pi}{8} = \frac{\sqrt{2 + \sqrt{2}}}{2}$$

Q015 Written

Solve the given half-angle exercises for the three quadrant intervals and the challenge with $\tan A = \frac{\sqrt{5}}{2}$, $4\pi < A < 5\pi$, $\cos B = -\frac{\sqrt{5}}{3}$, $5\pi < B < 6\pi$.

Solution

1. Halve the interval first; its quadrant controls every sign.
2. Apply the half-angle formula again for $(A/4)$ after finding $(\cos(A/2))$.

Final answer

Use the signed half-angle formulas; for the challenge, $\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6}$, $\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}$.

Worked Solutions

Advanced Theorems and Applications

Q016 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin 2\theta + \sin \theta = 0$, $b \cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Solution

1. Use $\sin 2\theta = 2 \sin \theta \cos \theta$ and factor.
2. Use $\cos 2\theta = 2 \cos^2 \theta - 1$, solve the quadratic in $c = \cos \theta$, and map the interval.

Final answer

$$a \theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}; \quad b \frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$$

Q017 Written

Solve the given equation $\sin 2\theta + \cos \theta = 0$ **on** $0 \leq \theta < 2\pi$.

Solution

1. Factor $\cos \theta (2 \sin \theta + 1) = 0$.
2. List the two cosine zeros and the two angles with $\sin \theta = -1/2$.

Final answer

$$\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$$

Worked Solutions

Advanced Theorems and Applications

Q018 Written

Solve all given double-angle equations and inequalities on $0 \leq \theta < 2\pi$, including the challenge $\cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Solution

1. Choose the identity that leaves one trig function only.
2. For inequalities, use a sign chart and include or exclude boundary points exactly as stated.

Final answer

See the numbered solution: replace double angles, solve in $s = \sin \theta$ or $c = \cos \theta$, and translate the accepted ranges.

Q019 Written

Rewrite $\sin \theta - \sqrt{3} \cos \theta$ as $r \sin(\theta + \alpha)$, $r > 0$, $-\pi < \alpha < \pi$.

Solution

1. Plot $P(1, -\sqrt{3})$, find $r = OP$, then read the angle α .

Final answer

$$r = 2, \quad \alpha = -\frac{\pi}{3}, \quad \sin \theta - \sqrt{3} \cos \theta = 2 \sin \theta - \frac{\pi}{3}$$

Worked Solutions

Advanced Theorems and Applications

Q020 Written

Rewrite the given expressions $\sin \theta - \cos \theta$, $\sin \theta + \cos \theta$, $3 \sin \theta - \sqrt{3} \cos \theta$, and $3 \sin \theta - \cos \theta$ in harmonic-addition form.

Solution

1. For $a \sin \theta + b \cos \theta$, use $r = \sqrt{a^2 + b^2}$, $\cos \alpha = a/r$, $\sin \alpha = b/r$.

Final answer

$$\sqrt{2} \sin \theta - \frac{\pi}{4}, \sqrt{2} \sin \theta + \frac{\pi}{4}, 2\sqrt{3} \sin \theta - \frac{\pi}{6}, \sqrt{10} \sin \theta - \arctan \frac{1}{3}$$

Q021 Written

Rewrite all given harmonic-addition exercises in the form $r \sin(\theta + \alpha)$, including $2 \sin \theta + 2 \cos \theta$ and the challenge $\sin \theta - \sqrt{3} \cos \theta$.

Solution

1. Keep the sign of the cosine coefficient when locating P(a,b).
2. Substitute the obtained (r,a) and expand once to verify both coefficients.

Final answer

See the numbered solution; every result has $r = \sqrt{a^2 + b^2} > 0$ and α chosen in $(-\pi, \pi)$.

Worked Solutions

Advanced Theorems and Applications

Q022 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin \theta - \cos \theta = 1$, $b \sin \theta + \cos \theta < 3/2$.**Solution**

1. Synthesize each left-hand side and account for the shifted angle range.
2. The maximum of $\sin \theta + \cos \theta$ is $\sqrt{2} < 3/2$, so the second inequality is always true.

Final answer

$$a \theta = \frac{\pi}{2}, \pi; \quad b \ 0 \leq \theta < 2\pi$$

Q023 Written

Solve $\sin \theta \leq \cos \theta$ on $0 \leq \theta < 2\pi$.**Solution**

1. Rewrite as $\sqrt{2} \sin(\theta - \pi/4) \leq 0$.
2. Read the accepted half-turns and retain the equality endpoints.

Final answer

$$0 \leq \theta \leq \frac{\pi}{4} \quad \text{or} \quad \frac{5\pi}{4} \leq \theta < 2\pi$$

Worked Solutions

Advanced Theorems and Applications

Q024 **Written**

Solve all given harmonic-addition equations and inequalities on $0 \leq \theta < 2\pi$, including $\sin \theta + \cos \theta + 1 = 0$ and $\sin \theta - \cos \theta + 3/2 > 0$.

Solution

1. After synthesis set $t = \theta + \alpha$ and write the exact t-range.
2. Use one full revolution in t, then subtract α and intersect with the original domain.

Final answer

See the numbered solution; synthesize first, then solve in the shifted variable over its full range.

Q025 **Written**

Find the maximum and minimum of $y = 2 \sin(\theta - \pi/3)$, and of $y = 2 \cos 2\theta + 4 \cos \theta + 1$, $0 \leq \theta < 2\pi$.

Solution

1. Use $-1 \leq \sin(t) \leq 1$ for the first function.
2. For the second, set $t = \cos \theta$ and complete the square after the double-angle substitution.

Final answer

First: max = 2, min = -2. Second: max = 7 at $\theta = 0$; min = -2 at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$

Worked Solutions

Advanced Theorems and Applications

Q026 Written

Find the maximum and minimum of the given functions $y = -\frac{4}{3} \sin^2 \theta + \cos \theta$ and $y = \cos 2\theta - 2 \cos \theta$.

Solution

1. Reduce double angles or $\sin^2 \theta$ to one variable.
2. Check the vertex and both endpoints of the allowed interval.

Final answer

For the first, $\max = 1$, $\min = -\frac{73}{48}$; for the second, use $t = \cos \theta$ and the endpoint/vertex test.

Q027 Written

Solve all given maximum/minimum problems in the final lesson, including absolute-value and double-angle forms, and state the corresponding angles where requested.

Solution

1. Use the same transformation rules as for equations and inequalities.
2. Never report a vertex outside $[-1, 1]$; compare it with both endpoints.

Final answer

See the numbered solution: synthesize $a \sin \theta + b \cos \theta$ or substitute $t = \cos \theta$, then test the full allowed range.

Worked Solutions

Advanced Theorems and Applications

Q028 MCQ

The substitution for $\sin^2 \theta + \cos \theta + 1$ should be:**Solution**1. Use $\sin^2 \theta = 1 - t^2$.

Correct option: B

Final answer $t = \cos \theta$

Q029 MCQ

The maximum of $\sin^2 \theta + \cos \theta + 1$ is:**Solution**

1. Complete the square.

Correct option: C

Final answer $9/4$

Worked Solutions

Advanced Theorems and Applications

Q030 MCQ

An extremal vertex is valid only if t lies in:**Solution**

1. Trig sine and cosine values are bounded.

Correct option: B**Final answer** $[-1, 1]$

Q031 MCQ

The identity for $\sin(a+b)$ is:**Solution**

1. Use the addition theorem.

Correct option: A**Final answer** $\sin a \cos b + \cos a \sin b$

Worked Solutions

Advanced Theorems and Applications

Q032 MCQ

sin 75 degrees equals:**Solution**

1. Rewrite 75 as 45+30.

Correct option: B

Final answer

$$\sqrt{6} + \sqrt{2}/4$$

Q033 MCQ

A missing trig value is selected using:**Solution**

1. The quadrant fixes the sign.

Correct option: A

Final answer

the quadrant

Worked Solutions

Advanced Theorems and Applications

Q034 MCQ

The slope of $y=mx+k$ is:**Solution**

1. The coefficient of x is the slope.

Correct option: B**Final answer** m

Q035 MCQ

The angle formula uses denominator:**Solution**

1. Use the tangent difference theorem.

Correct option: A**Final answer** $1 + m_1 m_2$

Worked Solutions

Advanced Theorems and Applications

Q036 MCQ

For an acute angle, if $\tan \theta_1$ is negative we use:

Solution

1. Choose the acute representative.

Correct option: B

Final answer

$\pi - \theta_1$

Q037 MCQ

$\sin(2a)$ equals:

Solution

1. Double-angle identity.

Correct option: B

Final answer

$2 \sin a \cos a$

Worked Solutions

Advanced Theorems and Applications

Q038 MCQ

cos(2a) can be written as:**Solution**

1. Use the standard formula.

Correct option: A**Final answer**

$$\cos^2 a - \sin^2 a$$

Q039 MCQ

For theta in QIV, sin theta is:**Solution**

1. Read the quadrant.

Correct option: B**Final answer**

negative

Worked Solutions

Advanced Theorems and Applications

Q040 MCQ

The half-angle formula for $\sin(a/2)$ uses:**Solution**1. Choose the sign from $a/2$.

Correct option: A

Final answer $\sqrt{(1 - \cos a)/2}$

Q041 MCQ

The sign of a half-angle root is chosen from:**Solution**

1. Halving changes the interval.

Correct option: A

Final answer

the original quadrant interval

Worked Solutions

Advanced Theorems and Applications

Q042 MCQ

If $\cos a = -1/3$ and $0 < a < \pi$, $\cos(a/2)$ is:

Solution

1. $a/2$ is in QI.

Correct option: B

Final answer

positive

Q043 MCQ

For double-angle equations, first rewrite in:

Solution

1. Use a double-angle identity.

Correct option: A

Final answer

one trig function

Worked Solutions

Advanced Theorems and Applications

Q044 MCQ

The factorisation of $\sin(2\theta) + \sin\theta$ is:**Solution**1. Factor after substituting $\sin(2\theta)$.

Correct option: A

Final answer $\sin\theta(2\cos\theta + 1)$

Q045 MCQ

A tangent asymptote is always:**Solution**

1. Tangent is undefined there.

Correct option: B

Final answer

excluded

Worked Solutions

Advanced Theorems and Applications

Q046 MCQ

For a $\sin \theta + b \cos \theta$, r is:**Solution**

1. It is the length OP.

Correct option: B**Final answer**

$$\sqrt{a^2 + b^2}$$

Q047 MCQ

The harmonic target form is:**Solution**

1. Match the coefficients.

Correct option: A**Final answer**

$$r \sin(\theta + a)$$

Worked Solutions

Advanced Theorems and Applications

Q048 MCQ

The angle a is located from point:**Solution**

1. The coefficient point determines direction.

Correct option: A

Final answer $P(a, b)$

Q049 MCQ

After synthesis, t is usually:**Solution**

1. Track the shifted range.

Correct option: A

Final answer $\theta + a$

Worked Solutions

Advanced Theorems and Applications

Q050 MCQ

If $\max(\sin \theta + \cos \theta) = \sqrt{2}$, then $\sin \theta + \cos \theta < 3/2$ is:

Solution

1. $\sqrt{2} < 3/2$.

Correct option: B

Final answer

always true

Q051 MCQ

In a strict inequality, equality endpoints are:

Solution

1. Use open endpoints.

Correct option: B

Final answer

excluded

Worked Solutions

Advanced Theorems and Applications

Q052 MCQ

The range of $2\sin(\theta - \pi/3)$ is:**Solution**

1. Multiply the sine range by 2.

Correct option: B

Final answer $[-2, 2]$

Q053 MCQ

For a double-angle maximum problem, use:**Solution**1. Reduce to $t = \sin \theta$ or $\cos \theta$.

Correct option: A

Final answer

the double-angle formula then a bounded variable

Worked Solutions

Advanced Theorems and Applications

Q054 MCQ

A vertex outside $[-1,1]$ is:**Solution**

1. It is not attainable by sine/cosine.

Correct option: B**Final answer**

rejected

Quiz MCQ 1 Concept Quiz · MCQ

The maximum of $2 \sin(\theta - \pi/3)$ is:**Solution**

1. Use the sine range.

Correct option: D**Final answer**

2

Worked Solutions

Advanced Theorems and Applications

Quiz MCQ 2 [Concept Quiz · MCQ](#)

The addition-theorem value $\sin 75^\circ$ is:

Solution

1. Use 45 degrees plus 30 degrees.

Correct option: A

Final answer

$$\sqrt{6} + \sqrt{2}/4$$

Quiz MCQ 3 [Concept Quiz · MCQ](#)

For $y = 2x^2 + 4x + 1$, the vertex x-value is:

Solution

1. Complete the square.

Correct option: A

Final answer

$$-2$$

Worked Solutions

Advanced Theorems and Applications

Quiz MCQ 4 **Concept Quiz · MCQ**

The angle between slopes 2 and -3 is:

Solution

1. The tangent magnitude is 1.

Correct option: B

Final answer

$$\pi/4$$

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the maximum and minimum of $y = \sin^2 \theta + \cos \theta + 1$ on $0 \leq \theta < 2\pi$.

Solution

1. Let $t = \cos \theta$ and complete the square.

Final answer

$$\max = \frac{9}{4}, \min = 0$$

Worked Solutions

Advanced Theorems and Applications

Quiz WRITTEN 6 Concept Quiz · Written

Find $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$ **when** $\sin \alpha = 1/3$ **and** α **is in** QII.

Solution

1. Use the quadrant to select $\cos \alpha$.

Final answer

$$-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$$

Quiz WRITTEN 7 Concept Quiz · Written

Rewrite $3 \sin \theta - \sqrt{3} \cos \theta$ **as** $r \sin(\theta + \alpha)$.

Solution

1. Find r and the coefficient angle from $P(3, -\sqrt{3})$.

Final answer

$$2\sqrt{3} \sin \theta - \frac{\pi}{6}$$

Worked Solutions

Advanced Theorems and Applications

Quiz WRITTEN 8 **Concept Quiz · Written****Solve** $\sin \theta \leq \cos \theta$ **on** $0 \leq \theta < 2\pi$.**Solution**1. Synthesize to $\sqrt{2} \sin(\theta - \pi/4) \leq 0$.**Final answer**

$$[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$$

Answer key

Use the question number shown on each page.

- Q001** $\max y = \frac{9}{4}$ at $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$; $\min y = 0$ at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$
- Q002** $\max y = \frac{5}{4}$ at $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$; $\min y = -1$ at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$
- Q003** See the numbered solution: substitute $t = \sin \theta$ or $t = \cos \theta$, complete the square, then test $t = \pm 1$.
- Q004** $\sin 75^\circ = \cos \pi/12 = \frac{\sqrt{6} + \sqrt{2}}{4}$
- Q005** $\sin \alpha + \beta = \frac{4 - 3\sqrt{3}}{10}$; $\cos \alpha - \beta = \frac{3 - 4\sqrt{3}}{10}$
- Q006** See the numbered solution; the three-angle challenge gives $\sin(\alpha + \beta + \gamma) = \frac{117}{125}$.
- Q007** $\tan 105^\circ = -2 + \sqrt{3}$; $\theta = \frac{\pi}{4}$
- Q008** The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.
- Q009** See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.
- Q010** $\sin 2\alpha = -\frac{4\sqrt{2}}{9}$, $\cos 2\alpha = \frac{7}{9}$, $\tan 2\alpha = -\frac{4\sqrt{2}}{7}$
- Q011** $\sin 2\alpha = \frac{4\sqrt{5}}{9}$, $\cos 2\alpha = \frac{1}{9}$, $\tan 2\alpha = 4\sqrt{5}$
- Q012** For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.
- Q013** $\sin \frac{\alpha}{2} = \frac{1}{\sqrt{5}}$, $\cos \frac{\alpha}{2} = \frac{2}{\sqrt{5}}$, $\tan \frac{\alpha}{2} = \frac{1}{2}$, $\cos \frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$
- Q014** $\sin \frac{\alpha}{2} = \sqrt{\frac{2}{3}}$, $\cos \frac{\alpha}{2} = \frac{1}{\sqrt{3}}$, $\tan \frac{\alpha}{2} = \sqrt{2}$, $\sin \frac{3\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$
- Q015** Use the signed half-angle formulas; for the challenge,
 $\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6}$,
 $\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}$.
- Q016** $a \theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$; $b \frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$
- Q017** $\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$
- Q018** See the numbered solution: replace double angles, solve in $s = \sin \theta$ or $c = \cos \theta$, and translate the accepted ranges.
- Q019** $r = 2$, $\alpha = -\frac{\pi}{3}$, $\sin \theta - \sqrt{3} \cos \theta = 2 \sin \theta - \frac{\pi}{3}$
- Q020** $\sqrt{2} \sin \theta - \frac{\pi}{4}$, $\sqrt{2} \sin \theta + \frac{\pi}{4}$, $2\sqrt{3} \sin \theta - \frac{\pi}{6}$, $\sqrt{10} \sin \theta - \arctan \frac{1}{2}$
- Q021** See the numbered solution; every result has $r = \sqrt{a^2 + b^2} > 0$ and α chosen in $(-\pi, \pi)$.
- Q022** $a \theta = \frac{\pi}{2}, \pi$; $b \ 0 \leq \theta < 2\pi$
- Q023** $0 \leq \theta \leq \frac{\pi}{4}$ or $\frac{5\pi}{4} \leq \theta < 2\pi$
- Q024** See the numbered solution; synthesize first, then solve in the shifted variable over its full range.
- Q025** First: $\max = 2$, $\min = -2$. Second: $\max = 7$ at $\theta = 0$; $\min = -2$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$
- Q026** For the first, $\max = 1$, $\min = -\frac{73}{48}$; for the second, use $t = \cos \theta$ and the endpoint/vertex test.
- Q027** See the numbered solution: synthesize $a \sin \theta + b \cos \theta$ or substitute $t = \cos \theta$, then test the full allowed range.
- Q028** B
- Q029** C
- Q030** B
- Q031** A
- Q032** B
- Q033** A
- Q034** B
- Q035** A
- Q036** B
- Q037** B
- Q038** A
- Q039** B
- Q040** A
- Q041** A
- Q042** B
- Q043** A
- Q044** A
- Q045** B
- Q046** B
- Q047** A
- Q048** A
- Q049** A
- Q050** B
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Quiz MCQ 1** D
- Quiz MCQ 2** A
- Quiz MCQ 3** A
- Quiz MCQ 4** B
- Quiz WRITTEN 5** $\max = \frac{9}{4}$, $\min = 0$
- Quiz WRITTEN 6** $-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$
- Quiz WRITTEN 7** $2\sqrt{3} \sin \theta - \frac{\pi}{6}$
- Quiz WRITTEN 8** $[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$

Concept 1 | Laws of Exponents and Roots

English Student Edition • Focused formulas and guided practice

Formula opener

Product

$$a^m a^n = a^{m+n}$$

Power

$$(a^m)^n = a^{mn}$$

Root

$$a^{1/n} = \sqrt[n]{a}$$

Worked Solutions

Laws of Exponents and Roots

Q001 Written

Find: (a) 10^0 , (b) 2^{-3} , (c) $(0.2)^{-2}$.**Solution**

1. Use $a^0 = 1$ and $a^{-n} = 1/a^n$; convert $0.2 = 1/5$.

Final answer

$$1, \frac{1}{8}, 25$$

Q002 Written

Let $a \neq 0, b \neq 0$. Simplify: (a) $a^{-5}a^8$, (b) $a^5 \div a^{-3}$, (c) $(a^{-1}b)^2$, (d) $(2^2)^3 \div 2^{-2} \times 2^{-3}$.**Solution**

1. Add exponents when multiplying, subtract when dividing, and multiply exponents for a power of a power.

Final answer

$$a^3, a^8, \frac{b^2}{a^2}, 32$$

Worked Solutions

Laws of Exponents and Roots

Q003 **Written****Find:** (a) 5^0 , (b) 3^{-2} , (c) $(0.25)^{-2}$.**Solution**1. Apply the zero and reciprocal-power laws; $0.25 = 1/4$.**Final answer** $1, \frac{1}{9}, 16$

Worked Solutions

Laws of Exponents and Roots

Q004 **Written**

Simplify the eight given expressions: a^5a^4 , $a^3 \div a^{-2}$, $(a^{-2}b^4)^{-1}$, $2^3 \times 2^{-7}$, $5^{-4} \div 5^{-6}$, $(a^3b^{-1})^3(a^2b^{-3})^{-2}$, $4^{-5}4^2 \div 4^{-2}$, and $2^4 \div (2^{-1})^3 \times 2^{-2}$.

Solution

1. Collect the exponent of each common base; distribute a negative outer power over every factor.

Final answer

$$a^9, a^5, a^2b^{-4}, \frac{1}{16}, 25, a^5b^3, \frac{1}{4}, 32$$

Worked Solutions

Laws of Exponents and Roots

Q005 Written

Find the nine values: $3^0, 11^0, (2/3)^0, 4^{-3}, 5^{-3}, 6^{-2}, (1/3)^{-3}, (3/5)^{-2}, (0.04)^{-2}$.**Solution**

1. Convert terminating decimals to fractions before applying the reciprocal rule.

Final answer $1, 1, 1, \frac{1}{64}, \frac{1}{125}, \frac{1}{36}, 27, \frac{25}{9}, 625$

Worked Solutions

Laws of Exponents and Roots

Q006 Written

Simplify the twelve same-base expressions in given Question 2: $a^2a^4, a^{-2}a^5, a^{-3}a^{-4}, a^3 \div a^5, a^8 \div a^{-3}, a^{-6} \div a^{-2}, (a^{-1})^3, (a^3b^4)^{-2}, (a^{-3}b^2)^{-3}, 3^53^{-3}, 4^44^{-3}, 5^55^{-5}$.

Solution

1. Use one exponent line for each common base and leave negative powers in reciprocal form when useful.

Final answer

$$a^6, a^3, a^{-7}, a^{-2}, a^{11}, a^{-4}, a^{-3}, a^{-6}b^{-8}, a^9b^{-6}, 9, 4, 1$$

Worked Solutions

Laws of Exponents and Roots

Q007 Written

Simplify given Question 3: $5^4 \div 5^2$, $3^2 \div 3^{-2}$, $2^{-4} \div 2^{-5}$, $(a^{-2}b)^3(a^2b^{-1})^2$, $(a^{-3}b)^{-3}(a^4b^2)^{-2}$, $(2a^3b^{-1})^3(-3ab^{-2})^{-3}$, $3^23^{-1} \div 3^3$, $5^{-4}5^5 \div 5^{-2}$, $2^42^{-8} \div 2^{-6}$, $5^3(5^{-2})^2 \div 5^{-4}$, $3^5 \div (3^{-2})^{-3} \times 3$, and $(a/2)^3a^{-4} \div (2a)^{-2}$.

Solution

1. Distribute powers first, then combine exponents base by base; keep coefficients separate from variables.

Final answer

$$25, 81, 2, \frac{b}{a^2}, \frac{a}{b^7}, -\frac{8a^6b^3}{27}, \frac{1}{9}, 125, 4, 125, 1, \frac{a}{2}$$

Worked Solutions

Laws of Exponents and Roots

Q008 Written

Simplify completely with positive exponents: $\left(\frac{2a^2}{b^{-3}}\right)^{-2} (4a^{-3}b^2)^3 \div \left(\frac{a}{2b}\right)^{-4} \div (a^{-2}b^2)$.

Solution

1. Rewrite the first factor as $(2a^2b^3)^{-2}$; distribute powers and combine all a- and b-exponents.

Final answer

$$\frac{1}{a^7b^6}$$

Q009 Written

Find $\sqrt[3]{64}$ and $\sqrt[3]{1/27}$.

Solution

1. Recognise $64 = 4^3$ and $1/27 = (1/3)^3$.

Final answer

$$4, \frac{1}{3}$$

Worked Solutions

Laws of Exponents and Roots

Q010 Written

Perform: $\sqrt[3]{12}\sqrt[3]{18}$, $\sqrt[3]{250}/\sqrt[3]{2}$, $(\sqrt[4]{8})^2$, $\sqrt[3]{\sqrt{729}}$, and $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Solution

1. Combine equal-index roots, factor perfect powers, then collect like cube-root terms.

Final answer $6, 5, 2, 3, 2\sqrt[3]{2}$

Q011 Written

Find $\sqrt[4]{81}$ and $\sqrt[4]{1/16}$.

Solution

1. Use perfect fourth powers.

Final answer $3, \frac{1}{2}$

Worked Solutions

Laws of Exponents and Roots

Q012 Written

Perform: $\sqrt[4]{8}\sqrt[4]{32}$, $\sqrt[4]{48}/\sqrt[4]{3}$, $(\sqrt[6]{100})^3$, $\sqrt[3]{\sqrt[3]{64}}$, and $\sqrt[3]{81} - \sqrt[3]{3} + \sqrt[3]{24}$.

Solution

- Combine roots with the same index and extract perfect powers; leave the nested radical exact.

Final answer

4, 2, 10, $\sqrt[3]{4}$, $4\sqrt[3]{3}$

Q013 Written

Find: $\sqrt[3]{216}$, $\sqrt[4]{625}$, $\sqrt[5]{243}$, $\sqrt[3]{1/64}$, $\sqrt[3]{1/125}$, $\sqrt[4]{1/81}$, $\sqrt[3]{1}$, $\sqrt[5]{100000}$, $\sqrt[4]{0.0001}$.

Solution

- Prime-factorise each radicand and match the index to a perfect power.

Final answer

6, 5, 3, $\frac{1}{4}$, $\frac{1}{5}$, $\frac{1}{3}$, 1, 10, 0.1

Worked Solutions

Laws of Exponents and Roots

Q014 Written

Perform the fifteen **given** calculations involving products, quotients, powers and sums of roots (including $\sqrt[3]{2}\sqrt[3]{4}$, $\sqrt[4]{27}\sqrt[4]{243}$, nested roots and the final three like-radical sums).

Solution

1. Convert every radicand to prime factors, combine equal-index roots, and collect only like radicals.

Final answer

$2, 4, 9, \sqrt[4]{27}, \frac{1}{4}, \frac{1}{5}, 7, 3, 4, 9, 4, \sqrt[3]{9}, \sqrt[3]{3}, 0, 0$

Worked Solutions

Laws of Exponents and Roots

Q015 Written

Perform $\sqrt[3]{\sqrt[4]{4096}} \div \sqrt{\sqrt[6]{4096}}$.

Solution

1. Since $4096 = 2^{12}$, both numerator and denominator equal 2.

Final answer

1

Q016 Written

Express in the form a^r : $\sqrt{5}$, $\sqrt[5]{2^4}$, and $1/\sqrt[5]{3^3}$.

Solution

1. Use $\sqrt[n]{a^m} = a^{m/n}$ and move a radical denominator to a negative exponent.

Final answer

$5^{1/2}$, $2^{4/5}$, $3^{-3/5}$

Worked Solutions

Laws of Exponents and Roots

Q017 Written

Find: $27^{2/3}$, $64^{-3/2}$, $(25/4)^{-3/2}$, and $[(16/9)^{-3/4}]^{2/3}$.**Solution**

1. Rewrite each base as a perfect power, multiply fractional exponents, and apply negative powers last.

Final answer

$$9, \frac{1}{512}, \frac{8}{125}, \frac{3}{4}$$

Q018 Written

Express in the form a^r : $\sqrt{a}$, $\sqrt[4]{a^3}$, $\sqrt[6]{a^5}$, $\sqrt[3]{2^2}$, $1/\sqrt{a}$, $1/\sqrt[3]{a^2}$, and $1/\sqrt[4]{5^3}$.**Solution**

1. Index over radicand exponent gives the rational power; reciprocals change the sign.

Final answer

$$a^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$$

Worked Solutions

Laws of Exponents and Roots

Q019 Written

Find $81^{3/4}$, $16^{-3/2}$, $(25/4)^{-3/2}$, **and** $[(27/125)^{1/2}]^{-4/3}$.**Solution**

1. Prime-factorise the bases before multiplying the fractional exponents.

Final answer

$$27, \frac{1}{64}, \frac{8}{125}, \frac{25}{9}$$

Q020 Written

Express the nine given expressions in the form a^r , including roots of a and numerical bases 2, 3, 5.**Solution**

1. Translate each root and reciprocal directly into a rational exponent.

Final answer

$$2^{1/2}, 6^{1/2}, 7^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$$

Worked Solutions

Laws of Exponents and Roots

Q021 Written

Evaluate the twelve given expressions with rational powers: $25^{3/2}$, $16^{3/4}$, $64^{2/3}$, $9^{-3/2}$, $125^{-2/3}$, $81^{-3/4}$, $(1/8)^{-1/3}$, $(125/8)^{-2/3}$, $(16/81)^{-3/4}$, $[(1/2)^{-1/3}]^{-2/3}$, $[(1/2)^{-1/3}]^{-2/3}$, $[(1/2)^{-1/3}]^{-2/3}$

Solution

1. Express every base as a power of a small integer, then multiply exponents exactly.

Final answer

$$125, 8, 16, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$$

Worked Solutions

Laws of Exponents and Roots

Q022 Written

Simplify $\left[\left(\left(1000/27\right)^{-2/3}\right)^{7/15}\right]^{-15/14}$.

Solution

1. Since $1000/27 = (10/3)^3$, the inner power is $(10/3)^{-2}$; multiply the remaining exponents.

Final answer

$$\frac{10}{3}$$

Q023 Written

Perform: (a) $2^{2/3}2^{1/2} \div 2^{7/6}$, (b) $6^{1/2}36^{1/4}$, (c) $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$, (d) $\sqrt{2} \div \sqrt[5]{32} \times \sqrt[3]{16}$.

Solution

1. Make the bases the same, then add/subtract rational exponents.

Final answer

1, 6, 6, 2

Worked Solutions

Laws of Exponents and Roots

Q024 **Written****Perform:** $5^{1/2}5^{1/3} \div 5^{-1/6}$, $4^{5/6}2^{4/3}$, $\sqrt{7}\sqrt[3]{7}\sqrt[6]{7}$, and $\sqrt[4]{27}\sqrt{27} \div \sqrt[4]{3}$.**Solution**

1. Rewrite powers of 4 and 27 with bases 2 and 3, then combine exponents.

Final answer

5, 8, 7, 9

Q025 **Written****For** $a > 0, b > 0$, **perform:** $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ and $\sqrt[3]{a}\sqrt{ab} \div \sqrt[6]{a^2/b^3}$.**Solution**

1. Translate every radical to exponents of a and b, then collect like bases.

Final answer $a^{1/3}, b\sqrt{a}$

Worked Solutions

Laws of Exponents and Roots

Q026 **Written**

Perform the twelve numerical calculations in given Question 1, including rational powers of 2, 3 and 5 and mixed radical products/quotients.

Solution

1. Make all bases equal before adding and subtracting rational exponents; extract perfect powers.

Final answer

$$1, 9, 5, 9, \sqrt{5}, 4, 5, 3, 2, 1, 8, \sqrt[3]{3}$$

Q027 **Written**

For $a > 0, b > 0$, simplify the six given expressions built from $\sqrt{a}, \sqrt[3]{a}, \sqrt[4]{a}, \sqrt[6]{a}$ and mixed a,b radicals.

Solution

1. Convert all radicals to rational powers and combine exponents separately for a and b.

Final answer

$$a^{1/3}, a^{7/6}, a^4, a^{2/3}b^{5/6}, a^{1/2}b^3, a^{13/6}b^{-5/6}$$

Worked Solutions

Laws of Exponents and Roots

Q028 Written

Simplify the ministry expression $\left(\frac{(243/32)^{-2/5}(9/4)^{-1}}{(125/64)^{1/3}(0.004)^{-1/2}} \right)^{-3/2}$.

Solution

1. The printed decimal is $0.004 = 1/250$, so the exact result is not rational despite the prompt's wording; retain the exact radical form.

Final answer

$$\left(\frac{2025\sqrt{10}}{64} \right)^{3/2}$$

Q029 Written

Assess the statement $\sqrt{a^2} = a$ **for all real a, using positive and negative examples.**

Solution

1. For $a = 3$ the value is 3; for $a = -3$ the value is 3, not -3.

Final answer

Sometimes: $\sqrt{a^2} = |a|$; it equals a only when $a \geq 0$.

Worked Solutions

Laws of Exponents and Roots

Q030 MCQ

The value of 2^{-3} is:**Solution**

1. Take the reciprocal of 2^3 .

Correct option: B**Final answer** $1/8$

Q031 MCQ

When multiplying equal bases, we:**Solution**

1. Use $a^m a^n = a^{m+n}$.

Correct option: B**Final answer**

add exponents

Worked Solutions

Laws of Exponents and Roots

Q032 MCQ

The expression $a^5 \div a^{-3}$ simplifies to:**Solution**1. Subtract: $5 - (-3) = 8$.

Correct option: B

Final answer a^8

Q033 MCQ

A negative exponent means:**Solution**1. Use $a^{-n} = 1/a^n$.

Correct option: B

Final answer

take the reciprocal power

Worked Solutions

Laws of Exponents and Roots

Q034 MCQ

The value of $(0.04)^{-2}$ is:**Solution**

1. $0.04 = 1/25$, so the reciprocal square is 25^2 .

Correct option: C

Final answer

625

Q035 MCQ

The exponent of a in $(a^{-2}b)^3(a^2b^{-1})^2$ is:**Solution**

1. Compute $-6 + 4 = -2$.

Correct option: A

Final answer

-2

Worked Solutions

Laws of Exponents and Roots

Q036 MCQ

The positive-exponent form of the 6-1 challenge is:**Solution**1. Combine the final powers as $a^{-7}b^{-6}$.

Correct option: B

Final answer

$$1/(a^7b^6)$$

Q037 MCQ

The value of $3^5 \div (3^{-2})^{-3} \times 3$ is:**Solution**1. The exponent is $5 - 6 + 1 = 0$.

Correct option: B

Final answer

$$1$$

Worked Solutions

Laws of Exponents and Roots

Q038 MCQ

 $\sqrt[3]{64}$ equals:**Solution**1. Since $4^3 = 64$.

Correct option: B

Final answer

4

Q039 MCQ

 $\sqrt[3]{12}\sqrt[3]{18}$ equals:**Solution**1. Combine to $\sqrt[3]{216} = 6$.

Correct option: C

Final answer

6

Worked Solutions

Laws of Exponents and Roots

Q040 MCQ

The fourth root of $1/16$ is:**Solution**

1. $(1/2)^4 = 1/16$.

Correct option: A**Final answer**

$$1/2$$

Q041 MCQ

The result of $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$ is:**Solution**

1. Collect $3\sqrt[3]{2} - 2\sqrt[3]{2} + \sqrt[3]{2}$.

Correct option: B**Final answer**

$$2\sqrt[3]{2}$$

Worked Solutions

Laws of Exponents and Roots

Q042 MCQ

 $\sqrt[4]{625}$ equals:

Solution

1. $5^4 = 625$.

Correct option: B

Final answer

5

Q043 MCQ

When combining roots with the same index, multiply:

Solution

1. Use $\sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$.

Correct option: B

Final answer

the radicands

Worked Solutions

Laws of Exponents and Roots

Q044 MCQ

The roots challenge evaluates to:**Solution**

1. Both nested roots are 2.

Correct option: B

Final answer

1

Q045 MCQ

 $\sqrt[5]{2^4}$ in rational-index form is:**Solution**

1. Index becomes the denominator.

Correct option: A

Final answer $2^{4/5}$

Worked Solutions

Laws of Exponents and Roots

Q046 MCQ

The value of $27^{2/3}$ is:**Solution**

1. $(3^3)^{2/3} = 3^2$.

Correct option: C**Final answer**

9

Q047 MCQ

A reciprocal square root is written as:**Solution**

1. A denominator changes the exponent sign.

Correct option: B**Final answer** $a^{-1/2}$

Worked Solutions

Laws of Exponents and Roots

Q048 MCQ

The value of $81^{-3/4}$ is:**Solution**1. $81^{3/4} = 27$, then take the reciprocal.

Correct option: C

Final answer

1/27

Q049 MCQ

The exponent of a in $\sqrt[4]{a^3}$ is:**Solution**

1. Radical index is the denominator.

Correct option: A

Final answer

3/4

Worked Solutions

Laws of Exponents and Roots

Q050 MCQ

The value of $[(64/27)^{-4/3}]^{-1/2}$ is:**Solution**

1. Use $64/27 = (4/3)^3$ and multiply exponents.

Correct option: C

Final answer

16/9

Q051 MCQ

The 6-3 challenge simplifies to:**Solution**

1. Collect the three rational exponents.

Correct option: B

Final answer

10/3

Worked Solutions

Laws of Exponents and Roots

Q052 MCQ

 $2^{2/3}2^{1/2} \div 2^{7/6}$ equals:**Solution**1. The exponent is $4/6 + 3/6 - 7/6 = 0$.

Correct option: B

Final answer

1

Q053 MCQ

 $4^{5/6}2^{4/3}$ equals:**Solution**1. Rewrite $4 = 2^2$, giving exponent 3.

Correct option: C

Final answer

8

Worked Solutions

Laws of Exponents and Roots

Q054 MCQ

 $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ equals:**Solution**1. Compute $3/2 - 1/3 - 5/6 = 1/3$.

Correct option: A

Final answer

$$a^{1/3}$$

Q055 MCQ

The numerical result of $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$ is:**Solution**

1. The exponents add to 1.

Correct option: C

Final answer

$$6$$

Worked Solutions

Laws of Exponents and Roots

Q056 MCQ

The given expression $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ tests:

Solution

1. Translate roots before combining.

Correct option: B

Final answer

combining rational exponents

Q057 MCQ

The printed decimal in the 6-4 challenge is:

Solution

1. Retain the ministry's printed value exactly.

Correct option: B

Final answer

0.004

Worked Solutions

Laws of Exponents and Roots

Q058 MCQ

For real a , $\sqrt{a^2}$ equals:

Solution

1. The principal square root is nonnegative.

Correct option: C

Final answer

$|a|$

Quiz MCQ 1 Concept Quiz · MCQ

Which law gives $a^m a^n = a^{m+n}$?

Solution

1. Add exponents for multiplication.

Correct option: A

Final answer

same-base multiplication

Worked Solutions

Laws of Exponents and Roots

Quiz MCQ 2 [Concept Quiz · MCQ](#)

The value of $\sqrt[3]{250}/\sqrt[3]{2}$ is:

Solution

1. Combine to $\sqrt[3]{125}$.

Correct option: B

Final answer

5

Quiz MCQ 3 [Concept Quiz · MCQ](#)

The value of $64^{-3/2}$ is:

Solution

1. $64^{3/2} = 512$, then take the reciprocal.

Correct option: C

Final answer

1/512

Worked Solutions

Laws of Exponents and Roots

Quiz MCQ 4 Concept Quiz · MCQ

For a negative real a , $\sqrt{a^2}$ is:

Solution

1. The principal root is $|a| = -a$ when a is negative.

Correct option: B

Final answer

$$-a$$

Quiz WRITTEN 5 Concept Quiz · Written

Simplify $a^{-3}a^7 \div a^2$.

Solution

1. The exponent is $-3 + 7 - 2 = 2$.

Final answer

$$a^2$$

Worked Solutions

Laws of Exponents and Roots

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Solution

1. Extract $3\sqrt[3]{2} - 2\sqrt[3]{2} + \sqrt[3]{2}$.

Final answer

$$2\sqrt[3]{2}$$

Quiz WRITTEN 7 Concept Quiz · Written

Evaluate $125^{-2/3}$.

Solution

1. $125^{2/3} = 25$, then take the reciprocal.

Final answer

$$1/25$$

Worked Solutions

Laws of Exponents and Roots

Quiz WRITTEN 8 Concept Quiz · Written

Explain why $\sqrt{a^2} = a$ is not true for every real a .

Solution

1. Use one positive and one negative counterexample.

Final answer

$\sqrt{a^2} = |a|$; e.g. $a = -3$ gives $3 \neq -3$

Answer key

Use the question number shown on each page.

Q001 $1, \frac{1}{8}, 25$

Q002 $a^3, a^8, \frac{b^2}{a^2}, 32$

Q003 $1, \frac{1}{9}, 16$

Q004 $a^9, a^5, a^2b^{-4}, \frac{1}{16}, 25, a^5b^3, \frac{1}{4}, 32$

Q005 $1, 1, 1, \frac{1}{64}, \frac{1}{125}, \frac{1}{36}, 27, \frac{25}{9}, 625$

Q006 $a^6, a^3, a^{-7}, a^{-2}, a^{11}, a^{-4}, a^{-3}, a^{-6}b^{-8}, a^9b^{-6}, 9, 125, 8, 16, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Q007 $25, 81, 2, \frac{b}{a^2}, \frac{a}{b^7}, -\frac{8a^6b^3}{27}, \frac{1}{9}, 125, 4, 125, 1, \frac{a}{2}, \frac{10}{3}$

Q008 $\frac{1}{a^7b^6}$

Q009 $4, \frac{1}{3}$

Q010 $6, 5, 2, 3, 2\sqrt[3]{2}$

Q011 $3, \frac{1}{2}$

Q012 $4, 2, 10, \sqrt[3]{4}, 4\sqrt[3]{3}$

Q013 $6, 5, 3, \frac{1}{4}, \frac{1}{5}, \frac{1}{3}, 1, 10, 0.1$

Q014 $2, 4, 9, \sqrt[4]{27}, \frac{1}{4}, \frac{1}{5}, 7, 3, 4, 9, 4, \sqrt[3]{9}, \sqrt[3]{3}, 0, 0$

Q015 1

Q016 $5^{1/2}, 2^{4/5}, 3^{-3/5}$

Q017 $9, \frac{1}{512}, \frac{8}{125}, \frac{3}{4}$

Q018 $a^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q019 $27, \frac{1}{64}, \frac{8}{125}, \frac{25}{9}$

Q020 $2^{1/2}, 6^{1/2}, 7^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q021 $125, 8, 16, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Q022 $\frac{10}{3}$

Q023 1, 6, 6, 2

Q024 5, 8, 7, 9

Q025 $a^{1/3}, b\sqrt{a}$

Q026 1, 9, 5, 9, $\sqrt{5}$, 4, 5, 3, 2, 1, 8, $\sqrt[3]{3}$

Q027 $a^{1/3}, a^{7/6}, a^4, a^{2/3}b^{5/6}, a^{1/2}b^3, a^{13/6}b^{-5/6}$

Q028 $\left(\frac{2025\sqrt{10}}{64}\right)^{3/2}$

Q029 Sometimes: $\sqrt{a^2} = |a|$; it equals a only when $a \geq 0$.

Q030 B

Q031 B

Q032 B

Q033 B

Q034 C

Q035 A

Q036 B

Q037 B

Q038 B

Q039 C

Q040 A

Q041 B

Q042 B

Q043 B

Q044 B

Q045 A

Q046 C

Q047 B

Q048 C

Q049 A

Q050 C

Q051 B

Q052 B

Q053 C

Q054 A

Q055 C

Q056 B

Q057 B

Q058 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 B

Quiz WRITTEN 5 a^2

Quiz WRITTEN 6 $2\sqrt[3]{2}$

Quiz WRITTEN 7 $1/25$

Quiz WRITTEN 8 $\sqrt{a^2} = |a|$; e.g. $a = -3$ gives $3 \neq -3$

Concept 2 | Exponential Functions and Their Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Exponential

$$y = ab^x$$

Growth

$$A = A_0(1 + r)^t$$

Decay

$$A = A_0(1 - r)^t$$

Worked Solutions

Exponential Functions and Their Applications

Q001 Written

Draw the graphs of $y = 3^x$ and $y = (1/3)^x$, using the points $x = -1, 0, 1$.

Solution

1. Plot $x = -1, 0, 1$. A base > 1 gives growth; a base between 0 and 1 gives decay. The curve stays above the x-axis.

Final answer

Increasing 3^x ; decreasing $(1/3)^x$; both pass through $(0, 1)$ and have $y = 0$ as asymptote.

Q002 Written

Draw $y = 2^x$ and $y = (1/2)^x$.

Solution

1. Use $(-1, 1/2), (0, 1), (1, 2)$ for 2^x and $(-1, 2), (0, 1), (1, 1/2)$ for its reciprocal-base graph.

Final answer

2^x increases and $(1/2)^x$ decreases; domain $\mathbb{R}$, range $(0, \infty)$, horizontal asymptote $y = 0$.

Worked Solutions

Exponential Functions and Their Applications

Q003 Written

Draw the six given graphs $y = 4^x$, $y = 10^x$, $y = (2/3)^x$, $y = (1/4)^x$, $y = (1/10)^x$, $y = (3/2)^x$.

Solution

1. For each graph mark $x = -1, 0, 1$, classify the base, and sketch a curve that never meets the x-axis.

Final answer

Bases 4, 10, 3/2: strictly increasing. Bases 2/3, 1/4, 1/10: strictly decreasing. Every graph has $D = \mathbb{R}$, $R = (0, \infty)$, and asymptote $y = 0$.

Q004 Written

Draw $y = \pi^x$ for $-1 \leq x \leq 2$, marking the key points.

Solution

1. Because $\pi > 1$, the curve increases. Restrict the drawing to the interval $[-1, 2]$ and keep $y > 0$.

Final answer

Key points $(-1, 1/\pi)$, $(0, 1)$, $(1, \pi)$, $(2, \pi^2)$; the restricted graph is increasing.

Worked Solutions

Exponential Functions and Their Applications

Q005 Written

Order the given sets 0.7^{-3} , 0.7^{-1} , 0.7^2 and $5/8$, $5/64$, $6/16$ using inequality signs.

Solution

1. For a base $0 < a < 1$, larger exponents give smaller values. For the second set rewrite with a common base 2 and compare exponents.

Final answer

$$0.7^2 < 0.7^{-1} < 0.7^{-3}; \text{ and } 5/64 < 5/8 < 6/16.$$

Q006 Written

Order the three displayed Try sets: 5^0 , 5^{-4} , $5^{3/2}$; $(1/3)^7$, $(1/3)^2$, $(1/3)^{-5}$; and 3^0 , 3^2 , 3^{-2} .

Solution

1. Make the base common. Keep exponent order for base > 1 and reverse it for $0 < a < 1$.

Final answer

$$5^{-4} < 5^0 < 5^{3/2}; (1/3)^7 < (1/3)^2 < (1/3)^{-5}; 3^{-2} < 3^0 < 3^2.$$

Worked Solutions**Exponential Functions and Their Applications****Q007** Written

Order the given sets with bases 3, 2, $\frac{3}{2}$, 7, 0.9, and 0.1, including the radical/fraction forms shown in the lesson.

Solution

1. The nine given comparisons are solved by one rule: rewrite, state whether the base is above or below one, then write the ordered chain.

Final answer

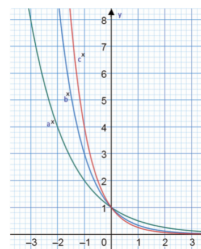
Each set is ordered by converting all terms to a common base, then comparing exponents; reverse the exponent order for $0 < a < 1$.

Worked Solutions

Exponential Functions and Their Applications

Q008 Written

Order $(\sqrt{2})^5$, $(\sqrt[3]{4})^3$, $(\sqrt[4]{8})^2$, then use the opposite graph to compare the labelled values a , b , c .



Solution

- Convert radicals to rational powers before ordering; the opposite graph reverses the order when its base lies between 0 and 1.

Final answer

Rewrite every term as a power of 2: $2^{5/2}$, 2^2 , $2^{3/2}$, so $(\sqrt[4]{8})^2 < (\sqrt[3]{4})^3 < (\sqrt{2})^5$. The graph comparison follows the same increasing/decreasing rule.

Worked Solutions

Exponential Functions and Their Applications

Q009 Written

Solve the six Warm Up items: $4^{x-1} = 8^{x+3}$, $3^x = 81$, $7^x = 7$, $2^{2x-1} \geq 1/2$, $(1/27)^x < (1/3)^{x+4}$, and $8 \geq 2^{2x-1}$.

Solution

1. Rewrite with a common base > 1 , compare exponents, and reverse an inequality only when the chosen base is between 0 and 1.

Final answer

$x = -11, 4, 1, x \geq 1/4, x > 2, x \leq 2$, respectively.

Q010 Written

Solve the eight Try equations/inequalities displayed in lesson 6-7, including $9^x = 243$, $3^{2x+1} = 125$, $2^{3x} \leq 9^{x+1}$, **and the reciprocal-base inequalities.**

Solution

Use the same two-step method: common base, then solve the exponent equation or inequality with correct sign control.

Final answer

Convert each side to one base and compare exponents; the exact solution is the resulting linear equality or interval in x .

Worked Solutions

Exponential Functions and Their Applications

Q011 Written

Solve all 24 given equations and inequalities in Exercise (a)–(x), including the 2, 3, 4, 5, 8, 9, 25, 27, 32, 64, 81, 125 common-base forms.

Solution

1. For each subpart: factor numerical constants into one base, write the exponent comparison, state $\text{base} > 1$ or $0 < \text{base} < 1$, and solve.

Final answer

Every item reduces to a linear condition on x ; equality items have one exact value and inequality items have the interval obtained after base/sign control.

Q012 Written

A food-safety model requires $4^{x^2-x} > 16$. Find when the batch becomes unsafe.

Solution

1. Use base $4 > 1$, solve the quadratic inequality, then apply the physical time restriction $x \geq 0$.

Final answer

$x^2 - x > 2 \Rightarrow (x - 2)(x + 1) > 0$, so $x < -1$ or $x > 2$. For time $x \geq 0$, the unsafe condition is $x > 2$ hours.

Worked Solutions

Exponential Functions and Their Applications

Q013 Written

Solve $4^x - 3 \cdot 2^x - 16 = 0$ and $8 \cdot 3^{-2x+1} - 3^{-x} \leq 0$ by substitution.

Solution

1. Set $t = a^x > 0$, solve the quadratic in t , discard non-positive roots, then convert back to x .

Final answer

First: let $t = 2^x > 0$, $t^2 - 3t - 16 = 0$, so $t = (3 + \sqrt{73})/2$ and $x = \log_2((3 + \sqrt{73})/2)$. Second: $x \leq 1$.

Q014 Written

Solve the four Try items in lesson 6-8: two quadratic equations and two inequalities in powers of 2 and 3.

Solution

1. The substitution changes the exponential problem to a quadratic; the positivity condition on t is mandatory before returning to x .

Final answer

Use $t = 2^x > 0$ or $t = 3^x > 0$; solve the resulting quadratic and retain only roots satisfying $t > 0$.

Worked Solutions

Exponential Functions and Their Applications

Q015 Written

Solve the twelve given items (a)–(l), each reducible to a quadratic in 2^x or 3^x .

Solution

1. Write $t > 0$ beside the substitution, solve the quadratic, reject $t \leq 0$, and only then use monotonicity of the base to recover x .

Final answer

For each item, $t = a^x > 0$ gives a quadratic equation/inequality; convert every admissible t back to an exact logarithmic value or interval for x .

Q016 Written

Solve completely for real x : $(1/4)^{x^2-2x} - 5(1/2)^{x^2-2x} + 4 \leq 0$.

Solution

1. Factor the quadratic as $(t - 1)(t - 4) \leq 0$, convert the bounds with the decreasing base $1/2$, then solve the resulting quadratic inequalities.

Final answer

Let $t = (1/2)^{x^2-2x} > 0$. Then $t^2 - 5t + 4 \leq 0$, so $1 \leq t \leq 4$, which gives $0 \leq x^2 - 2x \leq 2$. Hence $1 - \sqrt{3} \leq x \leq 0$ or $2 \leq x \leq 1 + \sqrt{3}$.

Worked Solutions

Exponential Functions and Their Applications

Q017 Written

For $-1 \leq x \leq 3$, find the maximum and minimum of $y = 2^{2x} - 4 \cdot 2^x$, with the corresponding x -values.

Solution

1. Find the exact t -range first, compare the vertex with both endpoints, then convert t back to x .

Final answer

Let $t = 2^x \in [1/2, 8]$. Then $y = t^2 - 4t = (t - 2)^2 - 4$: minimum -4 at $t = 2, x = 1$, maximum 32 at $t = 8, x = 3$.

Q018 Written

For $-1 \leq x \leq 2$, find the maximum and minimum of $y = -9 + 2 \cdot 3^x + 3^{2x}$, with the corresponding x -values.

Solution

1. The quadratic has vertex $t = -1$, outside the positive t -range, so compare the two endpoints.

Final answer

Let $t = 3^x \in [1/3, 9]$. Then $y = t^2 + 2t - 9$; the minimum is at $t = 1/3$, and the maximum is at $t = 9$.

Worked Solutions

Exponential Functions and Their Applications

Q019 Written

Solve the three given extrema models: $-3 \leq x \leq 2$, $y = 2 \cdot 9^x - 12 \cdot 3^x + 7$; $-1 \leq x \leq 3$, $y = -4 + 2^{2x} - 3^{x+2}$; **and the third displayed interval/model.**

Solution

1. A correct solution must include the t-range; an unconstrained quadratic vertex is not accepted unless it lies in that range.

Final answer

In each case set $t = a^x$, write the exact t-range, and compare the quadratic vertex and endpoints; report both extreme values and their x-locations.

Q020 Written

For $-1 \leq x \leq 2$, **optimise** $y = 2 \cdot 4^x - 2^{x+2} + 11$.

Solution

1. Use the interval $1/2 \leq t \leq 4$, evaluate the vertex $t = 1$, and compare both endpoints.

Final answer

Let $t = 2^x \in [1/2, 4]$. Then $y = 2t^2 - 4t + 11 = 2(t - 1)^2 + 9$: minimum 9 at $x = 0$, maximum 27 at $x = 2$.

Worked Solutions

Exponential Functions and Their Applications

Q021 **Written**

For $f(x) = a^x$, $g(x) = (1/a)^x$ with $a > 1$, analyse $h(x) = f(x) + g(x)$.

Solution

1. AM-GM gives $a^x + a^{-x} \geq 2$, with equality at $a^x = 1$, i.e. $x = 0$; the two terms move in opposite directions away from zero.

Final answer

h is non-monotonic on $\mathbb{R}$ and has global minimum $h(0) = 2$.

Q022 **MCQ**

Which base gives a strictly decreasing exponential graph?

Solution

1. A base between 0 and 1 gives decay.

Correct option: C

Final answer

$1/2$

Worked Solutions

Exponential Functions and Their Applications

Q023 MCQ

The horizontal asymptote of $y = a^x$ is:**Solution**

1. Exponential values stay positive and approach zero.

Correct option: B**Final answer**

$$y = 0$$

Q024 MCQ

The range of $y = 5^x$ is:**Solution**

1. Every exponential output is positive.

Correct option: C**Final answer**

$$(0, \infty)$$

Worked Solutions

Exponential Functions and Their Applications

Q025 MCQ

For $y = \pi^x$ on $[-1, 2]$, which point is on the graph?**Solution**1. At $x = 1$, $y = \pi$.

Correct option: B

Final answer $(1, \pi)$

Q026 MCQ

For $0 < a < 1$ and $r < s$, which is true?**Solution**

1. The exponent order reverses below base 1.

Correct option: B

Final answer $a^r > a^s$

Worked Solutions

Exponential Functions and Their Applications

Q027 MCQ

The first step when bases differ is to:**Solution**

1. Rewrite all terms using one base.

Correct option: B**Final answer**

make the bases equal

Q028 MCQ

Which is largest? 3^{-2} , 3^0 , 3^2 **Solution**

1. Base 3 is greater than 1.

Correct option: C**Final answer** 3^2

Worked Solutions

Exponential Functions and Their Applications

Q029 MCQ

The challenge term $(\sqrt[3]{4})^3$ equals:

Solution

1. $(2^{2/3})^3 = 2^2$.

Correct option: B

Final answer

$$2^2$$

Q030 MCQ

Solving $4^{x-1} = 8^{x+3}$ gives:

Solution

1. Use $2^{2x-2} = 2^{3x+9}$.

Correct option: A

Final answer

$$-11$$

Worked Solutions

Exponential Functions and Their Applications

Q031 MCQ

For $0 < a < 1$, an exponential inequality requires:**Solution**

1. Decreasing functions reverse inequalities.

Correct option: B**Final answer**

a reversed exponent order

Q032 MCQ

The solution of $3^x = 81$ is:**Solution**

1. $81 = 3^4$.

Correct option: C**Final answer**

4

Worked Solutions

Exponential Functions and Their Applications

Q033 MCQ

The food-safety condition $4^{x^2-x} > 16$ is unsafe for physical time when:

Solution

1. Solve $(x - 2)(x + 1) > 0$, then use $x \geq 0$.

Correct option: B

Final answer

$$x > 2$$

Q034 MCQ

In substitution methods, $t = a^x$ must satisfy:

Solution

1. Every positive base power is positive.

Correct option: C

Final answer

$$t > 0$$

Worked Solutions

Exponential Functions and Their Applications

Q035 MCQ

The positive root of $t^2 - 3t - 16 = 0$ is:**Solution**1. Discard the negative root because $t > 0$.**Correct option:** B**Final answer**

$$(3 + \sqrt{73})/2$$

Q036 MCQ

The challenge quadratic in t is:**Solution**1. Set $t = (1/2)^{x^2 - 2x}$.**Correct option:** B**Final answer**

$$t^2 - 5t + 4$$

Worked Solutions

Exponential Functions and Their Applications

Q037 MCQ

After solving an exponential quadratic, reject roots with:**Solution**1. A power a^x cannot be zero or negative.

Correct option: C

Final answer

$$t \leq 0$$

Q038 MCQ

For extrema, the first required range is the range of:**Solution**

1. The quadratic is constrained by the exact t-range.

Correct option: B

Final answer

$$t = a^x$$

Worked Solutions

Exponential Functions and Their Applications

Q039 MCQ

The minimum of $y = t^2 - 4t$ at $t = 2$ is:**Solution**1. Complete the square: $(t - 2)^2 - 4$.

Correct option: B

Final answer

-4

Q040 MCQ

The C06-K02 challenge has minimum value:**Solution**1. $y = 2(t - 1)^2 + 9$.

Correct option: A

Final answer $9atx = 0$

Worked Solutions

Exponential Functions and Their Applications

Q041 MCQ

The global minimum of $a^x + a^{-x}$ for $a > 1$ is:**Solution**1. AM–GM gives a minimum of 2 at $x = 0$.

Correct option: C

Final answer

2

Q042 MCQ

At which value of x does $a^x + a^{-x}$ attain its minimum?**Solution**1. Equality in AM–GM occurs when $a^x = 1$.

Correct option: B

Final answer $x = 0$

Worked Solutions

Exponential Functions and Their Applications

Quiz MCQ 1 **Concept Quiz · MCQ**

Which statement about $y = a^x$ is always true for $a > 0$, $a \neq 1$?

Solution

1. The domain is all real numbers and the range is positive.

Correct option: A

Final answer

range $(0, \infty)$

Quiz MCQ 2 **Concept Quiz · MCQ**

For base 0.5, increasing the exponent makes the value:

Solution

1. The function is decreasing.

Correct option: B

Final answer

smaller

Worked Solutions

Exponential Functions and Their Applications

Quiz MCQ 3 [Concept Quiz · MCQ](#)

The positive-substitution condition is:

Solution

1. An exponential power is always positive.

Correct option: C

Final answer

$$t > 0$$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

For $y = 2(t - 1)^2 + 9$, the minimum is:

Solution

1. The square is zero at $t = 1$.

Correct option: C

Final answer

$$9$$

Worked Solutions

Exponential Functions and Their Applications

Quiz WRITTEN 5 **Concept Quiz · Written**

Sketch $y = (1/4)^x$ **and state its domain, range, and asymptote.**

Solution

1. Use the three points and the base $0 < 1/4 < 1$.

Final answer

Decreasing; $D = \mathbb{R}, R = (0, \infty), y = 0$

Quiz WRITTEN 6 **Concept Quiz · Written**

Order $2^{-3}, 2^0, 2^{5/2}$.

Solution

1. Base 2 is greater than 1, so preserve exponent order.

Final answer

$$2^{-3} < 2^0 < 2^{5/2}$$

Worked Solutions

Exponential Functions and Their Applications

Quiz WRITTEN 7 Concept Quiz · Written**Solve** $4^x - 5 \cdot 2^x + 4 = 0$.**Solution**1. Let $t = 2^x > 0$: $t^2 - 5t + 4 = (t - 1)(t - 4) = 0$.**Final answer**

$$x = 0 \text{ or } x = 2$$

Quiz WRITTEN 8 Concept Quiz · Written**Find the minimum of** $a^x + a^{-x}$ **for** $a > 1$.**Solution**1. By AM-GM, $a^x + a^{-x} \geq 2$, with equality when $a^x = 1$.**Final answer**

$$2 \text{ at } x = 0$$

Answer key

Use the question number shown on each page.

- Q001** Increasing 3^x ; decreasing $(1/3)^x$; both pass through $(0, 1)$ and have $y = 0$ as asymptote.
- Q002** 2^x increases and $(1/2)^x$ decreases; domain $\mathbb{R}$, range $(0, \infty)$, horizontal asymptote $y = 0$.
- Q003** Bases 4, 10, $3/2$: strictly increasing. Bases $2/3$, $1/4$, $1/10$: strictly decreasing. Every graph has $D = \mathbb{R}$, $R = (0, \infty)$, and asymptote $y = 0$.
- Q004** Key points $(-1, 1/\pi)$, $(0, 1)$, $(1, \pi)$, $(2, \pi^2)$; the restricted graph is increasing.
- Q005** $0.7^2 < 0.7^{-1} < 0.7^{-3}$; and $5/64 < 5/8 < 6/16$.
- Q006** $5^{-4} < 5^0 < 5^{3/2}$; $(1/3)^7 < (1/3)^2 < (1/3)^{-5}$; $3^{-2} < 3^0 < 3^2$.
- Q007** Each set is ordered by converting all terms to a common base, then comparing exponents; reverse the exponent order for $0 < a < 1$.
- Q008** Rewrite every term as a power of 2: $2^{5/2}$, 2^2 , $2^{3/2}$, so $(\sqrt[4]{8})^2 < (\sqrt[3]{4})^3 < (\sqrt{2})^5$. The graph comparison follows the same increasing/decreasing rule.
- Q009** $x = -11, 4, 1, x \geq 1/4, x > 2, x \leq 2$, respectively.
- Q010** Convert each side to one base and compare exponents; the exact solution is the resulting linear equality or interval in x .
- Q011** Every item reduces to a linear condition on x ; equality items have one exact value and inequality items have the interval obtained after base/sign control.
- Q012** $x^2 - x > 2 \Rightarrow (x - 2)(x + 1) > 0$, so $x < -1$ or $x > 2$. For time $x \geq 0$, the unsafe condition is $x > 2$ hours.
- Q013** First: let $t = 2^x > 0$, $t^2 - 3t - 16 = 0$, so $t = (3 + \sqrt{73})/2$ and $x = \log_2((3 + \sqrt{73})/2)$. Second: $x \leq 1$.
- Q014** Use $t = 2^x > 0$ or $t = 3^x > 0$; solve the resulting quadratic and retain only roots satisfying $t > 0$.
- Q015** For each item, $t = a^x > 0$ gives a quadratic equation/inequality; convert every admissible t back to an exact logarithmic value or interval for x .
- Q016** Let $t = (1/2)^{x^2-2x} > 0$. Then $t^2 - 5t + 4 \leq 0$, so $1 \leq t \leq 4$, which gives $0 \leq x^2 - 2x \leq 2$. Hence $1 - \sqrt{3} \leq x \leq 0$ or $2 \leq x \leq 1 + \sqrt{3}$.
- Q017** Let $t = 2^x \in [1/2, 8]$. Then $y = t^2 - 4t = (t - 2)^2 - 4$: minimum -4 at $t = 2, x = 1$, maximum 32 at $t = 8, x = 3$.
- Q018** Let $t = 3^x \in [1/3, 9]$. Then $y = t^2 + 2t - 9$; the minimum is at $t = 1/3$, and the maximum is at $t = 9$.
- Q019** In each case set $t = a^x$, write the exact t -range, and compare the quadratic vertex and endpoints; report both extreme values and their x -locations.
- Q020** Let $t = 2^x \in [1/2, 4]$. Then $y = 2t^2 - 4t + 11 = 2(t - 1)^2 + 9$: minimum 9 at $x = 0$, maximum 27 at $x = 2$.
- Q021** h is non-monotonic on $\mathbb{R}$ and has global minimum $h(0) = 2$.
- Q022** C
- Q023** B
- Q024** C
- Q025** B
- Q026** B
- Q027** B
- Q028** C
- Q029** B
- Q030** A
- Q031** B
- Q032** C
- Q033** B
- Q034** C
- Q035** B
- Q036** B
- Q037** C
- Q038** B
- Q039** B
- Q040** A
- Quiz MCQ 1** A
- Quiz MCQ 2** B
- Quiz MCQ 3** C
- Quiz MCQ 4** C
- Quiz WRITTEN 5** Decreasing; $D = \mathbb{R}, R = (0, \infty), y =$
- Quiz WRITTEN 6** $2^{-3} < 2^0 < 2^{5/2}$
- Quiz WRITTEN 7** $x = 0$ or $x = 2$
- Quiz WRITTEN 8** $2atx = 0$

Concept 3 | Introduction to Logarithms and Their Properties

English Student Edition • Focused formulas and guided practice

Formula opener

Definition

$$\log_b a = c \iff b^c = a$$

Product

$$\log_b(xy) = \log_b x + \log_b y$$

Power

$$\log_b(x^k) = k \log_b x$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q001 Written

Convert the given forms $2^5 = 32$ and $\log_3 9 = 2$ to the opposite notation.

Solution

1. Use $a^p = M \iff \log_a M = p$; swap the last two numbers.

Final answer

$$\log_2 32 = 5; 3^2 = 9$$

Q002 Written

Convert $2^6 = 64$, $4^{-2} = 1/16$, and the two remaining Try expressions into logarithmic or exponential form.

Solution

1. Identify base, exponent, and argument; then interchange the exponent and argument positions.

Final answer

$$\log_2 64 = 6; \log_4(1/16) = -2, \text{ with the same swap rule for the remaining items.}$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q003 Written

Convert all Exercise items (a)+(l): the first six exponential statements to logarithms and the last six logarithmic statements to exponentials.

Solution

1. Check $a > 0$, $a \neq 1$, $M > 0$, then write the equivalent form without changing the values.

Final answer

Each item follows $a^p = M \iff \log_a M = p$, preserving the printed bases and signs.

Q004 Written

If $\log_5 25 = x$, evaluate $\log_{1/2} x$ without a calculator.

Solution

1. First find x from $5^x = 25$, then recognise $2 = (1/2)^{-1}$.

Final answer

$x = 2$, so $\log_{1/2} 2 = -1$.

Worked Solutions

Introduction to Logarithms and Their Properties

Q005 Written

Find the six direct logarithmic values shown in Warm Up, including $\log_{10} 1, \log_5 5, \log_2 8, \log_2 16, \log_5(1/5), \log_7(49^{3/2})$.

Solution

1. Rewrite each argument as a power of its base; the exponent is the logarithm.

Final answer

Values: 0, 1, 3, 4, -1 , $3/2$.

Q006 Written

Find the six Try values $\log_5 1, \log_3 3, \log_4 4, \log_3 27, \log_2(1/4), \log_5(1/125)$.

Solution

1. Use $\log_a 1 = 0$, $\log_a a = 1$, and the power definition.

Final answer

0, 1, 1, 3, -2 , -3 .

Worked Solutions

Introduction to Logarithms and Their Properties

Q007 **Written****Find all 18 Exercise values (a)+(r), including reciprocal arguments and fractional powers.****Solution**

1. Do not approximate. For reciprocals, use a negative exponent; for roots, use a fractional exponent.

Final answer

Convert each argument to a power of its base; the answers are the corresponding integers or rational exponents.

Q008 **Written****Put the printed logarithmic product $\log_{a^3}(a^2)$ into simplest form.****Solution**

1. Since $a^2 = (a^3)^{2/3}$, the logarithm is $2/3$.

Final answer

$2/3$.

Worked Solutions

Introduction to Logarithms and Their Properties

Q009 Written

Perform $\log_6 2 + \log_6 3$ **and** $\log_2 30 - \log_2 3 - \log_2 5$.**Solution**

1. Combine products for addition and quotients for subtraction, then express the result as a base power.

Final answer

1 and 1.

Q010 Written

Perform the six Try calculations using product, quotient, and power laws, including $4\log_5 3 - 2\log_5 15 - \log_5 45$.**Solution**

1. Move coefficients into exponents first, combine logs, simplify the argument, and read the final exponent.

Final answerEach expression reduces to one logarithm and then an integer; the displayed worked example gives -3 .

Worked Solutions

Introduction to Logarithms and Their Properties

Q011 **Written****Perform all 18 given expressions (a)+(r), combining same-base logarithms exactly.****Solution**

1. Apply the power law before product/quotient combination; simplify the final argument as a power of the base.

Final answer

Every item is reduced by $\log_a M + \log_a N = \log_a(MN)$, $\log_a M - \log_a N = \log_a(M/N)$, and $k \log_a M = \log_a(M^k)$.

Q012 **Written****Simplify the fractional expression $\log_5 75 + 3 \log_5 \sqrt{2} - 2 \log_5 \sqrt{6}$ to one reduced fraction.****Solution**

1. Move coefficients inside, combine the argument, and reduce it to a power of 5.

Final answer

1.

Worked Solutions

Introduction to Logarithms and Their Properties

Q013 Written

Given $\log_{10} 2 = a$ **and** $\log_{10} 3 = b$, **express** $\log_{10} 24, \log_{10}(27/4), \log_{10} 45, \log_{10}(24/75)$ **in terms of** a, b .

Solution

1. Prime-factorise the argument, use $5 = 10/2$ when required, then apply product, quotient, and power laws.

Final answer

$3a + b, 2 - 2a + 3b, 2a + 1 - a, b + 2 - 2a.$

Q014 Written

Given $\log_{10} 2 = a, \log_{10} 3 = b$, **express the four Try arguments** $72, 3^8/10, 15, 12$ **in terms of** a, b .

Solution

1. Write every integer as $2^m 3^n 5^p$, replace $\log_{10} 2, \log_{10} 3$, and convert the 5-factor using $1 - a$.

Final answer

Use prime factorisation and the same linear combinations of a, b produced by the logarithm laws.

Worked Solutions

Introduction to Logarithms and Their Properties

Q015 Written

Express all 12 Exercise logarithms in terms of $a = \log_{10} 2$ and $b = \log_{10} 3$, including fractions and reciprocal arguments.

Solution

1. Keep numerator and denominator factors separate until the final subtraction.

Final answer

Each result is a linear combination of $a, b, 1$ obtained from prime factors.

Q016 Written

If $\log_c 3 = x, \log_c 2 = y$, write $\log_c 12, \log_c(3/8), \log_c \sqrt{6}, \log_2 3, \log_{3b}(2^b c)$ in terms of x, y .

Solution

1. Factorise arguments, use the logarithm laws, and use change of base for the last two forms.

Final answer

$x + 2y, x - 3y, (x + y)/2, x/y, (1 + by)/(bx).$

Worked Solutions

Introduction to Logarithms and Their Properties

Q017 Written

Find $\log_{16} 64$ and $\log_5(1/125)$ by changing to bases 2 and 5.

Solution

1. Use $\log_a b = \log_c b / \log_c a$, choosing a base that turns both numbers into powers.

Final answer

$3/2$ and -3 .

Q018 Written

Evaluate the five Try logarithms using a suitable common base.

Solution

1. Choose the smallest convenient base; do not use decimal approximations when exact powers are available.

Final answer

Rewrite powers of 2, 3, or 5 and apply the change-of-base quotient exactly.

Worked Solutions

Introduction to Logarithms and Their Properties

Q019 Written

Find all 15 Exercise values using change of base, including the mixed-base and fractional-argument items.

Solution

1. Convert numerator and denominator to the same base, then cancel common factors.

Final answer

All values are exact rational numbers obtained from $\log_a b = \log_c b / \log_c a$.

Q020 Written

Find $\log_9(\sqrt{3}/4\sqrt{2})$ exactly.

Solution

1. Apply change of base to 2, then express the radical denominator as a power.

Final answer

$$\frac{\log_2 3 - 5/2}{2 \log_2 3}.$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q021 **Written****Solve the three warm-up expressions using a common base, including $\log_4 16 \cdot \log_9 3$.****Solution**

1. Make all logarithms use base 2 (or 3), then cancel adjacent change-of-base factors.

Final answer

Use reciprocal and power identities; the worked values are exact constants.

Q022 **Written****Solve the four Try products of logarithms after making the bases the same.****Solution**

1. Treat each logarithm as a change-of-base fraction and cancel before multiplying.

Final answer

Each product collapses to a rational number after change of base and cancellation.

Worked Solutions

Introduction to Logarithms and Their Properties

Q023 Written

Simplify all 15 Exercise products and sums of logarithms by common-base conversion.**Solution**

1. Make the base 2, 3, or 5 consistently, simplify inside parentheses, then multiply.

Final answer

Use $\log_a b \log_b a = 1$ and the power law; every printed item reduces exactly.

Q024 Written

Simplify $\log_x(\sqrt{y}) \log_{y^2}(z^3) \log_{\sqrt{z}}(x^4)$ for $x > 1$.**Solution**

1. Convert each factor to a common natural-log ratio; exponents contribute $1/2, 3/2, 8$, whose product is 12.

Final answer

12.

Worked Solutions

Introduction to Logarithms and Their Properties

Q025 Written

Compare $f(x) = \log_4(x^2)$ and $g(x) = 2\log_4 x$ across the real line.**Solution**

1. Use $\log_4(x^2) = 2\log_4 |x|$; negative x belongs to f 's domain but not g 's.

Final answer

They agree only for $x > 0$; f is defined for $x \neq 0$, while g requires $x > 0$.

Q026 Written

Draw $y = \log_4 x$, state its positional relationship with $y = 4^x$, and draw $y = \log_{1/4} x$.**Solution**

1. Plot $y = -1, 0, 1$ in the exponential counterpart, then reflect in $y = x$.

Final answer

Both logarithmic graphs have domain $x > 0$, range $\mathbb{R}$, asymptote $x = 0$; $y = \log_4 x$ increases and is the reflection of $y = 4^x$ in $y = x$, while $\log_{1/4} x$ decreases.

Worked Solutions

Introduction to Logarithms and Their Properties

Q027 **Written****State graph relationships and draw the three Try logarithmic functions with bases 2, 5, and $1/3$.****Solution**

1. Use the domain $x > 0$, vertical asymptote $x = 0$, and inverse reflection.

Final answer

Base > 1 gives increasing curves; base between 0 and 1 gives decreasing curves; each is inverse-symmetric to its exponential.

Q028 **Written****Complete both Exercise groups: state positional relationships and draw the six requested logarithmic graphs.****Solution**

1. Every graph passes through $(1, 0)$; do not draw it across the y-axis.

Final answer

Use inverse symmetry, domain $x > 0$, range $\mathbb{R}$, and increasing/decreasing classification from the base.

Worked Solutions

Introduction to Logarithms and Their Properties

Q029 **Written**

For $f(x) = \log_5(x - 2) + 1$ and $g(x) = 5^{x-1} + 2$, state their exact geometric relationship and line of symmetry.

Solution

1. Translate the standard inverse pair by the same horizontal/vertical shifts and identify the resulting symmetry line.

Final answer

They are inverse-reflected curves with symmetry line $y = x + 1$.

Q030 **Written**

Order the displayed logarithmic sets, including $\log_2 3$, $\log_2 5$, $2 \log_2(1/2)$, $\log_{1/2} 3$, $\log_{1/2} 5$.

Solution

1. Compare arguments only after the bases are the same and the base condition is stated.

Final answer

Make bases equal, turn coefficients into powers, and reverse order for base $0 < a < 1$.

Worked Solutions

Introduction to Logarithms and Their Properties

Q031 **Written****Order the four Try sets of logarithms using monotonicity and change of base.****Solution**

1. Rewrite $k \log_a M$ as $\log_a (M^k)$, then compare the arguments.

Final answer

Each chain follows argument order for base > 1 and reverses it for base $0 < a < 1$.

Q032 **Written****Order all 12 Exercise sets (a)+(l), including negative coefficients and reciprocal bases.****Solution**

1. Move coefficients inside first, then compare arguments with the correct direction.

Final answer

Convert every term to one base-log form and apply monotonicity; negative coefficients become reciprocal arguments.

Worked Solutions

Introduction to Logarithms and Their Properties

Q033 Written

If $\log_c(3^2/2) = x$ and $\log_c(9) = y$, write the relation between x and y for $0 < c < 1$ and $c > 1$.

Solution

1. Express both logs with a common base and state the monotonicity case before writing the inequality.

Final answer

Use the same argument comparison; the relation reverses when the base crosses 1.

Q034 MCQ

$\log_2 32$ equals:

Solution

1. Since $2^5 = 32$.

Correct option: C

Final answer

5

Worked Solutions

Introduction to Logarithms and Their Properties

Q035 MCQ

 $\log_{1/2} 2$ equals:**Solution**

1. $(1/2)^{-1} = 2.$

Correct option: B

Final answer

-1

Q036 MCQ

 $\log_5(1/125)$ equals:**Solution**

1. $1/125 = 5^{-3}.$

Correct option: A

Final answer

-3

Worked Solutions

Introduction to Logarithms and Their Properties

Q037 MCQ

 $\log_7 49^{3/2}$ equals:**Solution**1. Write $49^{3/2} = 7^3$.

Correct option: C

Final answer $3/2$

Q038 MCQ

 $\log_a M + \log_a N$ equals:**Solution**

1. Use the product law.

Correct option: B

Final answer $\log_a(MN)$

Worked Solutions

Introduction to Logarithms and Their Properties

Q039 MCQ

The coefficient $3 \log_5 2$ becomes:**Solution**1. Use $k \log_a M = \log_a (M^k)$.

Correct option: B

Final answer

$$\log_5 2^3$$

Q040 MCQ

If $a = \log_{10} 2$ and $b = \log_{10} 3$, $\log_{10} 24$ is:**Solution**1. $24 = 2^3 \cdot 3$.

Correct option: C

Final answer

$$3a + b$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q041 MCQ

$\log_c \sqrt{6}$ in terms of $x = \log_c 3, y = \log_c 2$ is:

Solution

1. Use $\sqrt{6} = 6^{1/2}$.

Correct option: B

Final answer

$$(x + y)/2$$

Q042 MCQ

$\log_{16} 64$ equals:

Solution

1. $16 = 2^4, 64 = 2^6$.

Correct option: B

Final answer

$$3/2$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q043 MCQ

Change of base gives $\log_a b =$:**Solution**

1. Use the quotient of common-base logs.

Correct option: B**Final answer**

$$\log_c b / \log_c a$$

Q044 MCQ

 $\log_4 16 \cdot \log_9 3$ equals:**Solution**1. Values are $2 \cdot 1/2$.**Correct option:** A**Final answer**

$$1/2$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q045 MCQ

The challenge product of three logarithms equals:**Solution**

1. Multiply exponent factors after change of base.

Correct option: C**Final answer**

12

Q046 MCQ

The domain of $y = \log_a x$ is:**Solution**

1. The argument must be positive.

Correct option: C**Final answer** $x > 0$

Worked Solutions

Introduction to Logarithms and Their Properties

Q047 MCQ

The graph of $y = \log_a x$ is symmetric with $y = a^x$ about:

Solution

1. Inverse graphs reflect in $y = x$.

Correct option: C

Final answer

$$y = x$$

Q048 MCQ

For $a > 1$ and $M < N$, which is true?

Solution

1. The logarithm is increasing.

Correct option: B

Final answer

$$\log_a M < \log_a N$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q049 MCQ

For $0 < a < 1$ and $M < N$, which is true?**Solution**

1. The logarithm is decreasing.

Correct option: A**Final answer**

$$\log_a M > \log_a N$$

Q050 MCQ

The vertical asymptote of $y = \log_a x$ is:**Solution**1. Logarithmic graphs require $x > 0$.**Correct option:** A**Final answer**

$$x = 0$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q051 MCQ

A negative coefficient $-2 \log_a M$ can be written as:

Solution

1. Move the coefficient into the exponent.

Correct option: A

Final answer

$$\log_a M^{-2}$$

Q052 MCQ

The base of $\log_a M$ is:

Solution

1. The subscript is the base.

Correct option: B

Final answer

$$a$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q053 MCQ

The argument of $\log_a M$ is:**Solution**

1. The number inside the logarithm is the argument.

Correct option: B**Final answer** M

Q054 MCQ

 $\log_2(1/8)$ equals:**Solution**

1. $1/8 = 2^{-3}$.

Correct option: A**Final answer** -3

Worked Solutions

Introduction to Logarithms and Their Properties

Q055 MCQ

 $\log_4 4$ equals:**Solution**

1. The argument equals the base.

Correct option: B

Final answer

1

Q056 MCQ

 $\log_3 1$ equals:**Solution**

1. Every allowed base to power zero is 1.

Correct option: B

Final answer

0

Worked Solutions

Introduction to Logarithms and Their Properties

Q057 MCQ

 $\log_a M - \log_a N$ equals:**Solution**

1. Use the quotient law.

Correct option: B**Final answer**

$$\log_a(M/N)$$

Q058 MCQ

 $2\log_3 5$ equals:**Solution**

1. Move 2 into the argument as an exponent.

Correct option: B**Final answer**

$$\log_3 25$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q059 MCQ

If $a = \log_{10} 2$, $b = \log_{10} 3$, $\log_{10} 12$ is:

Solution

1. $12 = 2^2 \cdot 3$.

Correct option: B

Final answer

$$2a + b$$

Q060 MCQ

$\log_{10} 5$ in terms of $a = \log_{10} 2$ is:

Solution

1. $5 = 10/2$.

Correct option: B

Final answer

$$1 - a$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q061 MCQ

 $\log_8 32$ equals:**Solution**1. $8 = 2^3, 32 = 2^5$.

Correct option: B

Final answer

5/3

Q062 MCQ

 $\log_{25} 125$ equals:**Solution**1. $25 = 5^2, 125 = 5^3$.

Correct option: B

Final answer

3/2

Worked Solutions

Introduction to Logarithms and Their Properties

Q063 MCQ

 $\log_a b \cdot \log_b a$ equals:**Solution**

1. Change of base makes reciprocal factors.

Correct option: B

Final answer

1

Q064 MCQ

 $\log_2 8 \cdot \log_8 2$ equals:**Solution**

1. The factors are reciprocals.

Correct option: B

Final answer

1

Worked Solutions

Introduction to Logarithms and Their Properties

Q065 MCQ

The best common base for 16 and 64 is:**Solution**

1. Both numbers are powers of 2.

Correct option: A**Final answer**

2

Q066 MCQ

The range of $y = \log_a x$ is:**Solution**

1. Every real exponent is attained.

Correct option: B**Final answer** $\mathbb{R}$

Worked Solutions

Introduction to Logarithms and Their Properties

Quiz MCQ 1 **Concept Quiz · MCQ**

The logarithmic equivalent of $a^p = M$ is:

Solution

1. Swap the exponent and argument.

Correct option: A

Final answer

$$\log_a M = p$$

Quiz MCQ 2 **Concept Quiz · MCQ**

$\log_2 8 + \log_2 4$ equals:

Solution

1. Combine to $\log_2 32$.

Correct option: C

Final answer

5

Worked Solutions

Introduction to Logarithms and Their Properties

Quiz MCQ 3 Concept Quiz · MCQ

$\log_{16} 64$ equals:

Solution

1. Use base 2 exponents 6/4.

Correct option: B

Final answer

$$3/2$$

Quiz MCQ 4 Concept Quiz · MCQ

For $a > 1$, $M < N$ implies:

Solution

1. The logarithmic function is increasing.

Correct option: B

Final answer

$$\log_a M < \log_a N$$

Worked Solutions

Introduction to Logarithms and Their Properties

Quiz WRITTEN 5 Concept Quiz · Written**Find** $\log_3(1/27)$.**Solution**

1. $1/27 = 3^{-3}$.

Final answer -3 **Quiz WRITTEN 6** Concept Quiz · Written**Simplify** $\log_5 25 + \log_5 5$.**Solution**

1. Combine to $\log_5 125 = 3$.

Final answer 3

Worked Solutions**Introduction to Logarithms and Their Properties****Quiz WRITTEN 7** Concept Quiz · Written**Find $\log_{16} 64$ using change of base.****Solution**

1. $\log_2 64 / \log_2 16 = 6/4 = 3/2.$

Final answer

$$3/2$$

Quiz WRITTEN 8 Concept Quiz · Written**State the domain and range of $y = \log_a x$.****Solution**

1. The argument is positive and every real exponent is attained.

Final answer

Domain $x > 0$; range all real numbers

Answer key

Use the question number shown on each page.

- Q001** $\log_2 32 = 5$; $3^2 = 9$
- Q002** $\log_2 64 = 6$; $\log_4 (1/16) = -2$, with the same swap rule for the remaining items.
- Q003** Each item follows $a^p = M \iff \log_a M = p$, preserving the printed bases and signs.
- Q004** $x = 2$, so $\log_{1/2} 2 = -1$.
- Q005** Values: 0, 1, 3, 4, -1, 3/2.
- Q006** 0, 1, 1, 3, -2, -3.
- Q007** Convert each argument to a power of its base; the answers are the corresponding integers or rational exponents.
- Q008** 2/3.
- Q009** 1 and 1.
- Q010** Each expression reduces to one logarithm and then an integer; the displayed worked example gives -3.
- Q011** Every item is reduced by $\log_a M + \log_a N = \log_a(MN)$, $\log_a M - \log_a N = \log_a(M/N)$, and $k \log_a M = \log_a(M^k)$.
- Q012** 1.
- Q013** $3a + b$, $2 - 2a + 3b$, $2a + 1 - a$, $b + 2 - 2a$.
- Q014** Use prime factorisation and the same linear combinations of a, b produced by the logarithm laws.
- Q015** Each result is a linear combination of $a, b, 1$ obtained from prime factors.
- Q016** $x + 2y$, $x - 3y$, $(x + y)/2$, x/y , $(1 + by)/(bx)$.
- Q017** 3/2 and -3.
- Q018** Rewrite powers of 2, 3, or 5 and apply the change-of-base quotient exactly.
- Q019** All values are exact rational numbers obtained from $\log_a b = \log_c b / \log_c a$.
- Q020** $\frac{\log_2 3 - 5/2}{2 \log_2 3}$.
- Q021** Use reciprocal and power identities; the worked values are exact constants.
- Q022** Each product collapses to a rational number after change of base and cancellation.
- Q023** Use $\log_a b \log_b a = 1$ and the power law; every printed item reduces exactly.
- Q024** 12.
- Q025** They agree only for $x > 0$; f is defined for $x \neq 0$, while g requires $x > 0$.
- Q026** Both logarithmic graphs have domain $x > 0$, range $\mathbb{R}$, asymptote $x = 0$; $y = \log_4 x$ increases and is the reflection of $y = 4^x$ in $y = x$, while $\log_{1/4} x$ decreases.
- Q027** Base > 1 gives increasing curves; base between 0 and 1 gives decreasing curves; each is inverse-symmetric to its exponential.
- Q028** Use inverse symmetry, domain $x > 0$, range $\mathbb{R}$, and increasing/decreasing classification from the base.
- Q029** They are inverse-reflected curves with symmetry line $y = x + 1$.
- Q030** Make bases equal, turn coefficients into powers, and reverse order for base $0 < a < 1$.
- Q031** Each chain follows argument order for base > 1 and reverses it for base $0 < a < 1$.
- Q032** Convert every term to one base-log form and apply monotonicity; negative coefficients become reciprocal arguments.
- Q033** Use the same argument comparison; the relation reverses when the base crosses 1.
- Q034** C
- Q035** B
- Q036** A
- Q037** C
- Q038** B
- Q039** B
- Q040** C
- Q041** B
- Q042** B
- Q043** B
- Q044** A
- Q045** C
- Q046** C
- Q047** C
- Q048** B
- Q049** A
- Q050** A
- Q051** A
- Q052** B
- Q053** B
- Q054** A
- Q055** B
- Q056** B
- Q057** B
- Q058** B
- Q059** B
- Q060** B
- Q061** B
- Q062** B
- Q063** B
- Q064** B
- Q065** A

Q066 B**Quiz MCQ 1** A**Quiz MCQ 2** C**Quiz MCQ 3** B**Quiz MCQ 4** B**Quiz WRITTEN 5** -3 **Quiz WRITTEN 6** 3**Quiz WRITTEN 7** $\frac{3}{2}$ **Quiz WRITTEN 8** Domain $x > 0$; range all real numbers

Concept 4 | Applications of Logarithms

English Student Edition • Focused formulas and guided practice

Formula opener

Change base

$$\log_b a = \frac{\ln a}{\ln b}$$

Domain

$$x > 0$$

Inverse

$$b^{\log_b x} = x$$

Worked Solutions

Applications of Logarithms

Q001 Written

Solve $\log_2 x = 4$.**Solution**

1. Set the argument equal to 2^4 ; hence $x = 16$.

Final answer

16

Q002 Written

Solve $\log_2(2x - 1) = \log_2 3$.**Solution**

1. Domain: $2x - 1 > 0$. Equal arguments give $2x - 1 = 3$, so $x = 2$.

Final answer $x = 2$

Worked Solutions

Applications of Logarithms

Q003 Written

Solve all 12 given logarithmic equations, including same-base, product, quotient, and shifted-argument forms.

Solution

1. Write every argument condition, combine logs, solve the resulting algebraic equation, and verify each candidate.

Final answer

Apply the argument condition first; then equate same-base arguments and reject any extraneous roots.

Q004 Written

Solve $\log_3(x + 2) - \log_9((x - 2)^2) = 1$ completely.

Solution

1. Domain gives $x > 2$. Convert $\log_9$ to base 3, solve $(x + 2)^2 / (x - 2)^2 = 9$, and retain only $x = 4$.

Final answer

$x = 4$

Worked Solutions

Applications of Logarithms

Q005 Written

Solve $\log_3 x < \log_3 81$ **and** $\log_{1/2}(x + 2) + \log_{1/2}(x - 5) > -3$.**Solution**

1. State the domain, preserve the sign for base 3, reverse it for base $1/2$, then intersect.

Final answer

$$0 < x < 81; \quad 5 < x < 6$$

Q006 Written

Solve the six given logarithmic inequalities using base monotonicity and argument conditions.**Solution**

1. For $a > 1$ keep the inequality direction; for $0 < a < 1$ reverse it before solving.

Final answer

Each solution is the intersection of the argument domain with the monotonicity interval.

Worked Solutions

Applications of Logarithms

Q007 Written

Solve all 24 given logarithmic inequalities, including shifted arguments, coefficients, and rational arguments.

Solution

1. Never omit the positive-argument condition; it can remove one branch of an algebraic solution.

Final answer

Use argument conditions, convert coefficients by the power law, then solve and intersect the intervals.

Q008 Written

Solve the given challenge inequality involving a reciprocal-base logarithmic sum and state the complete interval.

Solution

1. Combine the logarithms, solve the rational inequality, and retain only the common domain.

Final answer

Reverse the comparison after converting to a base below one, then intersect with both argument conditions.

Worked Solutions

Applications of Logarithms

Q009 Written

Solve $5(\log_2 x)^2 - 16\log_2 x + 3 < 0$.**Solution**

1. Let $t = \log_2 x$: $(5t - 1)(t - 3) < 0$, so $1/5 < t < 3$, then back-substitute.

Final answer

$$2^{1/5} < x < 8$$

Q010 Written

Solve the three given substitution problems: two quadratic logarithmic inequalities/equations and one exponential-log value.**Solution**

1. The substitution turns the problem into a quadratic; only then return to the original logarithm.

Final answerSubstitute $t = \log_a x$, solve in t , enforce $x > 0$, and back-substitute.

Worked Solutions

Applications of Logarithms

Q011 Written

Solve the given groups: three equations, three inequalities, and three exact logarithmic values.

Solution

1. Record the domain before setting $t = \log_a x$, and verify every returned value.

Final answer

All groups follow the same substitution, domain, solve, and back-substitution sequence.

Q012 Written

Solve $(\log_3 x)^2 - \log_3 x - 6 < 0$ for the positive wave-intensity multiplier x .

Solution

1. Let $t = \log_3 x$: $(t - 3)(t + 2) < 0$, so $-2 < t < 3$, hence $3^{-2} < x < 3^3$.

Final answer

$1/9 < x < 27$

Worked Solutions

Applications of Logarithms

Q013 Written

For $y = -(\log_3 x)^2 + 4 \log_3 x + 3$, $1 \leq x \leq 27$, find maximum/minimum values and the corresponding x .

Solution

1. Let $t = \log_3 x$, so $0 \leq t \leq 3$; complete the square $y = 7 - (t - 2)^2$.

Final answer

Maximum 7 at $x = 9$; minimum 3 at $x = 1$.

Q014 Written

Find the maximum and minimum of the given logarithmic quadratic on its stated interval, with the corresponding values of x .

Solution

1. Convert the x -interval to a t -interval before reading the quadratic range.

Final answer

Use the vertex in $t = \log_a x$ and compare it with both endpoints.

Worked Solutions

Applications of Logarithms

Q015 Written

Solve the three given extrema problems, each with a different base and interval.

Solution

1. The maximum/minimum in t must be translated back to the exact $x = a^t$.

Final answer

For every item, map the domain to t , locate the vertex, and check endpoints.

Q016 Written

For $y = -(\log_2 x)^2 + 2\log_2 x + 4$, $1 \leq x \leq 8$, find the maximum and minimum health ratings and their concentrations.

Solution

1. Here $0 \leq t \leq 3$ and $y = 5 - (t - 1)^2$; compare $t = 0, 1, 3$.

Final answer

Maximum 5 at $x = 2$; minimum 4 at $x = 1$.

Worked Solutions

Applications of Logarithms

Q017 Written

Using $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, determine the digits of 6^{30} and the first non-zero decimal place of $(2/3)^{50}$.

Solution

1. Take common logs: $30(\log 2 + \log 3) = 23.343$ and $50(\log 2 - \log 3) = -8.805$.

Final answer

24 digits; 9th decimal place.

Q018 Written

Complete the given digit-count and first-non-zero-decimal-place sets for the displayed powers and fractions.

Solution

1. Use the larger absolute exponent for a number between two negative powers of ten.

Final answer

Use $n - 1 < \log_{10} N < n$ for digits and $-n < \log_{10} N < -n + 1$ for the first non-zero place.

Worked Solutions

Applications of Logarithms

Q019 Written

Using the printed common logs, determine the number of digits and the leading digit of 12^{50} .

Solution

1. Compute $50 \log_{10} 12 = 50(2 \log 2 + \log 3) = 53.965$; then use the characteristic and mantissa.

Final answer

55 digits; leading digit 1.

Q020 Written

Solve $\log_x(x+2) \leq \log_x(4-x)$ for all real x .

Solution

1. Domain: $0 < x < 4, x \neq 1$. For $0 < x < 1$ reverse the sign and for $1 < x < 4$ keep it; both cases are impossible.

Final answer

No real solution.

Worked Solutions

Applications of Logarithms

Q021 MCQ

The argument condition for $\log_3(x - 5)$ is:**Solution**

1. A logarithm argument must be positive.

Correct option: C**Final answer**

$$x > 5$$

Q022 MCQ

If $\log_2(2x - 1) = \log_2 3$, then $x =$:**Solution**

1. Equate same-base arguments.

Correct option: B**Final answer**

$$2$$

Worked Solutions

Applications of Logarithms

Q023 MCQ

The valid solution of the challenge equation is:**Solution**

1. The domain $x > 2$ rejects the other algebraic root.

Correct option: C**Final answer**

4

Q024 MCQ

For same-base logarithms, equality of logs gives:**Solution**

1. The logarithm is one-to-one on its domain.

Correct option: A**Final answer**

arguments equal

Worked Solutions

Applications of Logarithms

Q025 MCQ

For base $a > 1$, $\log_a M < \log_a N$ means:

Solution

1. The logarithm is increasing.

Correct option: B

Final answer

$$M < N$$

Q026 MCQ

For base $0 < a < 1$, the inequality direction is:

Solution

1. A reciprocal-base logarithm is decreasing.

Correct option: B

Final answer

reversed

Worked Solutions

Applications of Logarithms

Q027 MCQ

The domain of $\log_{1/2}(x - 5)$ requires:**Solution**

1. The argument must be positive.

Correct option: C

Final answer

$$x > 5$$

Q028 MCQ

The warm-up intersection is:**Solution**1. Intersect the algebraic interval with $x > 5$.

Correct option: B

Final answer

$$5 < x < 6$$

Worked Solutions

Applications of Logarithms

Q029 MCQ

The best substitution for a quadratic logarithmic equation is:**Solution**

1. It turns the logarithm into a single variable.

Correct option: B**Final answer**

$$t = \log_a x$$

Q030 MCQ

The inequality $5t^2 - 16t + 3 < 0$ gives:**Solution**

1. Factor and use the sign chart.

Correct option: B**Final answer**

$$1/5 < t < 3$$

Worked Solutions

Applications of Logarithms

Q031 MCQ

The substitution variable must satisfy:**Solution**

1. A logarithm value can be any real number.

Correct option: B**Final answer**

$$t \in \mathbb{R}$$

Q032 MCQ

The challenge interval for $(\log_3 x)^2 - \log_3 x - 6 < 0$ is:**Solution**

1. Convert $-2 < t < 3$ back with base 3.

Correct option: B**Final answer**

$$1/9 < x < 27$$

Worked Solutions

Applications of Logarithms

Q033 MCQ

For $y = -(t - 2)^2 + 7$, the maximum occurs at:

Solution

1. The vertex gives the maximum.

Correct option: C

Final answer

$$t = 2$$

Q034 MCQ

The correct first step for extrema is:

Solution

1. Then map the domain to a t-range.

Correct option: B

Final answer

$$\text{set } t = \log_a x$$

Worked Solutions

Applications of Logarithms

Q035 MCQ

For $1 \leq x \leq 27$, $t = \log_3 x$ lies in:**Solution**1. Apply $\log_3$ to the endpoints.

Correct option: B

Final answer $[0, 3]$

Q036 MCQ

The water-tank challenge maximum health rating is:**Solution**1. The vertex is at $t = 1$, so $x = 2$.

Correct option: C

Final answer

5

Worked Solutions

Applications of Logarithms

Q037 MCQ

The number of digits of N is determined by:**Solution**

1. Use the characteristic of the common logarithm.

Correct option: B**Final answer**

$$[\log_{10} N] + 1$$

Q038 MCQ

If $-9 < \log_{10} N < -8$, the first non-zero digit is at:**Solution**

1. Use the larger absolute negative exponent.

Correct option: B**Final answer** 9^{th}

Worked Solutions

Applications of Logarithms

Q039 MCQ

The common logarithm base is:**Solution**

1. A common logarithm has base 10.

Correct option: C

Final answer

10

Q040 MCQ

The solution set of $\log_x(x + 2) \leq \log_x(4 - x)$ is:**Solution**

1. The increasing and decreasing cases both contradict their domains.

Correct option: D

Final answer

empty

Worked Solutions

Applications of Logarithms

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Solve $\log_5(x - 1) = 2$.

Solution

1. $x - 1 = 25$.

Correct option: A

Final answer

$$x = 26$$

Quiz MCQ 2 [Concept Quiz · MCQ](#)

For $0 < a < 1$ **and** $M < N$, **compare** $\log_a M$ **and** $\log_a N$.

Solution

1. The function is decreasing.

Correct option: B

Final answer

$$\log_a M > \log_a N$$

Worked Solutions

Applications of Logarithms

Quiz MCQ 3 Concept Quiz · MCQ

Solve $4(\log_2 x)^2 - 5\log_2 x + 1 = 0$.

Solution

1. Let $t = \log_2 x$: $(4t - 1)(t - 1) = 0$.

Correct option: C

Final answer

$$x = 2, \sqrt{2}$$

Quiz MCQ 4 Concept Quiz · MCQ

Find the maximum of $y = -(\log_3 x)^2 + 2\log_3 x + 4$ **on** $1 \leq x \leq 27$.

Solution

1. Let $t = \log_3 x \in [0, 3]$; vertex at $t = 1$.

Correct option: D

Final answer

$$5 \text{ at } x = 3$$

Worked Solutions

Applications of Logarithms

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $\log_3(x + 1) = 2$.**Solution**

1. $x + 1 = 9$.

Final answer

$$x = 8$$

Quiz WRITTEN 6 Concept Quiz · Written**Solve** $\log_2 x \geq 3$.**Solution**

1. Base 2 is increasing.

Final answer

$$x \geq 8$$

Worked Solutions

Applications of Logarithms

Quiz WRITTEN 7 Concept Quiz · Written**Solve** $(\log_5 x)^2 - 4 < 0$.**Solution**

1. $-2 < \log_5 x < 2$.

Final answer

$$1/25 < x < 25$$

Quiz WRITTEN 8 Concept Quiz · Written**How many digits does 6^{20} have using $\log_{10} 2 = 0.3010, \log_{10} 3 = 0.4771$?****Solution**

1. $20(0.3010 + 0.4771) = 15.562$.

Final answer

16 digits

Answer key

Use the question number shown on each page.

- Q001** 16
Q002 $x = 2$
Q003 Apply the argument condition first; then equate same-base arguments and reject any extraneous roots.
Q004 $x = 4$
Q005 $0 < x < 81$; $5 < x < 6$
Q006 Each solution is the intersection of the argument domain with the monotonicity interval.
Q007 Use argument conditions, convert coefficients by the power law, then solve and intersect the intervals.
Q008 Reverse the comparison after converting to a base below one, then intersect with both argument conditions.
Q009 $2^{1/5} < x < 8$
Q010 Substitute $t = \log_a x$, solve in t , enforce $x > 0$, and back-substitute.
Q011 All groups follow the same substitution, domain, solve, and back-substitution sequence.
Q012 $1/9 < x < 27$
Q013 Maximum 7 at $x = 9$; minimum 3 at $x = 1$.
Q014 Use the vertex in $t = \log_a x$ and compare it with both endpoints.
Q015 For every item, map the domain to t , locate the vertex, and check endpoints.
Q016 Maximum 5 at $x = 2$; minimum 4 at $x = 1$.
Q017 24 digits; 9th decimal place.
Q018 Use $n - 1 < \log_{10} N < n$ for digits and $-n < \log_{10} N < -n + 1$ for the first non-zero place.
Q019 55 digits; leading digit 1.
Q020 No real solution.
Q021 C
Q022 B
Q023 C
Q024 A
Q025 B
Q026 B
Q027 C
Q028 B
Q029 B
Q030 B
Q031 B
Q032 B
Q033 C
Q034 B
Q035 B
Q036 C
Q037 B
Q038 B
Q039 C
Q040 D
Quiz MCQ 1 A
Quiz MCQ 2 B
Quiz MCQ 3 C
Quiz MCQ 4 D
Quiz WRITTEN 5 $x = 8$
Quiz WRITTEN 6 $x \geq 8$
Quiz WRITTEN 7 $1/25 < x < 25$
Quiz WRITTEN 8 16 digits

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

English Student Edition • Focused formulas and guided practice

Formula opener

Derivative

$$\frac{d}{dx} x^n = nx^{n-1}$$

Product

$$(uv)' = u'v + uv'$$

Chain

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q001 Written

For $f(x) = x^2 + 2x$, find the average rate of change from -1 to 3 , and from 3 to $3 + h$.

Solution

1. Use $[f(b) - f(a)]/(b - a)$. For the h-form, factor $f(3 + h) - f(3) = h(8 + h)$.

Final answer

4; $8 + h$

Q002 Written

Find the average rate of change for the three given Try functions on their stated intervals.

Solution

1. Evaluate both endpoint values, subtract, and divide by the change in x; cancel h in the variable-endpoint case.

Final answer

2; $-5/3$; $h - 6$

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q003 Written

Find the average rate of change for all nine given Exercise functions, including fixed and h-form intervals.

Solution

1. Keep the endpoint order consistent and simplify before substituting numerical values.

Final answer

Apply $[f(b) - f(a)]/(b - a)$ to each printed item exactly.

Q004 Written

Find the average rate of change of a circle's area as its diameter changes from 10 cm to 18 cm.

Solution

1. With $A = \pi d^2/4$, compute $[A(18) - A(10)]/(18 - 10) = 7\pi$.

Final answer

$7\pi \text{ cm}^2/\text{cm}$

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q005 Written

Find the given limits by direct substitution and by simplifying the indeterminate fraction before substitution.

Solution

1. Direct substitution is valid when defined; otherwise factor/cancel first, then put the limiting value into the simplified expression.

Final answer

−2; 6

Q006 Written

Find the three given Try limits, including polynomial and h-quotient forms.

Solution

1. Test direct substitution, identify 0/0, simplify, and substitute again.

Final answer

Substitute after removing any common factor; polynomial limits are obtained directly.

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q007 Written

Find all nine given Exercise limits, including x-limits and h-limits.

Solution

1. For a polynomial use continuity; for a quotient, factor before taking the limit.

Final answer

Each limit is obtained by direct substitution or cancellation of the factor causing 0/0.

Q008 Written

If $\lim_{x \rightarrow 3} (x^2 - 5x + k)/(x - 3) = L$ is finite, find Lk .

Solution

1. Set the numerator to zero at 3: $k = 6$. Then cancel $x - 3$, giving $L = 1$, so $Lk = 6$.

Final answer

6

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q009 Written

Find the differential coefficient of $f(x) = 3x^2$ at $x = -2$ using the definition.

Solution

1. Use $\lim_{h \rightarrow 0} [f(-2 + h) - f(-2)]/h = \lim(-12 + 3h) = -12$.

Final answer

$$f'(-2) = -12$$

Q010 Written

Find the differential coefficients at the given values for the two given Try functions using the definition.

Solution

1. Expand the shifted polynomial fully before cancelling the common h.

Final answer

Substitute a into $[f(a + h) - f(a)]/h$, cancel h, and let h tend to zero.

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q011 Written

Find the differential coefficients for all six given Exercise functions at their stated values.

Solution

1. The resulting constant is the instantaneous slope at the given input.

Final answer

Use the definition at each specified point and simplify exactly.

Q012 Written

A dropped stone has displacement $s(t) = 4.9t^2$ metres. Find its instantaneous velocity at $t = 10$ seconds using the definition.

Solution

1. The differential coefficient is $s'(t) = 9.8t$; hence $s'(10) = 98$.

Final answer

98 m s^{-1}

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q013 Written

Differentiate $f(x) = x^2 + 3x$ using the definition of the derivative.**Solution**

1. Expand $f(x + h) - f(x) = 2xh + h^2 + 3h$, divide by h , and let h tend to zero.

Final answer

$$f'(x) = 2x + 3$$

Q014 Written

Differentiate the three given Try functions using the definition.**Solution**

1. Do not use a memorised rule; retain the h -limit structure until the cancellation is visible.

Final answer

Expand each $f(x + h) - f(x)$, cancel h , and take the limit.

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q015 **Written****Differentiate all nine given Exercise functions using the definition.****Solution**

1. Expand, collect the coefficient of h , divide by h , then set $h = 0$.

Final answerApply $f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)]/h$ to every printed polynomial.Q016 **Written****A regular hexagon has side length x . Write its area and differentiate it using the definition.****Solution**

1. Substitute the area formula into the difference quotient; the derivative of the quadratic factor is $3\sqrt{3}x$.

Final answer

$$A = \frac{3\sqrt{3}}{2}x^2, \quad A'(x) = 3\sqrt{3}x$$

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q017 Written

If $f(x) = 5$ for every x except that it is undefined at $x=2$, does $\lim_{x \rightarrow 2} f(x)$ exist?

Solution

1. A limit depends on nearby values, not on whether the function is defined at the point itself.

Final answer

Yes; the limit is 5.

Q018 MCQ

The average rate of change from a to b is:

Solution

1. Use the secant slope.

Correct option: B

Final answer

$$[f(b) - f(a)] / (b - a)$$

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q019 MCQ

For $f(x) = x^2 + 2x$ from -1 to 3, the average rate is:

Solution

1. Evaluate the two endpoints.

Correct option: C

Final answer

4

Q020 MCQ

The average area-rate for diameter 10 to 18 is:

Solution

1. Use $A = \pi d^2/4$.

Correct option: A

Final answer

7π

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q021 MCQ

The h-form average rate is found by:**Solution**

1. Simplify the difference quotient.

Correct option: B**Final answer**

cancelling the common h

Q022 MCQ

A direct-substitution limit is valid when:**Solution**

1. Use continuity where defined.

Correct option: A**Final answer**

the expression is defined

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q023 MCQ

The 0/0 form requires:**Solution**

1. Remove the common vanishing factor.

Correct option: A**Final answer**

factor/cancel first

Q024 MCQ

In the finite-limit challenge, k equals:**Solution**

1. The numerator must vanish at $x = 3$.

Correct option: C**Final answer**

6

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q025 MCQ

For a polynomial, the limit equals:**Solution**

1. Polynomials are continuous.

Correct option: A**Final answer**

its value at the point

Q026 MCQ

The definition of the differential coefficient at a is:**Solution**

1. Use the difference quotient.

Correct option: A**Final answer** $\lim [f(a+h)-f(a)]/h$

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q027 MCQ

For $f(x) = 3x^2$, $f'(-2)$ is:**Solution**

1. Expand and cancel h.

Correct option: A

Final answer

-12

Q028 MCQ

The stone's velocity at 10 s is:**Solution**1. Use $s'(t) = 9.8t$.

Correct option: B

Final answer

98

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q029 MCQ

After cancelling h , take the limit as:**Solution**

1. The definition uses h approaching zero.

Correct option: A**Final answer**

$$h \rightarrow 0$$

Q030 MCQ

The derivative function is:**Solution**

1. Associate the coefficient with each x .

Correct option: B**Final answer**the differential coefficient for every x

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q031 MCQ

The derivative of $x^2 + 3x$ from the definition is:**Solution**

1. Expand the difference quotient.

Correct option: B

Final answer

$$2x + 3$$

Q032 MCQ

The area of a regular hexagon of side x is:**Solution**

1. Use the standard hexagon area formula.

Correct option: B

Final answer

$$(3\sqrt{3}/2)x^2$$

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q033 MCQ

A function need not be defined at a for its limit to:**Solution**

1. Nearby values determine a limit.

Correct option: A**Final answer**

exist

Q034 MCQ

The derivative of a constant function is:**Solution**

1. The difference quotient of a constant is zero.

Correct option: A**Final answer**

0

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Find the average rate of change of x^2 from 1 to 3.

Solution

1. $[9 - 1]/2 = 4$.

Correct option: A

Final answer

4

Quiz MCQ 2 [Concept Quiz · MCQ](#)

Find $\lim_{x \rightarrow 2}(x^2 + 1)$.

Solution

1. Direct substitution gives 5.

Correct option: B

Final answer

5

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Quiz MCQ 3 [Concept Quiz · MCQ](#)

Find $f'(x)$ for $f(x) = x^2$ using the definition.

Solution

1. Expand $(x + h)^2 - x^2$, cancel h , and let h tend to zero.

Correct option: C

Final answer

$2x$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

If $s(t) = 4.9t^2$, find the velocity at $t=5$.

Solution

1. $s'(t) = 9.8t$.

Correct option: D

Final answer

49

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Quiz WRITTEN 5 Concept Quiz · Written

Find the average rate of change of $3x + 1$ from 2 to 7.

Solution

1. A linear function has constant slope 3.

Final answer

3

Quiz WRITTEN 6 Concept Quiz · Written

Evaluate $\lim_{h \rightarrow 0}(6 + 2h)$.

Solution

1. Substitute $h = 0$.

Final answer

6

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Quiz WRITTEN 7 Concept Quiz · Written

Find the differential coefficient of $2x^2$ at $x=3$ from the definition.

Solution

1. The difference quotient simplifies to $4x + 2h$, giving 12.

Final answer

12

Quiz WRITTEN 8 Concept Quiz · Written

State $\lim_{x \rightarrow 2} f(x)$ when $f(x)=5$ for all x except $x=2$ where it is undefined.

Solution

1. Nearby values are constantly 5.

Final answer

5

Answer key

Use the question number shown on each page.

Q001 4; $8 + h$	Q009 $f'(-2) = -12$	Q017 Yes; the limit is 5.	Q030 B
Q002 2; $-5/3$; $h - 6$	Q010 Substitute a into $[f(a + h) - f(a)]/h$	Q018 B	Q031 B
Q003 Apply $[f(b) - f(a)]/(b - a)$ to each printed item exactly.	cancel h , and let h tend to zero.	Q019 C	Q032 B
Q004 $7\pi \text{ cm}^2/\text{cm}$	Q011 Use the definition at each specified point and simplify exactly.	Q020 A	Q033 A
Q005 -2 ; 6	Q012 98 m s^{-1}	Q021 B	Q034 A
Q006 Substitute after removing any common factor; polynomial limits are obtained directly.	Q013 $f'(x) = 2x + 3$	Q022 A	Quiz MCQ 1 A
Q007 Each limit is obtained by direct substitution or cancellation of the factor causing $0/0$.	Q014 Expand each $f(x + h) - f(x)$, cancel h and take the limit.	Q023 A	Quiz MCQ 2 B
Q008 6	Q015 Apply $f'(x) = \lim_{h \rightarrow 0} [f(x + h) - f(x)]/h$ to every printed polynomial.	Q024 C	Quiz MCQ 3 C
	Q016 $A = \frac{3\sqrt{3}}{2}x^2$, $A'(x) = 3\sqrt{3}x$	Q025 A	Quiz MCQ 4 D
		Q026 A	Quiz WRITTEN 5 3
		Q027 A	Quiz WRITTEN 6 6
		Q028 B	Quiz WRITTEN 7 12
		Q029 A	Quiz WRITTEN 8 5

Concept 2 | Differentiation and Tangent Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Gradient

$$m = f'(x_1)$$

Tangent

$$y - y_1 = m(x - x_1)$$

Normal

$$m_n m = -1$$

Worked Solutions

Differentiation and Tangent Lines

Q001 Written

Differentiate the four Warm Up demands: $y = x^2 - 3x + 2$, $y = (2x + 1)^3$, $V = \pi r^2 h$ with respect to (r), and find $f'(-1)$ for $f(x) = x^3 - 3x + 1$.

Solution

1. Use $(x^n)' = nx^{n-1}$. For a composite cubic, expand or use the chain rule. Treat (h) as constant in dV/dr , then substitute $x = -1$ into $3x^2 - 3$.

Final answer

$$2x - 3; 6(2x + 1)^2; 2\pi rh; 0$$

Q002 Written

Differentiate the six printed Try functions, the two variable-labelled functions $S = \pi r^2$ and $h = vt - \frac{1}{2}gt^2$, and evaluate $f'(2)$, $f'(-3)$ for $f(x) = 4x^3 - 3x^2 + 2$.

Solution

1. Differentiate each term, keeping non-differentiated variables constant. For the values, first form $f'(x) = 12x^2 - 6x$, then substitute.

Final answer

$$3, -6x + 8, 2x + 3, 6x^2 - 8x + 6, 6(2x - 3)^2, 4x^2 + x - 1; 2\pi r, v - gt; 36, 126$$

Worked Solutions

Differentiation and Tangent Lines

Q003 Written

Differentiate every given Exercise function 18 polynomial/product forms, the six functions with variables indicated in brackets, and evaluate the two values in each of Exercises 3–5.

Solution

1. Bring down the exponent and subtract one. Expand products before differentiating when no chain rule is being used; then substitute the stated input.

Final answer

Apply the power rule term-by-term; for a product or power, expand first. The value sets are: Exercise 3 (33,65); Exercise 4 $-24, -3$; Exercise 5 (3,28).

Worked Solutions

Differentiation and Tangent Lines

Q004 Written

For $f(x) = \sqrt[3]{x^5} + \frac{1}{x^2}$, find $f'(1) + f'(-1)$.

Solution

1. Write $f(x) = x^{5/3} + x^{-2}$, so $f'(x) = \frac{5}{3}x^{2/3} - 2x^{-3}$. Hence $f'(1) = -\frac{1}{3}$, $f'(-1) = \frac{11}{3}$, and the sum is $\frac{10}{3}$.

Final answer

$$\frac{10}{3}$$

Q005 Written

Find the quadratic $f(x) = ax^2 + bx + c$ satisfying $f'(0) = -2$, $f'(2) = 2$, $f(1) = 2$.

Solution

1. Since $f'(x) = 2ax + b$, $b = -2$, $4a + b = 2$, and $a + b + c = 2$. Thus $a = 1$, $b = -2$, $c = 3$.

Final answer

$$f(x) = x^2 - 2x + 3$$

Worked Solutions

Differentiation and Tangent Lines

Q006 Written

Find the quadratic satisfying $f'(0) = 2$, $f'(1) = 4$, $f(2) = 6$.**Solution**

1. From $2ax + b$: $b = 2$, $2a + b = 4$, and $4a + 2b + c = 6$, giving $a = 1$, $b = 2$, $c = 0$.

Final answer

$$f(x) = x^2 + 2x$$

Q007 Written

Find the four quadratics satisfying the four given triples of derivative and function conditions.**Solution**

1. For each item set $f(x) = ax^2 + bx + c$, use $f'(x) = 2ax + b$, substitute the two derivative conditions and one function condition, then solve the three linear equations.

Final answer

$$x^2 - 4x + 8; -2x^2 + 3x + 3; -5x^2 + 26x - 14; -\frac{3}{2}x^2 + 2x + \frac{3}{2}$$

Worked Solutions

Differentiation and Tangent Lines

Q008 Written

Find the quadratic ($f(x)$) satisfying $f'(2) + f'(4) = 0$, $f(3) = 0$, $f(1) = 12$.

Solution

1. With $f'(x) = 2ax + b$, $12a + 2b = 0$, $9a + 3b + c = 0$, and $a + b + c = 12$. Solving gives $a = 3$, $b = -18$, $c = 27$.

Final answer

$$f(x) = 3x^2 - 18x + 27$$

Q009 Written

Find the tangent line to $y = x^2 - 4x + 1$ at $(3, -2)$.

Solution

1. The gradient is $f'(3) = 2$. Use $y + 2 = 2(x - 3)$, hence $y = 2x - 8$.

Final answer

$$y = 2x - 8$$

Worked Solutions

Differentiation and Tangent Lines

Q010 Written

Find the tangent line for the three given functions at their stated points: $y = x^2 + 3$ at $(-2, 7)$, $y = x^2 - 3x + 2$ at $((1, 0))$, and $y = -x^3 + 2x + 4$ at $(-2, 8)$.

Solution

1. Differentiate, evaluate the gradient at the given x-coordinate, and use $y - y_1 = m(x - x_1)$ for each point.

Final answer

$$y = -4x - 1; y = -x + 1; y = -10x - 12$$

Q011 Written

Find the tangent line for all nine given Exercise functions at their given points.

Solution

1. For each item compute $m = f'(a)$, insert $((a, f(a)))$ into $y - f(a) = m(x - a)$, then rearrange.

Final answer

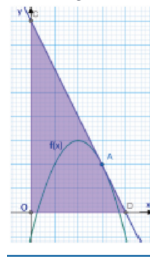
$$y = 4x - 5; y = 4x - 1; y = x - 4; y = x - 7; y = -4x + 3; y = 7x - 3; y = 9x - 16; y = -x - 2; y = x + 1$$

Worked Solutions

Differentiation and Tangent Lines

Q012 Written

For $f(x) = -x^2 + 4x - 1$, the tangent at x-coordinate 3 meets the axes at (D,C). Find the area of triangle (ODC).



Solution

- The tangent is $y = -2x + 8$, so $D = (4, 0)$ and $C = (0, 8)$. Therefore area $= \frac{1}{2}(4)(8) = 16$. The ministry graph is retained beside the demand.

Final answer

16 square units

Worked Solutions

Differentiation and Tangent Lines

Q013 Written

Solve the two given Warm Up demands: tangents from $C(0,2)$ to $y = x^2 - 3x + 6$, and tangents to $y = x^3 + 4x^2 - 3$ with gradient 3.

Solution

1. Let the point of tangency be $((a, f(a)))$. Equate the tangent line through the external point, or set $(f'(a))$ equal to the required gradient, then solve for (a) .

Final answer

$$y = -7x + 2, y = x + 2; \quad y = 3x + 15, y = 3x - \frac{95}{27}$$

Q014 Written

Solve the two given Try demands: tangents from $C(1, -1)$ to $y = x^2 - x + 3$, and tangents to $y = x^3 + 2x - 3$ with gradient 5.

Solution

1. Use $((a, f(a)))$ and $m = f'(a)$. The first condition gives $a = 3, -1$; the second gives $a = 1, -1$. Substitute each point into the tangent formula.

Final answer

$$y = 5x - 6, y = -3x + 2; \quad y = 5x - 5, y = 5x - 1$$

Worked Solutions

Differentiation and Tangent Lines

Q015 Written

Solve all six given Exercise demands involving external points or prescribed gradients.**Solution**

1. Let the contact point be $((a, f(a)))$, form $y - f(a) = f'(a)(x - a)$, and impose the external point or gradient condition. Retain every real value of (a) .

Final answer

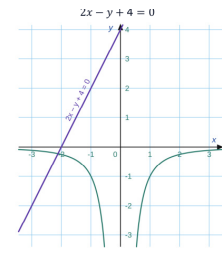
$$y = 6x - 8, y = -2x; y = x + 1, y = -7x + 25; y = -5x + 6, y = 3x - 2; y = 9x - 11, y = 9x + 21; y = -x; y = -2x + 16, y = -2x - 8$$

Worked Solutions

Differentiation and Tangent Lines

Q016 Written

For $f(x) = -1/x^2$, find the tangent line parallel to $2x - y + 4 = 0$.



Solution

1. The given line has gradient 2. Since $f'(x) = 2/x^3$, $2/a^3 = 2$ gives $a = 1$. The tangent at $(1, -1)$ is $y + 1 = 2(x - 1)$, so $y = 2x - 3$. The ministry graph is retained beside the demand.

Final answer

$$y = 2x - 3$$

Worked Solutions

Differentiation and Tangent Lines

Q017 MCQ

The derivative of x^n is:**Solution**

1. Bring down n and subtract 1 from the exponent.

Correct option: B**Final answer**

$$nx^{n-1}$$

Q018 MCQ

The derivative of $V = \pi r^2 h$ with respect to r is:**Solution**

1. Treat h as a constant.

Correct option: B**Final answer**

$$2\pi rh$$

Worked Solutions

Differentiation and Tangent Lines

Q019 MCQ

For $f(x) = x^3 - 3x + 1$, $f'(-1)$ is:

Solution

1. $f'(x) = 3x^2 - 3$.

Correct option: C

Final answer

0

Q020 MCQ

The challenge function can be written as:

Solution

1. Convert the cube root and reciprocal to powers.

Correct option: A

Final answer

$$x^{5/3} + x^{-2}$$

Worked Solutions

Differentiation and Tangent Lines

Q021 MCQ

For $f(x) = ax^2 + bx + c$, $(f'(x))$ is:**Solution**

1. Differentiate the quadratic.

Correct option: B**Final answer**

$$2ax + b$$

Q022 MCQ

The condition $f'(0) = -2$ gives:**Solution**

1. Substitute $x = 0$ into $2ax + b$.

Correct option: B**Final answer**

$$b = -2$$

Worked Solutions

Differentiation and Tangent Lines

Q023 MCQ

The warm-up quadratic is:**Solution**

1. Solve the three coefficient equations.

Correct option: A

Final answer

$$x^2 - 2x + 3$$

Q024 MCQ

The challenge coefficient a is:**Solution**1. Use $f'(2) + f'(4) = 0$, then the two function values.

Correct option: C

Final answer

3

Worked Solutions

Differentiation and Tangent Lines

Q025 MCQ

The tangent gradient at $x=a$ is:**Solution**

1. The derivative gives the instantaneous gradient.

Correct option: C**Final answer**

$$f'(a)$$

Q026 MCQ

The tangent to $y = x^2 - 4x + 1$ at $(3, -2)$ is:**Solution**

1. Use gradient 2 and point $3, -2$.

Correct option: B**Final answer**

$$y = 2x - 8$$

Worked Solutions

Differentiation and Tangent Lines

Q027 MCQ

In the triangle challenge, the x-intercept of the tangent is:**Solution**1. The tangent is $y = -2x + 8$.

Correct option: C

Final answer

4

Q028 MCQ

The area of triangle (ODC) in the challenge is:**Solution**

1. Use one half base times height.

Correct option: C

Final answer

16

Worked Solutions

Differentiation and Tangent Lines

Q029 MCQ

When the contact point is not given, set it to:**Solution**

1. Use an unknown contact x-coordinate.

Correct option: B**Final answer** $(a, f(a))$

Q030 MCQ

The gradient of $2x - y + 4 = 0$ is:**Solution**

1. Rearrange to $y = 2x + 4$.

Correct option: D**Final answer**

2

Worked Solutions

Differentiation and Tangent Lines

Q031 MCQ

For $f(x) = -1/x^2$, $(f'(x))$ is:**Solution**1. Write $f = -x^{-2}$ and differentiate.

Correct option: B

Final answer

$$2/x^3$$

Q032 MCQ

The parallel tangent in the challenge is:**Solution**1. The contact point is $a = 1$.

Correct option: B

Final answer

$$y = 2x - 3$$

Worked Solutions

Differentiation and Tangent Lines

Quiz WRITTEN 1 Concept Quiz · Written**Differentiate** $3x^3 - 2x$.**Solution**

1. Apply the power rule.

Final answer

$$9x^2 - 2$$

Quiz WRITTEN 2 Concept Quiz · Written**Find the quadratic with** $f'(0) = 1, f'(1) = 3, f(1) = 2$.**Solution**

1. Use $f'(x) = 2ax + b$ and solve for (a,b,c).

Final answer

$$x^2 + x$$

Worked Solutions

Differentiation and Tangent Lines

Quiz WRITTEN 3 Concept Quiz · Written

Find the tangent to $y = x^2$ at $((2,4))$.

Solution

1. Gradient $f'(2) = 4$, then use point-slope form.

Final answer

$$y = 4x - 4$$

Quiz WRITTEN 4 Concept Quiz · Written

For $f(x) = -1/x$, find the tangent parallel to $y = x + 2$.

Solution

1. Set $f'(a) = 1$, giving $a = 1$.

Final answer

$$y = x - 4$$

Worked Solutions

Differentiation and Tangent Lines

Quiz MCQ 5 **Concept Quiz · MCQ**

The derivative of a constant is:

Solution

1. A constant has zero rate of change.

Correct option: A

Final answer

0

Quiz MCQ 6 **Concept Quiz · MCQ**

For $f(x) = ax^2 + bx + c$, the coefficient equations come from:

Solution

1. There are three unknown coefficients.

Correct option: B

Final answer

two derivatives and one function value

Worked Solutions

Differentiation and Tangent Lines

Quiz MCQ 7 [Concept Quiz · MCQ](#)

The tangent equation in point-slope form is:

Solution

1. Use the given point and tangent gradient.

Correct option: A

Final answer

$$y - y_1 = mx - x_1$$

Quiz MCQ 8 [Concept Quiz · MCQ](#)

A tangent parallel to $y = 3x - 1$ has gradient:

Solution

1. Parallel lines have equal gradients.

Correct option: C

Final answer

3

Answer key

Use the question number shown on each page.

Q001 $2x - 3; 6(2x + 1)^2; 2\pi rh; 0$

Q002 $3, -6x + 8, 2x + 3, 6x^2 - 8x + 6, 6(2x - 3)^2, 4x^2 + x - 1; 2\pi r, v - gt; 36, 20$

Q003 Apply the power rule term-by-term; for a product or power, expand first. The value sets are: Exercise 3 (33,65); Exercise 4 $-24, -3$; Exercise 5 (3,28).

Q004 $\frac{10}{3}$

Q005 $f(x) = x^2 - 2x + 3$

Q006 $f(x) = x^2 + 2x$

Q007 $x^2 - 4x + 8; -2x^2 + 3x + 3; -5x^2 + 26x - 14; -\frac{3}{2}x^2 + 2x - \frac{3}{2}$

Q008 $f(x) = 3x^2 - 18x + 27$

Q009 $y = 2x - 8$

Q010 $y = -4x - 1; y = -x + 1; y = -10x - 12$

Q011 $y = 4x - 5; y = 4x - 1; y = x - 4; y = x - 7; y = -4x + 3; y = 7x - 3; y = 9x - 16; y = -x - 2; y = -4x - 1$

Q012 16 square units

Q013 $y = -7x + 2, y = x + 2; y = 3x + 15, y = 3x - \frac{85}{27}$

Q014 $y = 5x - 6, y = -3x + 2; y = 5x - 5, y = 5x - 1$

Q015 $y = 6x - 8, y = -2x; y = x + 1, y = -7x + 25; y = -5x + 6, y = 3x - 2; y = 9x - 11, y = 9x + 21; y = -x; y = -2x + 16, y = -2x - 4$

Q016 $y = 2x - 3$

Q017 B

Q018 B

Q019 C

Q020 A

Q021 B

Q022 B

Q023 A

Q024 C

Q025 C

Q026 B

Q027 C

Q028 C

Q029 B

Q030 D

Q031 B

Q032 B

Quiz WRITTEN 1 $9x^2 - 2$

Quiz WRITTEN 2 $x^2 + x$

Quiz WRITTEN 3 $y = 4x - 4$

Quiz WRITTEN 4 $y = x - 4$

Quiz MCQ 5 A

Quiz MCQ 6 B

Quiz MCQ 7 A

Quiz MCQ 8 C

Concept 3 | Applications of Differentiation: Polynomial Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Stationary

$$f'(x) = 0$$

Second derivative

$$f''(x) > 0 \Rightarrow \text{minimum}$$

Rate

$$\frac{dy}{dx}$$

Eng Eslam Ahmed | 01120009622

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q001 Written

Find the extreme values and sketch the graphs of the three cubic functions printed in the Warm Up.

Solution

1. Differentiate, solve $f'(x) = 0$, make a sign table, and substitute each stationary x-value into $f(x)$.

Final answer

Use the derivative sign table: for $x^3 - 6x^2 + 16$, local maximum 16 at $x = 0$ and local minimum -16 at $x = 4$; classify the other two from their stationary points.

Q002 Written

Find the extreme values and sketch the three given Try cubics.

Solution

1. Record the sign of f' from left to right and label each change $+$ $\rightarrow$ $-$ as a local maximum and $-$ $\rightarrow$ $+$ as a local minimum.

Final answer

Apply the same derivative sign-table test; a stationary point where the sign of f' does not change is not an extreme value.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q003 Written

Find the extreme values and draw the graphs for all nine given cubic functions in Exercise.

Solution

1. Use exact stationary coordinates and keep the graph's increasing/decreasing intervals consistent with the sign table.

Final answer

Differentiate each polynomial, solve the quadratic derivative, use the sign table, and evaluate f at every genuine stationary point.

Q004 Written

For $P(x) = x^3 - 3ax^2 + (3a^2 - 3)x + 4$, $a > 0$, find the flow rates giving the maximum and minimum peak efficiency.

Solution

1. $P'(x) = 3[(x - a)^2 - 1]$, so the critical points are $a - 1$ and $a + 1$. The derivative changes $+$ $\rightarrow$ $-$ then $-$ $\rightarrow$ $+$.

Final answer

Local maximum at $x = a - 1$; local minimum at $x = a + 1$.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q005 **Written**

On $[-3, 4]$, find the maximum and minimum values of $y = -x^3 + 12x + 5$.

Solution

1. Compare $f(-3)$, $f(4)$ with the stationary values $f(-2) = -11$ and $f(2) = 21$.

Final answer

Maximum 21 at $x = 2$; minimum -11 at $x = -2$.

Q006 **Written**

Find the maximum and minimum values of the two given Try cubics on their stated closed intervals.

Solution

1. A local extreme is not automatically absolute: compare it with both domain endpoints.

Final answer

Evaluate both endpoints and every stationary point inside the interval; the greatest value is the absolute maximum and the least is the absolute minimum.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q007 Written

Find the maximum and minimum values of all six given cubics on their given closed intervals.

Solution

1. State the extreme value together with its x-coordinate and retain only points in the specified domain.

Final answer

For each function, compare the endpoint values with the values at all critical x-values lying in the interval.

Q008 Written

For $f(x) = x^3 - 6x$ on $[-3, 3]$, find the absolute deepest dive and highest ascent.

Solution

1. Critical points are $x = \pm\sqrt{2}$, with values $\mp 4\sqrt{2}$; compare these with $f(-3) = -9$ and $f(3) = 9$.

Final answer

Absolute minimum -9 at $x = -3$; absolute maximum 9 at $x = 3$.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q009 Written

If $f(x) = -x^3 + ax + b$ has a local maximum of 9 at $x = 2$, find a , b , and the local minimum.

Solution

1. Use $f'(2) = 0$ to get $a = 12$, then $f(2) = 9$ to get $b = -7$. The other stationary point is $x = -2$.

Final answer

$a = 12, b = -7$; local minimum -23 at $x = -2$.

Q010 Written

Solve the two given Try problems: determine the coefficients from a local minimum/maximum and another stationary point.

Solution

1. Differentiate first, use each stated extreme point, then evaluate the remaining extreme value from the resulting cubic.

Final answer

Use $f'(a) = 0$, $f(a) = M$, and the second stationary condition to form three simultaneous equations in the coefficients.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q011 Written

Solve all six given Exercise problems asking for cubic coefficients from extrema.

Solution

1. Do not skip the condition $f'(a) = 0$: it identifies the stationary point before the function-value equation is used.

Final answer

Each set is determined by two derivative roots/conditions and one extreme-value condition; solve the resulting simultaneous equations.

Q012 Written

For $f(x) = x^3 + ax^2 + bx + c$, the graph has a local peak 178 at $x = -2$ and a local trough at $x = 4$. Find a, b, c .

Solution

1. The derivative roots are $-2, 4$, so $f'(x) = 3(x + 2)(x - 4) = 3x^2 - 6x - 24$. Thus $a = -3, b = -24$. Substitution of $f(-2) = 178$ gives $c = 150$.

Final answer

$a = -3, b = -24, c = 150$.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q013 Written

If $f(x) = x^3 - 3kx^2 + kx + 2$ has no extreme values, find the range of k .

Solution

1. The derivative is $f'(x) = 3x^2 - 6kx + k$. No two distinct stationary roots means $D = 36k^2 - 12k \leq 0$, hence $0 \leq k \leq \frac{1}{3}$.

Final answer

$$0 \leq k \leq \frac{1}{3}.$$

Q014 Written

Determine the parameter ranges in the given Try problems for a cubic to have no extreme values or to have both extremes.

Solution

1. Differentiate, calculate the quadratic discriminant, and solve the resulting parameter inequality exactly.

Final answer

Use $D \leq 0$ for no extremes and $D > 0$ for two distinct stationary points.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q015 Written

Determine the parameter ranges for all six given Exercise cubics according to whether they have extreme values.

Solution

1. Remember: $D > 0$ gives two distinct real stationary roots; $D \leq 0$ gives no pair of distinct extremes.

Final answer

Apply the derivative-discriminant criterion separately to each printed cubic.

Q016 Written

For $f(x) = x^3 + 3kx^2 + 12kx + 10$, find the exact k -range that guarantees both a local maximum and minimum.

Solution

1. $f'(x) = 3x^2 + 6kx + 12k$ has two distinct real roots when $36k^2 - 144k > 0$, i.e. $k(k - 4) > 0$.

Final answer

$k < 0$ or $k > 4$.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q017 Written

If $f(x) = x^3 - 3kx^2 + (k^2 + 5)x + 3$ is always monotonically increasing, find the range of k .

Solution

1. For a positive leading coefficient, monotonic increase requires the derivative quadratic to have no two distinct real roots: $D = 24k^2 - 60 \leq 0$.

Final answer

$$-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}.$$

Q018 Written

Find the parameter ranges in the two given Try problems for the cubic to be monotonically increasing.

Solution

1. Monotonicity is checked from $f'(x) \geq 0$ for every real x ; use the discriminant condition for the quadratic derivative.

Final answer

Set the derivative discriminant $D \leq 0$, with the derivative leading coefficient positive.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q019 Written

Find the parameter ranges in all six given Exercise problems for monotonic increase.

Solution

1. Complete the discriminant test and include equality when the derivative has a repeated zero.

Final answer

Differentiate each cubic and impose that its quadratic derivative never becomes negative.

Q020 Written

For $f(x) = \frac{1}{3}x^3 - kx^2 + (k + 6)x + 10$, find the k-range for monotonic increase on $x \in [1, \infty)$.

Solution

1. $f'(x) = x^2 - 2kx + k + 6$. If $k \leq 1$, its minimum on $[1, \infty)$ is $7 - k \geq 0$; if $k > 1$, the vertex condition gives $k + 6 - k^2 \geq 0$, hence $k \leq 3$.

Final answer

$k \leq 3$.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q021 Written

Find the number of distinct real roots of $x^3 - 6x^2 + 9x - 3 = 0$.**Solution**

1. The stationary values are $f(1) = 1 > 0$ and $f(3) = -3 < 0$, so the cubic crosses the x-axis three times.

Final answer

Three distinct real roots.

Q022 Written

Find the number of distinct real roots for the three given Try equations.**Solution**

1. Draw $y = f(x)$, find its critical points, and compare the local maximum/minimum with zero.

Final answer

Use the cubic graph: count x-axis intersections from the stationary points and end behavior.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q023 Written

Find the number of distinct real roots for all nine given cubic equations.

Solution

1. Differentiate, locate the turning points, and use their signs relative to zero to decide between one, two, or three distinct roots.

Final answer

Each answer is obtained from the number of x-axis intersections of its corresponding cubic graph.

Q024 Written

Find the number of distinct real roots of $-h^3 + 12h^2 - 36h + 40 = 0$.

Solution

1. The derivative is $-3(h - 2)(h - 6)$; both stationary values are positive, so the graph crosses the axis only once.

Final answer

One distinct real root.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q025 Written

If $-x^3 + 3x^2 + 9x + a = 0$ has three distinct real roots, find the range of a .

Solution

1. Rewrite as $x^3 - 3x^2 - 9x = a$. The local values are 5 and -27 ; a horizontal line intersects the cubic three times strictly between them.

Final answer

$$-27 < a < 5.$$

Q026 Written

Solve the two given Try parameter problems for three or two distinct real roots.

Solution

1. Strict inequalities give three roots; equality at a stationary value produces two distinct roots and must be handled separately.

Final answer

Rewrite each equation as $g(x) = a$, then compare the parameter line with the local maximum and minimum of g .

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q027 Written

Solve all four given Exercise parameter problems for the required number of distinct roots.

Solution

1. Keep the endpoint cases: tangency at an extreme value counts as two distinct roots, not three.

Final answer

Use the local extrema of the parameter-free cubic and count intersections with the horizontal parameter line.

Q028 Written

For $f(x) = x^3 - 6x^2 + 9x - 2a^2 + 7a$, find all real a such that $f(x) = 2$ has exactly three distinct real roots.

Solution

1. Set $g(x) = x^3 - 6x^2 + 9x$, whose extrema are 0 and 4. The horizontal level is $2a^2 - 7a + 2$; require $0 < 2a^2 - 7a + 2 < 4$.

Final answer

$$\frac{7 - \sqrt{65}}{4} < a < \frac{7 - \sqrt{33}}{4} \text{ or } \frac{7 + \sqrt{33}}{4} < a < \frac{7 + \sqrt{65}}{4}.$$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q029 **Written****Prove the given inequalities using an auxiliary cubic and a derivative sign table.****Solution**

1. Differentiate the auxiliary function, make the sign table on the domain, and substitute the minimum point or endpoint.

Final answer

Set $f(x) = \text{LHS} - \text{RHS}$, prove $f(x) \geq 0$ on the stated domain, and identify its minimum.

Q030 **Written****Prove all three given cubic inequalities on their stated domains.****Solution**

1. The proof is complete only after the derivative sign table and the minimum value are stated.

Final answer

Move the right-hand side to the left, then show the resulting auxiliary cubic has non-negative minimum on the domain.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q031 Written

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ for every $x \in [-1, 3]$.

Solution

1. The maximum of $x^3 - 3x^2 - 3x + 3$ on the interval is $4\sqrt{2} - 2$ at $x = 1 - \sqrt{2}$. Hence $k + 4\sqrt{2} - 2 \leq 0$.

Final answer

$k \leq 2 - 4\sqrt{2}$; the largest value is $2 - 4\sqrt{2}$.

Q032 Written

Find the extreme values and sketch the two given quartic functions.

Solution

1. Differentiate the quartic, use a sign table for the cubic derivative, and substitute the critical values.

Final answer

For $3x^4 + 4x^3 - 12x^2$: local maximum 0 at $x = 0$, local minima -32 at $x = -2$ and -5 at $x = 1$. For $-x^4 - 4x^3 - 8$: maximum 19 at $x = -3$; $x = 0$ is stationary but not an extreme.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q033 Written

Find the extreme values and graphs for all six given quartic functions.

Solution

1. Repeated derivative roots may be stationary without being extrema; check the sign change before labelling a point.

Final answer

Differentiate each quartic, construct the derivative sign table, and evaluate each genuine turning point.

Q034 Written

For $f(x) = x^4 - 4x^3 + k$, find the exact k -range for exactly two distinct real roots.

Solution

1. The global minimum of $x^4 - 4x^3 + k$ is $k - 27$ at $x = 3$. Exactly two crossings require this minimum to be negative.

Final answer

$k < 27$.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q035 MCQ

At a local maximum, the sign of f' changes:

Solution

1. Use the first derivative sign change.

Correct option: B

Final answer

$+to-$

Q036 MCQ

For $P'(x) = 3[(x - a)^2 - 1]$, the local maximum occurs at:

Solution

1. The derivative changes + to - at $a - 1$.

Correct option: A

Final answer

$a - 1$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q037 MCQ

Absolute extrema on a closed interval require comparing:**Solution**

1. Closed intervals include both.

Correct option: C**Final answer**

stationary points and endpoints

Q038 MCQ

The absolute maximum of $x^3 - 6x$ on $[-3, 3]$ is:**Solution**

1. Compare the endpoints with the critical values.

Correct option: B**Final answer**

9

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q039 MCQ

If $f(x) = -x^3 + ax + b$ has a stationary point at 2, then a is:

Solution

1. Set $f'(2) = -12 + a = 0$.

Correct option: C

Final answer

12

Q040 MCQ

The derivative roots in the challenge are:

Solution

1. Use the two turning-point x -values.

Correct option: B

Final answer

-2 and 4

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q041 MCQ

No two distinct extrema for a cubic means the derivative discriminant is:**Solution**

1. A repeated root or no real roots gives no pair.

Correct option: C**Final answer**

$$D \leq 0$$

Q042 MCQ

The challenge range for both extrema is:**Solution**

1. Require $36k^2 - 144k > 0$.

Correct option: B**Final answer**

$$k < 0 \text{ or } k > 4$$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q043 MCQ

For a positive-leading cubic to be monotone increasing, its derivative must be:

Solution

1. Use $f'(x) \geq 0$.

Correct option: B

Final answer

non-negative everywhere

Q044 MCQ

The challenge domain in 7-13 is:

Solution

1. Use the restricted domain in the vertex test.

Correct option: B

Final answer

[1,infinity)

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q045 MCQ

Three distinct roots occur when the x-axis lies:**Solution**

1. Then the graph crosses the axis three times.

Correct option: B**Final answer**

between the extrema

Q046 MCQ

The 7-14 challenge has how many distinct real roots?**Solution**

1. Both turning values are above zero.

Correct option: B**Final answer**

1

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q047 MCQ

Three intersections with $y = a$ require the level to lie:**Solution**

1. Tangency is an endpoint case.

Correct option: A

Final answer

strictly between the extrema

Q048 MCQ

The 7-15 warm-up range is:**Solution**

1. Use the two local values.

Correct option: A

Final answer $-27 < a < 5$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q049 MCQ

To prove $LHS \geq RHS$, define:**Solution**

1. Then prove $f \geq 0$.

Correct option: B**Final answer**

LHS-RHS

Q050 MCQ

The largest k in the challenge is:**Solution**

1. Bound k by the maximum of the k -free part.

Correct option: A**Final answer** $2-4\sqrt{2}$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q051 MCQ

A repeated root of a derivative may be:**Solution**

1. Check the sign change.

Correct option: C**Final answer**

stationary but not an extreme

Q052 MCQ

Exactly two distinct roots for $x^4 - 4x^3 + k = 0$ require:**Solution**

1. The global minimum must be below zero.

Correct option: B**Final answer** $k < 27$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q053 MCQ

A local minimum is identified by the change:**Solution**

1. A negative derivative followed by a positive derivative gives a local minimum.

Correct option: B**Final answer** $-to+$

Q054 MCQ

For $y = x^3 - 6x^2 + 16$, the critical x-values are:**Solution**

1. Solve $3x^2 - 12x = 0$.

Correct option: A**Final answer**

0 and 4

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q055 MCQ

The derivative sign table is used to determine:**Solution**

1. Signs of (f') control monotonicity.

Correct option: B**Final answer**

increase/decrease and extrema

Q056 MCQ

On a closed interval, an endpoint can be:**Solution**

1. Compare both endpoints with stationary values.

Correct option: B**Final answer**

an absolute extreme

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q057 MCQ

For $y = -x^3 + 12x + 5$, the stationary x-values are:

Solution

1. Solve $-3x^2 + 12 = 0$.

Correct option: A

Final answer

-2 and 2

Q058 MCQ

The absolute minimum of $x^3 - 6x$ on $[-3, 3]$ is at:

Solution

1. Compare all candidates; $f(-3) = -9$.

Correct option: A

Final answer

$x = -3$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q059 MCQ

To determine a cubic from an extreme value, use both $f(a) = M$ and:

Solution

1. An extreme point is stationary.

Correct option: B

Final answer

$$f'(a) = 0$$

Q060 MCQ

In the warm-up $f(x) = -x^3 + 12x - 7$, the local minimum occurs at:

Solution

1. The derivative changes from negative to positive at -2.

Correct option: A

Final answer

$$x = -2$$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q061 MCQ

For $f(x) = x^3 - 3kx^2 + kx + 2$, no two distinct stationary roots requires:

Solution

1. Use $D = 12k(3k - 1) \leq 0$.

Correct option: A

Final answer

$$0 \leq k \leq 1/3$$

Q062 MCQ

Two distinct stationary points occur when the derivative discriminant is:

Solution

1. A positive discriminant gives two distinct real derivative roots.

Correct option: C

Final answer

$$D > 0$$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q063 MCQ

For a positive-leading quadratic derivative, $D \leq 0$ implies it is:

Solution

1. The quadratic has no two distinct real zeros.

Correct option: A

Final answer

non-negative for all x

Q064 MCQ

In the 7-13 challenge, monotonicity is required on:

Solution

1. Use the operational interval.

Correct option: A

Final answer

[1,infinity)

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q065 MCQ

A cubic with both stationary values above the x-axis has:**Solution**

1. The graph crosses the x-axis once.

Correct option: B**Final answer**

one distinct real root

Q066 MCQ

To count cubic roots, compare the extrema with:**Solution**

1. Roots are x-axis intersections.

Correct option: B**Final answer**

the x-axis

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q067 MCQ

At an extremal parameter level, a cubic and a horizontal line have:

Solution

1. One contact is tangent and the other crossing remains.

Correct option: B

Final answer

two distinct intersections

Q068 MCQ

For $x^4 - 4x^3 + k$, the global minimum occurs at:

Solution

1. The derivative is $4x^2(x - 3)$.

Correct option: C

Final answer

$x = 3$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Quiz WRITTEN 1 Concept Quiz · Written

Find the local maximum and minimum of $x^3 - 3x^2 + 1$.

Solution

1. Use $f'(x) = 3x(x - 2)$ and a sign table.

Final answer

Maximum 1 at $x = 0$; minimum -3 at $x = 2$

Quiz WRITTEN 2 Concept Quiz · Written

Find a, b if $f(x) = -x^3 + ax + b$ has a local maximum 4 at $x=1$.

Solution

1. Use $f'(1) = 0$ and $f(1) = 4$.

Final answer

$a = 3, b = 2$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Quiz WRITTEN 3 Concept Quiz · Written

Find the k-range for $x^3 - 3kx^2 + kx$ to have no two distinct stationary points.

Solution

1. Set the derivative discriminant ≤ 0 .

Final answer

$$-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$$

Quiz WRITTEN 4 Concept Quiz · Written

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ on $[-1, 3]$.

Solution

1. Use the maximum of the k-free cubic on the interval.

Final answer

$$2 - 4\sqrt{2}$$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Quiz MCQ 5 Concept Quiz · MCQ

At a local minimum, f' changes:

Solution

1. Use the first derivative sign change.

Correct option: B

Final answer

$-to+$

Quiz MCQ 6 Concept Quiz · MCQ

A cubic with a positive-leading derivative and $D < 0$ is:

Solution

1. The derivative quadratic stays positive.

Correct option: A

Final answer

always increasing

Worked Solutions

Applications of Differentiation: Polynomial Functions

Quiz MCQ 7 [Concept Quiz · MCQ](#)

The 7-15 challenge requires the level to satisfy:

Solution

1. Three distinct intersections lie strictly between extrema.

Correct option: B

Final answer

$$0 < t < 4$$

Quiz MCQ 8 [Concept Quiz · MCQ](#)

The global minimum of $x^4 - 4x^3 + k$ is:

Solution

1. Evaluate at the quartic's global minimum $x = 3$.

Correct option: A

Final answer

$$k - 27$$

Answer key

Use the question number shown on each page.

- Q001** Use the derivative sign table: for $x^3 - 6x^2 + 16$, local maximum 16 at $x = 0$ and local minimum -16 at $x = 4$; classify the other two from their stationary points.
- Q002** Apply the same derivative sign-table test; a stationary point where the sign of f' does not change is not an extreme value.
- Q003** Differentiate each polynomial, solve the quadratic derivative, use the sign table, and evaluate f at every genuine stationary point.
- Q004** Local maximum at $x = a - 1$; local minimum at $x = a + 1$.
- Q005** Maximum 21 at $x = 2$; minimum -11 at $x = -2$.
- Q006** Evaluate both endpoints and every stationary point inside the interval; the greatest value is the absolute maximum and the least is the absolute minimum.
- Q007** For each function, compare the endpoint values with the values at all critical x-values lying in the interval.
- Q008** Absolute minimum -9 at $x = -3$; absolute maximum 9 at $x = 3$.
- Q009** $a = 12, b = -7$; local minimum -23 at $x = -2$.
- Q010** Use $f'(a) = 0, f(a) = M$, and the second stationary condition to form three simultaneous equations in the coefficients.
- Q011** Each set is determined by two derivative roots/conditions and one extreme-value condition; solve the resulting simultaneous equations.
- Q012** $a = -3, b = -24, c = 150$.
- Q013** $0 \leq k \leq \frac{1}{3}$.
- Q014** Use $D \leq 0$ for no extremes and $D > 0$ for two distinct stationary points.
- Q015** Apply the derivative-discriminant criterion separately to each printed cubic.
- Q016** $k < 0$ or $k > 4$.
- Q017** $-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$.
- Q018** Set the derivative discriminant $D \leq 0$, with the derivative leading coefficient positive.
- Q019** Differentiate each cubic and impose that its quadratic derivative never becomes negative.
- Q020** $k \leq 3$.
- Q021** Three distinct real roots.
- Q022** Use the cubic graph: count x-axis intersections from the stationary points and end behavior.
- Q023** Each answer is obtained from the number of x-axis intersections of its corresponding cubic graph.
- Q024** One distinct real root.
- Q025** $-27 < a < 5$.
- Q026** Rewrite each equation as $g(x) = a$, then compare the parameter line with the local maximum and minimum of g .
- Q027** Use the local extrema of the parameter-free cubic and count intersections with the horizontal parameter line.
- Q028** $\frac{7-\sqrt{65}}{4} < a < \frac{7-\sqrt{33}}{4}$ or $\frac{7+\sqrt{33}}{4} < a < \frac{7+\sqrt{65}}{4}$.
- Q029** Set $f(x) = \text{LHS} - \text{RHS}$, prove $f(x) \geq 0$ on the stated domain, and identify its minimum.
- Q030** Move the right-hand side to the left, then show the resulting auxiliary cubic has non-negative minimum on the domain.
- Q031** $k \leq 2 - 4\sqrt{2}$; the largest value is $2 - 4\sqrt{2}$.
- Q032** For $3x^4 + 4x^3 - 12x^2$: local maximum 0 at $x = 0$, local minima -32 at $x = -2$ and -5 at $x = 1$. For $-x^4 - 4x^3 - 8$: maximum 19 at $x = -3$; $x = 0$ is stationary but not an extreme.
- Q033** Differentiate each quartic, construct the derivative sign table, and evaluate each genuine turning point.
- Q034** $k < 27$.
- Q035** B
- Q036** A
- Q037** C
- Q038** B
- Q039** C
- Q040** B
- Q041** C
- Q042** B
- Q043** B
- Q044** B
- Q045** B
- Q046** B
- Q047** A
- Q048** A
- Q049** B
- Q050** A
- Q051** C
- Q052** B

Q053 B

Q054 A

Q055 B

Q056 B

Q057 A

Q058 A

Q059 B

Q060 A

Q061 A

Q062 C

Q063 A

Q064 A

Q065 B

Q066 B

Q067 B

Q068 C

Quiz WRITTEN 1 Maximum 1 at $x = 0$; minimum -3 at
 $x = 2$

Quiz WRITTEN 2 $a = 3, b = 2$ Quiz WRITTEN 3 $-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$ Quiz WRITTEN 4 $2 - 4\sqrt{2}$

Quiz MCQ 5 B

Quiz MCQ 6 A

Quiz MCQ 7 B

Quiz MCQ 8 A

Concept 4 | Integration: Definite Integrals and Applications

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Formula opener

Antiderivative

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Definite

$$\int_a^b f(x) dx = F(b) - F(a)$$

Area

$$A = \int_a^b |f(x)| dx$$

Worked Solutions

Integration: Definite Integrals and Applications

Q001 Written

Find the two indefinite integrals in the Warm Up.

Solution

1. Use $\int x^n dx = \frac{x^{n+1}}{n+1} + C$, then differentiate your result to check.

Final answer

Integrate term by term and add C to each primitive.

Q002 Written

Find the six indefinite integrals in the given Try section, expanding a squared bracket first when needed.

Solution

1. For $(x - 1)^2$, write $x^2 - 2x + 1$ before applying the power rule.

Final answer

Expand products or powers, collect like powers, then integrate every term.

Worked Solutions

Integration: Definite Integrals and Applications

Q003 Written

Solve the first ten given indefinite-integral exercises (constants, linear and quadratic polynomials).

Solution

1. A constant integrates to a constant times x ; combine the terms only after integration.

Final answer

Apply linearity and the power rule; retain the constant of integration.

Q004 Written

Solve the remaining eight given integrals, including products and squared brackets.

Solution

1. Do not integrate a product factor-by-factor; expand first, then use the power rule.

Final answer

Expand each product or square, simplify, and integrate term by term.

Worked Solutions

Integration: Definite Integrals and Applications

Q005 **Written****Find $F(x)$ when $F'(x) = 3x^2 + 8x - 1$ and $F(1) = 7$.****Solution**

1. Integrate $F'(x)$, substitute $x = 1$, and solve the resulting equation for C .

Final answer

$$F(x) = x^3 + 4x^2 - x + 3.$$

Q006 **Written****Find the two given Try functions from their derivatives and one given value of F .****Solution**

1. The initial value fixes the otherwise undetermined constant of integration.

Final answerIntegrate the derivative and use the given point to determine C .

Worked Solutions

Integration: Definite Integrals and Applications

Q007 **Written****Find $F(x)$ for all six given Exercise conditions.****Solution**

1. Keep the derivative and the initial condition as two separate equations before solving for C .

Final answerFor each item, integrate $F'(x)$, then impose the stated value of F .Q008 **Written****Find the curve $y = f(x)$ through $(3, 2)$ whose tangent gradient is $3x^2 - 2x + 1$.****Solution**

1. The gradient is $f'(x)$; integrate it and use $f(3) = 2$ to determine C .

Final answer

$$f(x) = x^3 - x^2 + x - 19.$$

Worked Solutions

Integration: Definite Integrals and Applications

Q009 Written

Solve the given Try tangent-gradient problem through $(1, 1)$.**Solution**

1. Replace the gradient by $f'(x)$, integrate, then substitute the known coordinate.

Final answer

Integrate the given gradient and use the point on the curve to determine the constant.

Q010 Written

Solve the three given tangent-gradient exercises, finding each function from its point and gradient.**Solution**

1. A point (x_0, y_0) gives $f(x_0) = y_0$, which fixes C .

Final answer

Integrate the polynomial gradient and apply the given point condition in each case.

Worked Solutions

Integration: Definite Integrals and Applications

Q011 Written

Evaluate the three definite integrals in the Warm Up using the Fundamental Theorem of Calculus.

Solution

1. Write $[F(x)]_a^b = F(b) - F(a)$ before substituting the limits.

Final answer

Find an antiderivative F and calculate $F(b) - F(a)$; equal limits give zero.

Q012 Written

Evaluate the six given Try definite integrals, simplifying the integrand before applying the limits.

Solution

1. Linearity allows sums and differences of integrals to be combined before integration.

Final answer

Expand products, integrate, and evaluate the antiderivative at the upper and lower limits.

Worked Solutions

Integration: Definite Integrals and Applications

Q013 Written

Evaluate the first twelve given Exercise definite integrals, including reversed and equal limits.

Solution

1. Check the order of the limits before evaluating the antiderivative.

Final answer

Use the Fundamental Theorem; a reversed interval changes the sign and equal limits give 0.

Q014 Written

Evaluate the remaining twelve given Exercise integrals, including sums, differences and products.

Solution

1. Combine only integrals with the same limits; otherwise split or evaluate separately.

Final answer

Use linearity, expand products, then apply the endpoint-substitution rule.

Worked Solutions

Integration: Definite Integrals and Applications

Q015 Written

Use the four properties of definite integrals to simplify the three Warm Up expressions.

Solution

1. Choose the property that makes the integrand or limits match a known form.

Final answer

Scale constants, reverse limits, split intervals, or use the factored-interval identity as appropriate.

Q016 Written

Solve the four given Try definite-integral transformations.

Solution

1. For $(x - a)(x - b)$, first make the coefficient of x equal to 1.

Final answer

Swap limits when necessary, join adjacent intervals, and normalize a linear factor before using the identity.

Worked Solutions

Integration: Definite Integrals and Applications

Q017 **Written****Solve the first six given Exercise questions using interval splitting and reversal.****Solution**

1. A minus sign before an integral can be absorbed by reversing its limits.

Final answer

Rewrite each expression until adjacent intervals have matching integrands and compatible limits.

Q018 **Written****Solve the final six given Exercise questions using the factorized definite-integral identity.****Solution**

1. Identify (a,b) from the two linear factors and account for every constant multiplier.

Final answerScale to $\int_a^b (x - a)(x - b) dx = -\frac{(b-a)^3}{6}$.

Worked Solutions

Integration: Definite Integrals and Applications

Q019 Written

Find $f(x)$ from the given equation $f(x) = x^2 + \int_0^1 f(t) dt$.

Solution

1. Replace the definite integral by a constant, substitute the resulting $f(t)$ back into it, and solve.

Final answer

Let $A = \int_0^1 f(t) dt$; then $A = \frac{1}{3} + A$, so $A = -\frac{1}{3}$ and $f(x) = x^2 - \frac{1}{3}$.

Q020 Written

Solve the two given Try integral equations for $f(x)$.

Solution

1. The unknown integral is a number independent of x ; determine it before writing the final function.

Final answer

Introduce a constant for the definite integral, evaluate it using the proposed $f(t)$, and solve the linear equation.

Worked Solutions

Integration: Definite Integrals and Applications

Q021 **Written****Find $f(x)$ for all five given integral equations.****Solution**

1. Check the final function by integrating it over the stated interval and substituting back.

Final answer

Let the definite integral be A , substitute the expression for $f(t)$, solve for A , then recover $f(x)$.

Q022 **Written****Differentiate $\int_3^x (2t^2 + 3t - 5) dt$ with respect to x .****Solution**

1. By the Fundamental Theorem, $\frac{d}{dx} \int_a^x f(t) dt = f(x)$.

Final answer

$2x^2 + 3x - 5$.

Worked Solutions

Integration: Definite Integrals and Applications

Q023 Written

Solve the two given Try problems involving differentiation of a variable-limit integral and an unknown lower limit.

Solution

1. A constant lower limit disappears under differentiation; it is recovered from the original equation.

Final answer

Replace t by x when differentiating, then use equal limits or the given condition to determine a .

Q024 Written

Differentiate the three given expressions with respect to x .

Solution

1. After differentiation, replace t with the variable upper limit x .

Final answer

Apply the variable-limit Fundamental Theorem directly to each integrand.

Worked Solutions

Integration: Definite Integrals and Applications

Q025 **Written****Find $f(x)$ and a for the three given equations with a variable upper limit.****Solution**

1. Use $0 = \int_a^x f(t) dt$ in the original equation after finding $f(x)$.

Final answer

Differentiate to get $f(x)$, then set the integral to zero at $x = a$ to solve for a .

Q026 **MCQ****The constant of integration is required because:****Solution**

1. Differentiation removes additive constants.

Correct option: B**Final answer**

primitives differ by a constant

Worked Solutions

Integration: Definite Integrals and Applications

Q027 MCQ

The integral of x^n for $n \neq -1$ is:**Solution**

1. Use the power rule for integration.

Correct option: B

Final answer

$$x^{(n+1)/(n+1)} + C$$

Q028 MCQ

Before integrating $(x - 1)^2$, first:**Solution**1. Write $x^2 - 2x + 1$.

Correct option: B

Final answer

expand it

Worked Solutions

Integration: Definite Integrals and Applications

Q029 MCQ

The integral of $3x^2 - 2x + 1$ is:**Solution**

1. Integrate term by term.

Correct option: A**Final answer**

$$x^3 - x^2 + x + C$$

Q030 MCQ

To find $F(x)$ from $F'(x)$, you should first:**Solution**

1. Integration reverses differentiation.

Correct option: B**Final answer**integrate $F'x$

Worked Solutions

Integration: Definite Integrals and Applications

Q031 MCQ

The condition $F(1) = 7$ determines:**Solution**

1. Use the given function value after integrating.

Correct option: C

Final answer

C

Q032 MCQ

If $F'(x) = 6x - 2$ and $F(0) = 1$, then $F(x)$ is:**Solution**1. Integrate and use $F(0) = 1$.

Correct option: A

Final answer $3x^2 - 2x + 1$

Worked Solutions

Integration: Definite Integrals and Applications

Q033 MCQ

The gradient of the tangent at (x, y) is:**Solution**

1. By definition, the derivative is the tangent gradient.

Correct option: B**Final answer**

$$f'x$$

Q034 MCQ

A point (x_0, y_0) on $y = f(x)$ gives:**Solution**

1. Use the point as an initial condition.

Correct option: B**Final answer**

$$f(x_0) = y_0$$

Worked Solutions

Integration: Definite Integrals and Applications

Q035 MCQ

If $f'(x) = 3x^2 - 2x + 1$, an antiderivative is:

Solution

1. Integrate each term.

Correct option: B

Final answer

$$x^3 - x^2 + x + C$$

Q036 MCQ

The Fundamental Theorem gives $\int_a^b f(x)dx =$:

Solution

1. Evaluate an antiderivative at the upper minus lower limit.

Correct option: B

Final answer

$$Fb - Fa$$

Worked Solutions

Integration: Definite Integrals and Applications

Q037 MCQ

When the limits are equal, $\int_a^a f(x)dx$ equals:**Solution**

1. There is no interval of accumulation.

Correct option: C**Final answer**

0

Q038 MCQ

A reversed definite integral satisfies:**Solution**

1. Reversing the limits changes the sign.

Correct option: B**Final answer**

negative reversed integral

Worked Solutions

Integration: Definite Integrals and Applications

Q039 MCQ

To evaluate a definite integral of a product, first:**Solution**

1. Then apply the power rule and endpoint substitution.

Correct option: B

Final answer

expand the product

Q040 MCQ

Adjacent integrals with the same integrand can be:**Solution**

1. Use interval additivity.

Correct option: B

Final answer

combined over the joined interval

Worked Solutions

Integration: Definite Integrals and Applications

Q041 MCQ

To use $\int_{\alpha}^{\beta} (x - \alpha)(x - \beta)dx$, the factors must have:

Solution

1. Match each factor to its endpoint.

Correct option: B

Final answer

roots a and b

Q042 MCQ

The identity for the factored integral is:

Solution

1. Use the signed cubic interval length.

Correct option: B

Final answer

$-(b - a)^3/6$

Worked Solutions

Integration: Definite Integrals and Applications

Q043 MCQ

A minus sign before an integral can be removed by:

Solution

1. $-\int_a^b f = \int_b^a f.$

Correct option: B

Final answer

reversing its limits

Q044 MCQ

In $f(x) = p(x) + \int_a^b f(t)dt$, the definite integral is:

Solution

1. The limits contain no x .

Correct option: B

Final answer

a constant

Worked Solutions

Integration: Definite Integrals and Applications

Q045 MCQ

The best first step for an integral equation is to let:**Solution**

1. Replace the repeated definite integral by one symbol.

Correct option: B**Final answer** $A = \text{integral}$

Q046 MCQ

For the warm-up $f(x) = x^2 + \int_0^1 f(t)dt$, the constant integral is:**Solution**

1. Solve $A = 1/3 + A$ after substitution.

Correct option: B**Final answer** $-1/3$

Worked Solutions

Integration: Definite Integrals and Applications

Q047 MCQ

$\frac{d}{dx} \int_a^x f(t)dt$ equals:

Solution

1. Apply the variable-upper-limit theorem.

Correct option: B

Final answer $f x$

Q048 MCQ

When differentiating an integral, the dummy variable t is replaced by:

Solution

1. The derivative evaluates the integrand at the moving limit.

Correct option: B

Final answer

upper-limit variable

Worked Solutions

Integration: Definite Integrals and Applications

Q049 MCQ

If $\int_a^a f(t)dt = 0$, this can determine:

Solution

1. Substitute the equal limits into the original equation.

Correct option: B

Final answer

the unknown lower limit a

Q050 MCQ

For $\int_3^x (2t^2 + 3t - 5)dt$, the derivative is:

Solution

1. Use the Fundamental Theorem directly.

Correct option: A

Final answer

$2x^2 + 3x - 5$

Worked Solutions

Integration: Definite Integrals and Applications

Quiz WRITTEN 1 Concept Quiz · Written

Find $\int (4x^3 - 6x + 2)dx$.

Solution

1. Apply the power rule term by term.

Final answer

$$x^4 - 3x^2 + 2x + C$$

Quiz WRITTEN 2 Concept Quiz · Written

Find $F(x)$ if $F'(x) = 2x - 5$ and $F(2) = 1$.

Solution

1. Integrate and use $F(2) = 1$.

Final answer

$$F(x) = x^2 - 5x + 7$$

Worked Solutions

Integration: Definite Integrals and Applications

Quiz WRITTEN 3 Concept Quiz · Written

Evaluate $\int_{-1}^2 (3x^2 - 4x + 1)dx$.

Solution

1. Use an antiderivative and subtract the endpoint values.

Final answer

6

Quiz WRITTEN 4 Concept Quiz · Written

Find $f(x)$ if $f(x) = x^2 + \int_0^1 f(t)dt$.

Solution

1. Let $A = \int_0^1 f(t)dt$; solve $A = 1/3 + A$ after substitution.

Final answer

$f(x) = x^2 - 1/3$

Worked Solutions

Integration: Definite Integrals and Applications

Quiz MCQ 5 [Concept Quiz · MCQ](#)

The derivative of $\int_0^x (3t^2 + 1)dt$ is:

Solution

1. Evaluate the integrand at $t = x$.

Correct option: A

Final answer

$$3x^2 + 1$$

Quiz MCQ 6 [Concept Quiz · MCQ](#)

A primitive of $5x^4$ is:

Solution

1. Divide the power coefficient by the new exponent.

Correct option: A

Final answer

$$x^5 + C$$

Worked Solutions

Integration: Definite Integrals and Applications

Quiz MCQ 7 **Concept Quiz · MCQ**

For $F'(x) = 4x + 2$ and $F(0) = 3$, $F(x)$ is:

Solution

1. Integrate and apply $F(0) = 3$.

Correct option: A

Final answer

$$2x^2 + 2x + 3$$

Quiz MCQ 8 **Concept Quiz · MCQ**

$\int_1^3 f(x)dx + \int_3^5 f(x)dx$ equals:

Solution

1. Join adjacent intervals with the same integrand.

Correct option: A

Final answer

integral from 1 to 5

Answer key

Use the question number shown on each page.

- Q001 Integrate term by term and add C to each primitive. Q012 Expand products, integrate, and evaluate the antiderivative at the upper and lower limits. Q020 Introduce a constant for the definite integral, evaluate it using the proposed $f(t)$, and solve the linear equation. Q036 B
- Q002 Expand products or powers, collect like powers, then integrate every term. Q013 Use the Fundamental Theorem; a reversed interval changes the sign and equal limits give 0. Q021 Let the definite integral be A , substitute the expression for $f(t)$, solve for A , then recover $f(x)$. Q037 C
- Q003 Apply linearity and the power rule; retain the constant of integration. Q014 Use linearity, expand products, then apply the endpoint-substitution rule. Q022 $2x^2 + 3x - 5$. Q038 B
- Q004 Expand each product or square, simplify, and integrate term by term. Q015 Scale constants, reverse limits, split intervals, or use the factored-interval identity as appropriate. Q023 Replace t by x when differentiating, then use equal limits or the given condition to determine a . Q039 B
- Q005 $F(x) = x^3 + 4x^2 - x + 3$. Q016 Swap limits when necessary, join adjacent intervals, and normalize a linear factor before using the identity. Q024 Apply the variable-limit Fundamental Theorem directly to each integrand. Q040 B
- Q006 Integrate the derivative and use the given point to determine C . Q017 Rewrite each expression until adjacent intervals have matching integrands and compatible limits. Q025 Differentiate to get $f(x)$, then set the integral to zero at $x = a$ to solve for a . Q041 B
- Q007 For each item, integrate $F'(x)$, then impose the stated value of F . Q018 Scale to $\int_a^b (x-a)(x-b) dx = -\frac{(b-a)^3}{6}$. Q026 B
- Q008 $f(x) = x^3 - x^2 + x - 19$. Q019 Let $A = \int_0^1 f(t) dt$; then $A = \frac{1}{3} + A$, so $A = -\frac{1}{3}$ and $f(x) = x^2 - \frac{1}{3}$. Q027 B
- Q009 Integrate the given gradient and use the point on the curve to determine the constant. Q028 B
- Q010 Integrate the polynomial gradient and apply the given point condition in each case. Q029 A
- Q011 Find an antiderivative F and calculate $F(b) - F(a)$; equal limits give zero. Q030 B
- Q031 C
- Q032 A
- Q033 B
- Q034 B
- Q035 B
- Q042 B
- Q043 B
- Q044 B
- Q045 B
- Q046 B
- Q047 B
- Q048 B
- Q049 B
- Q050 A
- Quiz WRITTEN 1 $x^4 - 3x^2 + 2x + C$
- Quiz WRITTEN 2 $F(x) = x^2 - 5x + 7$
- Quiz WRITTEN 3 6
- Quiz WRITTEN 4 $f(x) = x^2 - 1/3$
- Quiz MCQ 5 A
- Quiz MCQ 6 A
- Quiz MCQ 7 A
- Quiz MCQ 8 A

Concept 5 | Applications of Integration to Geometry

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Formula opener

Area

$$A = \int_a^b (y_{top} - y_{bottom}) dx$$

Volume

$$V = \pi \int_a^b y^2 dx$$

Bounds

$$a \leq x \leq b$$

Worked Solutions

Applications of Integration to Geometry

Q001 Written

Find the area bounded by $y = 2x^2 + 1$, the x -axis, $x = -2$, and $x = 1$.

Solution

1. The parabola is above the x -axis on the interval.
2. $S = \int_{-2}^1 (2x^2 + 1) dx = 9$.

Final answer

9

Q002 Written

Find the area bounded by $y = x^2 + x - 6$, the x -axis, $x = -2$, and $x = 1$.

Solution

1. The curve is below the x -axis on $[-2, 1]$.
2. Use $S = -\int_{-2}^1 (x^2 + x - 6) dx = \frac{33}{2}$.

Final answer

 $\frac{33}{2}$

Worked Solutions

Applications of Integration to Geometry

Q003 Written

Find the area bounded by $y = -x^2 + 4$ and the x -axis.**Solution**

1. The intercepts are $x = -2, 2$.
2. Integrate the positive curve from -2 to 2 .

Final answer

$$\frac{32}{3}$$

Q004 Written

Find the area under $y = -x^2 - 2x + 3$ and above the x -axis.**Solution**

1. The roots are -3 and 1 .
2. Integrate $-x^2 - 2x + 3$ between the roots.

Final answer

$$\frac{32}{3}$$

Worked Solutions

Applications of Integration to Geometry

Q005 Written

Find the area between $y = x^2 - 2$ and $y = -2x + 1$.**Solution**

1. Intersections give $x = -3, 1$; the line is upper.
2. Integrate upper minus lower over $[-3, 1]$.

Final answer

$$\frac{32}{3}$$

Q006 Written

Find the area between $y = x^2 + 2x - 4$ and $y = -x^2 + 2x + 4$.**Solution**

1. The intersections are $x = -2, 2$.
2. Use the upper-minus-lower integral $\int_{-2}^2 (8 - 2x^2) dx$.

Final answer

$$\frac{64}{3}$$

Worked Solutions

Applications of Integration to Geometry

Q007 Written

Find the area between $y = -x^2 + 4x$ and $y = x$.**Solution**

1. Intersections are $x = 0, 3$.
2. Integrate $-x^2 + 3x$ from 0 to 3.

Final answer

$$\frac{9}{2}$$

Q008 Written

Find the area between $y = -x^2 + 8$ and $y = 2x$ for $-2 \leq x \leq 3$.**Solution**

1. The curves meet at $x = 2$ within the interval.
2. Split the absolute difference at $x = 2$; the two areas are $80/3$ and $10/3$.

Final answer

$$30$$

Worked Solutions

Applications of Integration to Geometry

Q009 Written

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x -axis.**Solution**

1. Roots are (0,2,3); the curve changes sign at 2.
2. Add the positive area on $[0, 2]$ to the absolute area on $[2, 3]$.

Final answer

$$\frac{37}{12}$$

Q010 Written

Find the area bounded by $y = -x^3 + 4x$ and the x -axis.**Solution**

1. Roots are (-2,0,2); the two lobes have equal area.
2. Double $\int_0^2 (-x^3 + 4x) dx = 4$.

Final answer

$$8$$

Worked Solutions

Applications of Integration to Geometry

Q011 Written

Solve the given cubic area exercises by finding every x -intercept and adding the absolute signed pieces.

Solution

1. Factor the cubic first.
2. Mark the sign on each interval before integrating.

Final answer

Use the numbered solution for each given cubic.

Q012 Written

Evaluate $\int_{-2}^1 |x + 1| dx$.

Solution

1. The sign changes at $x = -1$.
2. Split into two ordinary integrals and add the positive areas.

Final answer

$$\frac{5}{2}$$

Worked Solutions

Applications of Integration to Geometry

Q013 Written

Evaluate $\int_0^3 |x^2 - 4| dx$.**Solution**

1. The root in the interval is $x = 2$.
2. Use $-\int_0^2 (x^2 - 4) dx + \int_2^3 (x^2 - 4) dx$.

Final answer

$$\frac{23}{3}$$

Q014 Written

Evaluate $\int_0^2 |x^2 + 3x - 4| dx$.**Solution**

1. The relevant root is $x = 1$.
2. Reverse the sign on $[0, 1]$, then add the integral on $[1, 2]$.

Final answer

$$5$$

Worked Solutions

Applications of Integration to Geometry

Q015 **Written****Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.****Solution**

1. The tangents are $y = -x$ and $y = 3x - 4$, meeting at $x = 1$.
2. Split the area at $x = 1$ and integrate curve minus tangent.

Final answer

$$\frac{3}{2}$$

Q016 **Written****Find the area bounded by $y = x^3 + x^2 - 3x + 6$ and its tangent at $(1, 5)$.****Solution**

1. The tangent is $y = 2x + 3$; the intersections are $x = -3, 1$.
2. Factor the difference as $(x - 1)^2(x + 3)$ and integrate.

Final answer

$$\frac{64}{3}$$

Worked Solutions

Applications of Integration to Geometry

Q017 Written

For $y = x^2$, tangents through $(3, 8)$ touch at $x = 2, 4$. Find the enclosed area.

Solution

1. The tangent equations are $y = 4x - 4$ and $y = 8x - 16$.
2. Integrate the two squared gaps on $[2, 3]$ and $[3, 4]$.

Final answer

$$\frac{2}{3}$$

Q018 Written

If $y = kx$ bisects the area under $y = 2x - x^2$ and above the x -axis, find k .

Solution

1. The total area under the parabola is 3.
2. Set the area cut off by $y = kx$ equal to $3/2$, giving $(2 - k)^3 = 3/4$.

Final answer

$$k = 2 - \sqrt[3]{\frac{3}{4}}$$

Worked Solutions

Applications of Integration to Geometry

Q019 **Written**

Solve the given bisection exercises for the given parabolas and lines, using the required area ratio.

Solution

1. Find the total parabola area first.
2. Integrate between the line and curve, then impose the stated ratio.

Final answer

Use the numbered solution for each given ratio.

Q020 **MCQ**

When the graph is below the x-axis, area is:

Solution

1. Reverse the sign.

Correct option: B

Final answer

negative of the integral

Worked Solutions

Applications of Integration to Geometry

Q021 MCQ

A bounded area under a curve is found by first checking:**Solution**

1. Draw the graph.

Correct option: A**Final answer**

the graph and sign

Q022 MCQ

The area under $y = -x^2 + 4$ between its roots is:**Solution**

1. Integrate from -2 to 2.

Correct option: C**Final answer**

32/3

Worked Solutions

Applications of Integration to Geometry

Q023 MCQ

Between two curves, the integrand is:**Solution**

1. Identify the upper graph.

Correct option: B

Final answer

upper minus lower

Q024 MCQ

If curves cross inside the interval, area must be:**Solution**

1. The sign of the difference changes.

Correct option: A

Final answer

split at the crossing

Worked Solutions

Applications of Integration to Geometry

Q025 MCQ

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Solution

1. Use the intersection limits.

Correct option: B

Final answer

32/3

Q026 MCQ

For cubic area, the sign changes at:

Solution

1. Mark all roots.

Correct option: B

Final answer

each x-intercept

Worked Solutions

Applications of Integration to Geometry

Q027 MCQ

The area for $y = -x^3 + 4x$ is:**Solution**

1. The two lobes are equal.

Correct option: C

Final answer

8

Q028 MCQ

A negative lobe contributes:**Solution**

1. Area is non-negative.

Correct option: B

Final answer

positive absolute area

Worked Solutions

Applications of Integration to Geometry

Q029 MCQ

For an absolute-value integral, first locate:**Solution**

1. Roots set the sign intervals.

Correct option: A**Final answer**

the roots

Q030 MCQ

The value of $\int_0^2 |x^2 + 3x - 4| dx$ is:**Solution**

1. Split at $x = 1$.

Correct option: C**Final answer**

5

Worked Solutions

Applications of Integration to Geometry

Q031 MCQ

If f changes sign at c , split the integral:**Solution**

1. The absolute value changes form.

Correct option: A

Final answer atc

Q032 MCQ

A tangent line at $x=a$ has slope:**Solution**

1. Differentiate the curve.

Correct option: B

Final answer $f'a$

Worked Solutions

Applications of Integration to Geometry

Q033 MCQ

For two tangent lines, split at their:**Solution**

1. The upper tangent changes there.

Correct option: A**Final answer**

intersection x-coordinate

Q034 MCQ

The area in the given parabola-tangent Warm Up is:**Solution**

1. Integrate the two gaps.

Correct option: C**Final answer** $\frac{3}{2}$

Worked Solutions

Applications of Integration to Geometry

Q035 MCQ

Bisection means each part has:**Solution**

1. Set each part to half the total.

Correct option: A**Final answer**

the same area

Q036 MCQ

For $y=2x-x^2$, the total area above the x-axis is:**Solution**

1. Integrate from 0 to 2.

Correct option: C**Final answer**

3

Worked Solutions

Applications of Integration to Geometry

Q037 MCQ

The line $y=kx$ must satisfy the stated:**Solution**

1. Use the given bisection condition.

Correct option: A**Final answer**

area ratio

Q038 MCQ

In a bisection problem, the cut-off area is:**Solution**

1. Set the two parts equal.

Correct option: B**Final answer**

half the total area

Worked Solutions

Applications of Integration to Geometry

Quiz MCQ 1 [Concept Quiz · MCQ](#)

The area under $y = -x^2 + 4$ and above the x -axis is:

Solution

1. Integrate from -2 to 2 .

Correct option: C

Final answer

$$\frac{32}{3}$$

Quiz MCQ 2 [Concept Quiz · MCQ](#)

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Solution

1. Use upper minus lower from -3 to 1 .

Correct option: B

Final answer

$$\frac{32}{3}$$

Worked Solutions

Applications of Integration to Geometry

Quiz MCQ 3 [Concept Quiz · MCQ](#)

Evaluate $\int_{-2}^1 |x + 1| dx$:

Solution

1. Split at $x = -1$.

Correct option: B

Final answer

$$\frac{5}{2}$$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

The area enclosed by $y = x^2$ and tangents through $(3, 8)$ is:

Solution

1. Integrate the two squared gaps.

Correct option: C

Final answer

$$\frac{2}{3}$$

Worked Solutions

Applications of Integration to Geometry

Quiz WRITTEN 5 Concept Quiz · Written

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x -axis.

Solution

1. Use roots (0,2,3) and split at 2.

Final answer

$$\frac{37}{12}$$

Quiz WRITTEN 6 Concept Quiz · Written

Evaluate $\int_0^2 |x^2 + 3x - 4| dx$.

Solution

1. Split at the root $x = 1$.

Final answer

$$5$$

Worked Solutions

Applications of Integration to Geometry

Quiz WRITTEN 7 Concept Quiz · Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Solution

1. Find both tangents and split at their intersection.

Final answer

$$\frac{3}{2}$$

Quiz WRITTEN 8 Concept Quiz · Written

If $y = kx$ bisects the area under $y = 2x - x^2$, find k .

Solution

1. Set the cut-off area equal to half of 3.

Final answer

$$2 - \sqrt[3]{\frac{3}{4}}$$

Answer key

Use the question number shown on each page.

Q001 9

Q002 $\frac{33}{2}$

Q003 $\frac{32}{3}$

Q004 $\frac{32}{3}$

Q005 $\frac{32}{3}$

Q006 $\frac{64}{3}$

Q007 $\frac{9}{2}$

Q008 30

Q009 $\frac{37}{12}$

Q010 8

Q011 Use the numbered solution for each given cubic.

Q012 $\frac{5}{2}$

Q013 $\frac{23}{3}$

Q014 5

Q015 $\frac{3}{2}$

Q016 $\frac{64}{3}$

Q017 $\frac{2}{3}$

Q018 $k = 2 - \sqrt[3]{\frac{3}{4}}$

Q019 Use the numbered solution for each given ratio.

Q020 B

Q021 A

Q022 C

Q023 B

Q024 A

Q025 B

Q026 B

Q027 C

Q028 B

Q029 A

Q030 C

Q031 A

Q032 B

Q033 A

Q034 C

Q035 A

Q036 C

Q037 A

Q038 B

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 C

Quiz WRITTEN 5 $\frac{37}{12}$

Quiz WRITTEN 6 5

Quiz WRITTEN 7 $\frac{3}{2}$

Quiz WRITTEN 8 $2 - \sqrt[3]{\frac{3}{4}}$

Concept 1 | Fundamentals of Random Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Expectation

$$E(X) = \sum x p(x)$$

Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Probability

$$\sum p(x) = 1$$

Worked Solutions

Fundamentals of Random Variables

Q001 Written

For the sum of two dice, construct the probability distribution and find $P5 \leq X \leq 8$.

Solution

1. List the 36 equally likely pairs, count each sum, and divide each frequency by 36.

Final answer

Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; $P5 \leq X \leq 8 = 20/36 = 5/9$.

Q002 Written

Find the probability distributions for the absolute difference of two dice and the number of white balls drawn from the stated bag; evaluate the requested interval probabilities.

Solution

1. Check that all table probabilities add to 1 before evaluating an interval event.

Final answer

Use the sample space or combinations to list every possible X and sum the relevant probabilities.

Worked Solutions

Fundamentals of Random Variables

Q003 Written

Construct both given probability tables (four coins; three balls from 4 red and 2 blue) and evaluate the given ranges.

Solution

1. A probability distribution is a table of every possible value of X and its probability.

Final answer

Use binomial or hypergeometric counts, then add probabilities for the requested values.

Q004 Written

Find EX for tails in four coins and for red balls in a draw of three from a bag with 4 red and 5 white.

Solution

1. Build the table, multiply each value by its probability, and add the products.

Final answer

For four fair coins $EX=2$; for the hypergeometric draw $EX=4/3$.

Worked Solutions

Fundamentals of Random Variables

Q005 Written

Find expected values for tails in three coins, the absolute difference of two dice, and white balls in the stated bag.

Solution

1. Keep the random variable definition unchanged while forming the table.

Final answer

Use $EX = \sum xPX = x$ after constructing each distribution.

Q006 Written

Find EX for all six given random-variable experiments (cards, dice, balls, tickets, coins and repeated die rolls).

Solution

1. The expected number of successes in repeated identical trials can be checked with n times p .

Final answer

Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice.

Worked Solutions**Fundamentals of Random Variables****Q007** Written**For tails in four fair coins, find $E(3X+2)$, $E(-4X+1)$, and EX^2 .****Solution**

1. Constants shift the mean; a multiplier scales the mean by the same factor.

Final answer

Use $E(aX + b) = aEX + b$; calculate EX^2 directly from the table.

Q008 Written**For white balls drawn from 4 red and 3 white, find the four requested transformed expected values.****Solution**

1. Do not replace EX^2 by EX^2 ; they are different quantities.

Final answer

Apply linearity for $aX+b$ and use the weighted square sum for X^2 .

Worked Solutions

Fundamentals of Random Variables

Q009 Written

Solve both given Exercise groups of transformed expectations for coins and for balls.**Solution**

1. Write the transformation before substituting the known mean.

Final answer

For every expression use $E(aX + b) = aEX + b$; evaluate powers from the distribution table.

Q010 Written

For black balls drawn from the stated box, find EX and VX .**Solution**

1. The distribution is $PX = 0 = 2/7$, $PX = 1 = 4/7$, $PX = 2 = 1/7$. Hence $EX = 6/7$ and $VX = EX^2 - (EX)^2 = 20/49$.

Final answer

$EX = 6/7$; $VX = 20/49$.

Worked Solutions

Fundamentals of Random Variables

Q011 Written

Find the mean and variance for the two given Try random variables (a numbered card and winning tickets).**Solution**

1. Variance measures spread around the mean, not the mean itself.

Final answerConstruct each probability table, then use $VX = EX^2 - EX^2$.

Q012 Written

Find EX and VX for all five given Exercise random variables.**Solution**

1. For sampling without replacement, use combinations to obtain the exact probabilities.

Final answer

Use the distribution table for each experiment and the variance identity.

Worked Solutions

Fundamentals of Random Variables

Q013 Written

Find the standard deviation of tails when two biased coins have $P(\text{head})=4/7$ and $P(\text{tail})=3/7$.

Solution

1. Find VX first, then take the positive square root.

Final answer

The table gives $VX=24/49$, so $\sigma X=2\sqrt{6}/7$.

Q014 Written

Find the standard deviation from each given probability table and for white balls drawn from 4 red and 3 white.

Solution

1. Standard deviation has the same units as X and is always non-negative.

Final answer

Use $\sigma X=\sqrt{V(X)}$ after evaluating the variance from the table.

Worked Solutions

Fundamentals of Random Variables

Q015 Written

Find the standard deviation for both given probability-table exercises.

Solution

1. Keep the positive square root only.

Final answer

Compute EX , EX^2 , then $\sigma_X = \sqrt{E(X^2) - E(X)^2}$.

Q016 Written

For the number of winning tickets in the given draw, find VY and σ_Y for $Y=2X+1$ and $Y=-3X-2$.

Solution

1. The constant b changes neither variance nor standard deviation.

Final answer

Use $V(aX + b) = a^2 VX$ and $\sigma(aX+b)=|a|\sigma_X$.

Worked Solutions

Fundamentals of Random Variables

Q017 **Written**

For red balls drawn from the stated bag, find variance and standard deviation for the three linear transformations of X .

Solution

1. Write both transformation rules before substituting numbers.

Final answer

Apply the squared multiplier to VX and the absolute multiplier to σX .

Q018 **Written**

For a single die, find VY and σY for $Y=X+2$, $Y=3X+1$ and $Y=-2X+4$.

Solution

1. A translation changes the mean but leaves spread unchanged.

Final answer

Use the variance and standard-deviation transformation formulas for each Y .

Worked Solutions

Fundamentals of Random Variables

Q019 Written

Solve the two given transformation problems: a net prize after an entry fee, and finding a, b from $E(aX+b)=7/3$ and $\sigma(aX+b)=3$.

Solution

1. The mean equation determines b after the standard-deviation equation fixes a .

Final answer

Use $E(aX + b) = aEX + b$ and $\sigma(aX+b)=|a|\sigma X$, with $a>0$.

Q020 Written

Solve the two given Try problems involving net winnings and determining a, b for a die transformation.

Solution

1. Entry fees are negative constants; they shift the mean only.

Final answer

Model the payout as $Y = aX + b$, then apply the mean and standard-deviation rules.

Worked Solutions

Fundamentals of Random Variables

Q021 **Written**

Solve all four given transformation exercises, including the number-line coin walk, games, and the calibrated die model.

Solution

1. Translate the words into $aX+b$ before calculating.

Final answer

Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule.

Q022 **MCQ**

A probability distribution must have probabilities that sum to:

Solution

1. Check the table total.

Correct option: B

Final answer

1

Worked Solutions

Fundamentals of Random Variables

Q023 MCQ

For two fair dice, $PX = 2$ is:**Solution**

1. Only the pair (1,1) gives sum 2.

Correct option: B

Final answer

1/36

Q024 MCQ

The number of heads in three coins can be:**Solution**

1. Count heads in the sample space.

Correct option: B

Final answer

0, 1, 2, 3

Worked Solutions

Fundamentals of Random Variables

Q025 MCQ

The expected value is calculated as:**Solution**

1. Multiply each value by its probability.

Correct option: B**Final answer**

$$\sum xPX = x$$

Q026 MCQ

The mean number of tails in four fair coins is:**Solution**

1. Use symmetry or the table.

Correct option: B**Final answer**

2

Worked Solutions

Fundamentals of Random Variables

Q027 MCQ

For a fair die, EX is:**Solution**

1. The values 1 to 6 average to 3.5.

Correct option: B

Final answer

3.5

Q028 MCQ

 $E(aX+b)$ equals:**Solution**

1. Use linearity of expectation.

Correct option: B

Final answer $aEX + b$

Worked Solutions

Fundamentals of Random Variables

Q029 MCQ

In general, EX^2 is:**Solution**

1. Square values before weighting.

Correct option: C**Final answer**

weighted square sum

Q030 MCQ

If $EX=2$, then $E(3X+2)$ is:**Solution**

1. Calculate $3 \times 2 + 2$.

Correct option: C**Final answer**

8

Worked Solutions

Fundamentals of Random Variables

Q031 MCQ

The variance identity is:**Solution**

1. Subtract the squared mean.

Correct option: B

Final answer

$$EX^2 - EX^2$$

Q032 MCQ

Variance can never be:**Solution**

1. It is an expected square.

Correct option: C

Final answer

negative

Worked Solutions

Fundamentals of Random Variables

Q033 MCQ

The first step in a variance problem is to:**Solution**

1. The table supplies EX and EX^2 .

Correct option: A**Final answer**

draw the distribution

Q034 MCQ

Standard deviation is:**Solution**

1. Take the positive square root.

Correct option: B**Final answer** $\sqrt{V(X)}$

Worked Solutions

Fundamentals of Random Variables

Q035 MCQ

Standard deviation has units:**Solution**

1. The square root restores the units.

Correct option: B

Final answer

same units as X

Q036 MCQ

 $V(aX+b)$ equals:**Solution**

1. Constants do not affect spread.

Correct option: B

Final answer

a squared VX

Worked Solutions

Fundamentals of Random Variables

Q037 MCQ

 $\sigma(aX+b)$ equals:**Solution**

1. Standard deviation is non-negative.

Correct option: B**Final answer** $|a| \sigma X$

Q038 MCQ

For $Y=X+3$, the variance is:**Solution**

1. A translation leaves variance unchanged.

Correct option: B**Final answer** VX

Worked Solutions

Fundamentals of Random Variables

Q039 MCQ

A net prize after a fixed fee is modeled as:**Solution**1. The fee is the constant b .

Correct option: B

Final answer

$$aX + b$$

Q040 MCQ

If $a > 0$ and $\sigma(aX + b) = 3$, then a is:**Solution**1. Solve $a \sigma X = 3$.

Correct option: B

Final answer

$$3/\sigma X$$

Worked Solutions

Fundamentals of Random Variables

Q041 MCQ

Changing b in $aX+b$ changes:**Solution**

1. b is a translation.

Correct option: C**Final answer**

the mean only

Q042 MCQ

If $EX=2$ and $EX^2=7$, then VX is:**Solution**

1. Calculate $7 - 2^2$.

Correct option: A**Final answer**

3

Worked Solutions

Fundamentals of Random Variables

Quiz WRITTEN 1 Concept Quiz · Written

Find EX and VX for the number of heads in three fair coins.

Solution

1. Use the binomial distribution.

Final answer

$EX=3/2$ and $VX=3/4$

Quiz WRITTEN 2 Concept Quiz · Written

For $Y=4X-1$ with $VX=2$, find VY and σY .

Solution

1. Use $V(aX + b) = a^2 VX$.

Final answer

$VY=32$ and $\sigma Y=4 \sqrt{2}$

Worked Solutions

Fundamentals of Random Variables

Quiz WRITTEN 3 Concept Quiz · Written

For a fair die, find $E(2X+5)$.

Solution

1. $EX=7/2$, then use linearity.

Final answer

12

Quiz WRITTEN 4 Concept Quiz · Written

If $\sigma X=2$ and $EX=3$, find $\sigma(-5X+7)$.

Solution

1. Multiply standard deviation by $|a|$.

Final answer

10

Worked Solutions

Fundamentals of Random Variables

Quiz MCQ 5 [Concept Quiz · MCQ](#)

If $VX=5$, $V(3X+1)$ is:

Solution

1. Square the multiplier.

Correct option: D

Final answer

45

Quiz MCQ 6 [Concept Quiz · MCQ](#)

For a fair die EX is:

Solution

1. Average 1 through 6.

Correct option: C

Final answer

3.5

Worked Solutions

Fundamentals of Random Variables

Quiz MCQ 7 [Concept Quiz · MCQ](#)

A probability table total must be:

Solution

1. Add all probabilities.

Correct option: B

Final answer

1

Quiz MCQ 8 [Concept Quiz · MCQ](#)

If $\sigma X=4$, $\sigma(X-9)$ is:

Solution

1. Translations do not change spread.

Correct option: B

Final answer

4

Answer key

Use the question number shown on each page.

- | | | | |
|---|---|---|--|
| Q001 Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; $P5 \leq X \leq 8 = 20/36 = 5/9$. | Q009 For every expression use $E(aX + b) = aEX + b$; evaluate powers from the distribution table. | Q019 Use $E(aX + b) = aEX + b$ and $\sigma(aX + b) = a \sigma X$, with $a > 0$. | Q034 B |
| Q002 Use the sample space or combination to list every possible X and sum the relevant probabilities. | Q010 $EX = 6/7$; $VX = 20/49$. | Q020 Model the payout as $Y = aX + b$, then apply the mean and standard-deviation rules. | Q035 B |
| Q003 Use binomial or hypergeometric counts, then add probabilities for the requested values. | Q011 Construct each probability table, then use $VX = EX^2 - EX^2$. | Q021 Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule. | Q036 B |
| Q004 For four fair coins $EX = 2$; for the hypergeometric draw $EX = 4/3$. | Q012 Use the distribution table for each experiment and the variance identity. | Q022 B | Q037 B |
| Q005 Use $EX = \sum xPX = x$ after constructing each distribution. | Q013 The table gives $VX = 24/49$, so $\sigma X = 2\sqrt{6/7}$. | Q023 B | Q038 B |
| Q006 Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice. | Q014 Use $\sigma X = \sqrt{V(X)}$ after evaluating the variance from the table. | Q024 B | Q039 B |
| Q007 Use $E(aX + b) = aEX + b$; calculate EX^2 directly from the table. | Q015 Compute EX , EX^2 , then $\sigma X = \sqrt{E(X^2) - E(X)^2}$. | Q025 B | Q040 B |
| Q008 Apply linearity for $aX + b$ and use the weighted square sum for X^2 . | Q016 Use $V(aX + b) = a^2 VX$ and $\sigma(aX + b) = a \sigma X$. | Q026 B | Q041 C |
| | Q017 Apply the squared multiplier to VX and the absolute multiplier to σX . | Q027 B | Q042 A |
| | Q018 Use the variance and standard-deviation transformation formulas for each Y. | Q028 B | Quiz WRITTEN 1 $EX = 3/2$ and $VX = 3/4$ |
| | | Q029 C | Quiz WRITTEN 2 $VY = 32$ and $\sigma Y = 4\sqrt{2}$ |
| | | Q030 C | Quiz WRITTEN 3 12 |
| | | Q031 B | Quiz WRITTEN 4 10 |
| | | Q032 C | Quiz MCQ 5 D |
| | | Q033 A | Quiz MCQ 6 C |
| | | | Quiz MCQ 7 B |
| | | | Quiz MCQ 8 B |

Concept 2 | Expected Value and Variance of Multiple Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$E(aX + b) = aE(X) + b$$

Variance

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Independent

$$\text{Var}(X + Y) = \text{Var } X + \text{Var } Y$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q001 Written

Given the probability distributions of X and Y in the given tables, find $E(X+Y)$.

Solution

1. Find EX and EY separately, then add them.

Final answer

$$EX = 35/3, EY = 9/2, \text{ so } E(X + Y) = 97/6.$$

Q002 Written

Three coins are tossed once; let X , Y and Z be the head indicators. Find $E(X+Y+Z)$.

Solution

1. Each fair head indicator has expected value $1/2$.

Final answer

$$E(X + Y + Z) = 1/2 + 1/2 + 1/2 = 3/2.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q003 Written

For the two given tables with $X=1,2,3$ and $Y=4,5,6$, find $E(X+Y)$.

Solution

1. The expectation of a sum is the sum of the expectations.

Final answer

$$E(X + Y) = EX + EY = 2 + 5 = 7.$$

Q004 Written

For the two bags in the given exercise, find the expected total number of red balls drawn.

Solution

1. Use $E(\text{red drawn}) = \text{sample size} \times \text{the red proportion in each bag}$.

Final answer

Bag A: $6/7$; Bag B: $6/7$; total $E = 12/7$.

Worked Solutions

Expected Value and Variance of Multiple Variables

Q005 Written

Three independent subsystems each receive two signals with success probability 0.5. Find $E(X+Y+Z)$.

Solution

1. Add the three binomial means.

Final answer

Each mean is $2(0.5) = 1$, so $E(X + Y + Z) = 3$.

Q006 Written

Two 500-point coins and three 100-point coins are tossed. Find the expected total score Z .

Solution

1. Model the score as $aX+bY$, then apply linearity.

Final answer

$Z = 500X + 100Y$, so $EZ = 5001 + 1003/2 = 650$.

Worked Solutions

Expected Value and Variance of Multiple Variables

Q007 **Written****Two 500-point coins and two 100-point coins are tossed. Find EZ .****Solution**

1. The expected heads in each two-coin group is 1.

Final answer

$$EZ = 500(1) + 100(1) = 600.$$

Q008 **Written****A die is thrown twice and the score is $Z=3X+5Y$. Find EZ .****Solution**

1. Both die rolls have mean $7/2$.

Final answer

$$EZ = 3(7/2) + 5(7/2) = 28.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q009 Written

For three 100-point coins and one 50-point coin, find the expected score from heads.

Solution

1. Use the expected score of each fair coin and add.

Final answer

$$EZ = 350 + 25 = 175.$$

Q010 Written

In two independent raffles, buy two tickets from a 2/10 winner raffle and two from a 1/6 winner raffle. Find the expected prizes.

Solution

1. Expected winners are sample size times the winning proportion in each raffle.

Final answer

$$EZ = 22/10 + 21/6 = 11/15.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q011 Written

Three coins are tossed and a die is rolled; X is heads and Y is the die score. Find $E(XY)$.

Solution

1. Check independence before multiplying expectations.

Final answer

X and Y are independent, so $E(XY) = EXEY = 3/2 \times 7/2 = 21/4$.

Q012 Written

For independent X with probabilities $2/5, 2/5, 1/5$ at 1,2,3 and Y uniform on 2,4,6, find $E(XY)$.

Solution

1. Compute each mean from its table, then multiply.

Final answer

$EX = 8/5, EY = 4$, so $E(XY) = 32/5$.

Worked Solutions

Expected Value and Variance of Multiple Variables

Q013 Written

Ahmed tosses three coins and Mohamed tosses two coins. Let X and Y be their head counts. Find $E(XY)$.

Solution

1. The two experiments are independent.

Final answer

$$E(XY) = EXEY = 3/2 \cdot 3/2 = 9/4.$$

Q014 Written

A card numbered 1 to 4 is replaced and a second card is drawn. Find $E(XY)$.

Solution

1. Replacement makes the two draws independent.

Final answer

$$EX = EY = 5/2, \text{ so } E(XY) = 25/4.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q015 Written

For independent X and Y with $EX=1$ and $EY=2/7$, find $E(XY)$.

Solution

1. Use the product rule after checking independence.

Final answer

$$E(XY) = EXEY = 2/7.$$

Q016 Written

A card X is drawn from 1,2,3 and three-coin heads Y is independent. Find $V(X+Y)$ and $\sigma(X+Y)$.

Solution

1. For independent variables, add variances, then take the positive square root.

Final answer

$$VX=2/3, VY=3/4, \text{ so } V(X + Y) = 17/12 \text{ and } \sigma=\sqrt{17/12}.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q017 Written

Two independent dice are thrown. Find the variance and standard deviation of their sum.

Solution

1. Each fair die has variance $35/12$.

Final answer

$$V(X + Y) = 35/12 + 35/12 = 35/6; \sigma = \sqrt{35/6}.$$

Q018 Written

An independent coin head count X and die score Y are combined. Find $V(X+Y)$ and $\sigma(X+Y)$.

Solution

1. Use $V(\text{Bernoulli}(1/2)) = 1/4$ and $V(\text{die}) = 35/12$.

Final answer

$$V = 1/4 + 35/12 = 19/6; \sigma = \sqrt{19/6}.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q019 Written

Ahmed tosses two coins and Mohamed tosses three coins. Find the variance of the total head count.

Solution

1. Variances of independent Bernoulli counts add.

Final answer

$$V = 2/4 + 3/4 = 5/4.$$

Q020 Written

Independent X is uniform on 1 to 4 and Y on 1 to 6. Find $V(X+Y)$ and $\sigma(X+Y)$.

Solution

1. Use the uniform discrete variance for each scale and add.

Final answer

$$V = 5/4 + 35/12 = 25/6; \sigma = 5/\sqrt{6}.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q021 Written

Three independent special dice take values 2,4,6 equally. Find $E(XYZ)$ and $V(X+Y+Z)$.

Solution

1. For a product multiply means; for a sum add variances.

Final answer

$EX=4$, $VX=8/3$; $E(XYZ) = 64$ and $V(sum) = 8$.

Q022 Written

Three independent dice take values 1,3,5 equally. Find $E(XYZ)$ and $V(X+Y+Z)$.

Solution

1. All three distributions are identical and independent.

Final answer

$EX=3$, $VX=8/3$; $E(XYZ) = 27$ and $V(sum) = 8$.

Worked Solutions

Expected Value and Variance of Multiple Variables

Q023 Written

Three ordinary dice are thrown. Find the expected product and the variance of their sum.

Solution

1. Use independence for the product and variance addition for the sum.

Final answer

$$E(\text{product}) = 7/2^3 = 343/8; V(\text{sum}) = 335/12 = 35/4.$$

Q024 Written

Three friends have 5, 4 and 3 independent rolls to score a six. Find $E(XYZ)$ and $V(X+Y+Z)$.

Solution

1. For Binomial($n, 1/6$), use $E = np$ and $V = np(1 - p)$.

Final answer

$$\text{Means are } 5/6, 2/3, 1/2; E(\text{product}) = 5/18 \text{ and } V(\text{sum}) = 5/3.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q025 **Written**

If X, Y and Z are independent head counts from 4, 3 and 2 fair coins, find $E(XYZ)$.

Solution

1. Multiply the three expected head counts.

Final answer

$$E(XYZ) = 23/21=3.$$

Q026 **MCQ**

$E(X+Y)$ equals:

Solution

1. Add the two expectations.

Correct option: B

Final answer

$$EX + EY$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q027 MCQ

For three variables, $E(X+Y+Z)$ equals:**Solution**

1. Extend linearity to three terms.

Correct option: B

Final answer

$$EX + EY + EZ$$

Q028 MCQ

The expected heads in one fair coin toss is:**Solution**

1. Use the Bernoulli mean.

Correct option: B

Final answer

$$1/2$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q029 MCQ

For two signals with $p=0.5$, EX is:**Solution**1. Use np .**Correct option:** B**Final answer**

1

Q030 MCQ

If $EX=2$ and $EY=5$, $E(X+Y)$ is:**Solution**

1. Add the means.

Correct option: B**Final answer**

7

Worked Solutions

Expected Value and Variance of Multiple Variables

Q031 MCQ

 $E(aX+bY)$ equals:**Solution**

1. Apply linearity.

Correct option: A

Final answer

$$aEX + bEY$$

Q032 MCQ

The expected score of one fair 100-point coin is:**Solution**

1. Half the value is expected.

Correct option: B

Final answer

50

Worked Solutions

Expected Value and Variance of Multiple Variables

Q033 MCQ

For a fair die, $E(3X+5Y)$ with two rolls is:**Solution**

1. Use both means $7/2$.

Correct option: C**Final answer**

28

Q034 MCQ

A fixed entry fee changes:**Solution**

1. It is a constant term.

Correct option: B**Final answer**

the mean only

Worked Solutions

Expected Value and Variance of Multiple Variables

Q035 MCQ

The expected number of winners in 2 of 10 tickets with 2 winners is:

Solution

1. Use sample size times winner proportion.

Correct option: B

Final answer

2/5

Q036 MCQ

$E(XY) = E(X)E(Y)$ requires:

Solution

1. The product rule needs mutual independence.

Correct option: B

Final answer

independence

Worked Solutions

Expected Value and Variance of Multiple Variables

Q037 MCQ

For three fair coins, EX is:**Solution**

1. Use np.

Correct option: C**Final answer** $3/2$

Q038 MCQ

For a fair die, EY is:**Solution**

1. Average 1 through 6.

Correct option: C**Final answer**

3.5

Worked Solutions

Expected Value and Variance of Multiple Variables

Q039 MCQ

With replacement, two card draws are:**Solution**

1. Replacement restores the same distribution and independence.

Correct option: B

Final answer

independent

Q040 MCQ

If $EX=2$ and $EY=3$ for independent variables, $E(XY)$ is:**Solution**

1. Multiply the means.

Correct option: B

Final answer

6

Worked Solutions

Expected Value and Variance of Multiple Variables

Q041 MCQ

For independent X, Y , $V(X+Y)$ equals:**Solution**

1. Add independent variances.

Correct option: B**Final answer**

$$VX + VY$$

Q042 MCQ

The variance of a fair die is:**Solution**

1. Use the standard die variance.

Correct option: B**Final answer**

$$35/12$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q043 MCQ

Sigma(X+Y) is:**Solution**

1. Take the positive root.

Correct option: B

Final answer $\sqrt{V(X + Y)}$

Q044 MCQ

The variance of the sum of two independent fair dice is:**Solution**

1. Double 35/12.

Correct option: B

Final answer $35/6$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q045 MCQ

For independent sums, a constant shift changes:**Solution**

1. Translations do not change spread.

Correct option: C**Final answer**

neither spread measure

Q046 MCQ

For independent X, Y, Z , $E(XYZ)$ equals:**Solution**

1. Multiply the three means.

Correct option: B**Final answer** $E(X)E(Y)E(Z)$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q047 MCQ

For independent X, Y, Z , $V(X+Y+Z)$ equals:**Solution**

1. Add the three variances.

Correct option: B

Final answer

$$VX + VY + VZ$$

Q048 MCQ

For a binomial count with $n=5$ and $p=1/6$, EX is:**Solution**1. Use np .

Correct option: B

Final answer

$$5/6$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q049 MCQ

For three values 2,4,6 equally likely, EX is:**Solution**

1. Average the three values.

Correct option: C**Final answer**

4

Q050 MCQ

For three independent fair dice, $V(\text{sum})$ is:**Solution**

1. Add three copies of $35/12$.

Correct option: C**Final answer** $35/4$

Worked Solutions

Expected Value and Variance of Multiple Variables

Quiz WRITTEN 1 Concept Quiz · Written

If $EX=4$ and $EY=7$, find $E(3X-2Y)$.

Solution

1. Use linearity: 34-27.

Final answer

-2

Quiz WRITTEN 2 Concept Quiz · Written

For independent variables with $EX=2$ and $EY=5$, find $E(XY)$.

Solution

1. Multiply the means.

Final answer

10

Worked Solutions

Expected Value and Variance of Multiple Variables

Quiz WRITTEN 3 Concept Quiz · Written

Two independent fair dice are thrown. Find $V(X+Y)$ and $\sigma(X+Y)$.

Solution

1. Add the two die variances.

Final answer

$$V = 35/6, \sigma = \sqrt{35/6}$$

Quiz WRITTEN 4 Concept Quiz · Written

Three independent Binomial counts have $(n,p)=(2,1/2),(3,1/2),(4,1/2)$. Find $V(X+Y+Z)$.

Solution

1. Add $np(1-p)$ for the three counts.

Final answer

$$9/4$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Quiz MCQ 5 [Concept Quiz · MCQ](#)

If $EX=3$ and $EY=4$, $E(2X+Y)$ is:

Solution

1. Compute $2 \times 3 + 4$.

Correct option: B

Final answer

10

Quiz MCQ 6 [Concept Quiz · MCQ](#)

For independent variables $EX=2$, $EY=5$, $E(XY)$ is:

Solution

1. Multiply the means.

Correct option: B

Final answer

10

Worked Solutions

Expected Value and Variance of Multiple Variables

Quiz MCQ 7 [Concept Quiz · MCQ](#)

For two independent die rolls, $V(X+Y)$ is:

Solution

1. Add the two variances.

Correct option: B

Final answer

35/6

Quiz MCQ 8 [Concept Quiz · MCQ](#)

For three independent variables, $V(X+Y+Z)$ is:

Solution

1. Extend variance addition.

Correct option: B

Final answer

the sum of variances

Answer key

Use the question number shown on each page.

Q001 $EX = 35/3, EY = 9/2, \text{ so } E(X + Y) = 97/6.$ Q016 $VX=2/3, VY=3/4, \text{ so } V(X + Y) = 17/12.$ Q025 $E(XYZ) = 23/21=3.$

Q002 $E(X + Y + Z) = 1/2 + 1/2 + 1/2 = 3/2.$ and $\sigma = \sqrt{17/12}.$ Q026 B

Q003 $E(X + Y) = EX + EY = 2 + 5 = 7.$ Q017 $V(X + Y) = 35/12 + 35/12 = 35/6;$ Q027 B

Q004 Bag A: $6/7$; Bag B: $6/7$; total $E = 12/7.$ $\sigma = \sqrt{35/6}.$ Q028 B

Q005 Each mean is $2(0.5) = 1$, so $E(X + Y + Z) = 3.$ Q018 $V = 1/4 + 35/12 = 19/6;$ Q029 B
 $\sigma = \sqrt{19/6}.$ Q030 B

Q006 $Z = 500X + 100Y, \text{ so } EZ = 5001 + 1003/2 = 6001.$ Q019 $V = 21/4 + 31/4 = 5/4.$ Q031 A

Q007 $EZ = 5001 + 1001 = 6001.$ Q020 $V = 5/4 + 35/12 = 25/6;$ Q032 B

Q008 $EZ = 37/2 + 57/2 = 28.$ $\sigma = 5/\sqrt{6}.$ Q033 C

Q009 $EZ = 350 + 25 = 175.$ Q021 $EX=4, VX=8/3; E(XYZ) = 64 \text{ and } V(sum) = 8.$ Q034 B

Q010 $EZ = 22/10 + 21/6 = 11/5.$ Q022 $EX=3, VX=8/3; E(XYZ) = 27 \text{ and } V(sum) = 8.$ Q035 B

Q011 X and Y are independent, so $E(XY) = EXEY = 3/27 \cdot 2 = 21/4.$ Q023 $E(product) = 7/2^3 = 343/8;$ Q036 B
 $V(sum) = 335/12 = 35/4.$ Q037 C

Q012 $EX = 8/5, EY = 4, \text{ so } E(XY) = 32/5.$ Q024 Means are $5/6, 2/3, 1/2;$ Q038 C

Q013 $E(XY) = EXEY = 3/21 = 3/2.$ $E(product) = 5/18 \text{ and } V(sum) = 5/3.$ Q039 B

Q014 $EX = EY = 5/2, \text{ so } E(XY) = 25/4.$ Q040 B

Q015 $E(XY) = EXEY = 2/7.$ Q041 B

Q042 B

Q043 B

Q044 B

Q045 C

Q046 B

Q047 B

Q048 B

Q049 C

Q050 C

Quiz WRITTEN 1 -2

Quiz WRITTEN 2 10

Quiz WRITTEN 3 $V = 35/6, \sigma = \sqrt{35/6}$

Quiz WRITTEN 4 $9/4$

Quiz MCQ 5 B

Quiz MCQ 6 B

Quiz MCQ 7 B

Quiz MCQ 8 B